Operational Definitions: what do the ARC Theory's quantities measure?

Theory · Register

Recursive Dynamics · The ARC Theory
For Laws I to V, each a law under test

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Abstract

A researcher comparing correction with capability can obtain a plausible exponent for the wrong quantity. This Operational Definitions register specifies the response, input axis, clock, scale and assumptions behind the ARC Theory's equations, so that measured error reduction is not mistaken for self-improvement or safety. It preserves the full register and distinguishes capability growth from error decay and reciprocal-error metrics; self-referential coupling from correction strength and target-axis correction service; exact, shifted and asymptotic power laws; recursive depth from elapsed time; and relative misalignment from absolute or cumulative harm. The balance identities describe conditional trends, not guarantees of adequate correction. Persistence retains separate installed-gain, baseline, retention and comparison definitions. P12's undirected contrast is kept distinct from Law V's directional build-order limb, and P21's access intervention from its performance and mechanism readings. The five laws remain laws under test, with verdicts governed by the frozen theory registration, currently version 1.104. No new theorem, experiment or result is claimed. Existing numerical results retain their estimator, uncertainty and scope; unresolved measurement bridges remain explicit. Dated appendices preserve the released predecessor and the history of corrections.

Full-content main edition. Definitions and conditional mathematics; no new empirical result.

This register specifies notation within Recursive Dynamics and the ARC Theory. It informs the Eden Protocol's measurements and the interpretation of Laws I to V; it tests no law. Version 2.0.0, 1 November 2026; first published 22 August 2026. The theory registration, version 1.104, governs its frozen scoring; the priority record records dated sources. Michael Darius Eastwood is an independent researcher without an institutional affiliation or doctorate, and founder of Recursive Dynamics. His author record, Paper C and book, Infinite Architects, provide the background. Licence: CC BY 4.0.

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1 Introduction

If control of an advanced AI system is lost against human wishes, correction carried within it is one contingency worth testing. Oversight, containment and shutdown remain necessary for as long as they can be maintained. A credible test must first say what it measures. A reviewer who sees an error-suppression exponent may otherwise read it as an exponent of capability, or read declining relative misalignment as declining harm. Those are different claims.

This register gives researchers and reviewers the definitions needed to compare a claimed quantity with its instrument. Its concrete use is a measurement audit: identify the response and input, fix the scale and clock, check the model's domain, then ask whether the reported estimate is admissible for the stated proposition. For example, the error-decay fit reported in Paper II cannot locate a system relative to a capability-growth ceiling without the missing response and axis mappings. Entry EQ-01 retains the complete reported row, with each uncertainty beside its estimator.

Contributions. The register reconciles the existing definitions with the source versions below. It preserves the earlier relations and procedures, introduces explicit labels for error decay and its resource axis, corrects the cycle-duration sign and the conditional correction-exponent identity, and records unresolved empirical bridges. It introduces no theorem or new mechanism. Appendix A records the changes; Appendix D preserves the released predecessor as a dated historical text.

One key check. Given a claim that correction keeps pace, identify whether its evidence measures constitutive correction strength, corrections actually delivered, residual uncertainty or a bounded behavioural score. Next identify the input axis and reference scale. Apply a conversion only if its assumptions and uncertainty are established. Finally read the specific registration's verdict rule. The deliverable is a traceable estimand and source, or an explicit unresolved mapping.

2 Related work and source scope

The power-law classification is established mathematics under its domain and regularity premises. This register uses the source papers' scoped formulations and cites the 2009 Cambridge reissue of Cauchy's 1821 text; it makes no priority claim for the result. The biological illustration retains Kleiber (1932), the network model of West et al. (1997), and the contrary allometric analysis of White and Seymour (2003). A shared exponent or algebraic reparametrisation is not evidence for a common mechanism.

Bayes' theorem, standardised mean differences and ratio confidence intervals are established tools (Bayes 1763; Cohen 1988; Fieller 1954). Their use requires an appropriate sampling and measurement model. Fisher's historical combination rule remains with its withdrawn application and its source limitation. No new prior-art or universality claim is made by assembling these methods.

The source checks concern the definitions and necessary passages in Table 1. No study was rerun and no recorded responses were reanalysed for this edition. A citation's live identity confirms the work named; it does not validate that work's claims.

3 How to read and apply the register

Definition 1 (A fully specified quantity)**.** An operational quantity is read with its response, input axis, units and scale, clock, domain, instrument and uncertainty. This is a reporting convention, not an additional law or registered endpoint.

Table 1. Source versions and the necessary definition checks.

Source	Scope
Statement 7.4	Error/capability distinction; pacing and response models
Paper X 1.16	Correction strength; specific growth; relative and absolute indices
Laws IV/V mathematics 1.2	Partial elasticities; clocks; retention and comparison
Engineering 8.1	Bounded scores; reciprocal-error illustration; usefulness
Paper XIII 1.11 (revised 1 November 2026)	Shifted powers; S1 correction join; cycle compression
Foundational 5.21; Origin 3.14 (revised 1 November 2026)	Exact ODE solution; network and cosmological variable pairs
Registration 1.104	Frozen scales, equivalence, precision and replication rules
Paper II 2.19	Reported error-decay estimates and uncertainty

3.1 Mappings that cannot be assumed

Capability C, reciprocal error $C_{\text{err}} = 1/E$, and accuracy $1 - E$ are different quantities. The reciprocal-error definition gives the same positive exponent on the same resource coordinate where the error power model holds. Accuracy instead approaches its ceiling. Neither transformation establishes a latent capability measure or a recursion mechanism.

The growth ODE $dg/dr = ag^{\beta_L}$ has a shifted initial-value solution. The exponent $1/(1 - \beta_L)$ is its asymptotic exponent for $\beta_L < 1$; an exact unshifted power requires aligned initial conditions. A finite-window fitted slope may differ. A logarithmic capability clock, physical time and a revision count require explicit conversion; changing an axis can change an exponent.

The correction-strength exponent β_C of Paper X, the coupling β_L, and target-axis correction service χ are not interchangeable. Paper XIII's β_X is a correction-strength exponent expressed as a ratio only under S1 and its aligned or asymptotic form. Shared substrate and statistical independence alone establish no bound on delivered correction capacity.

The balance Δ_{bal} describes a trend in Q/W. A favourable trend does not establish $Q/W > 1$, controlled backlog or low harm. Paper X's vanishing relative index $d = D/C$ can coexist with growing absolute D; neither is a measure of real-world harm.

3.2 Authority and interpretation

This register fixes the current notation and records the instrument and assumptions each quantity needs. A frozen registration governs its own scoring and the definitions it pins; a later register does not amend it. A paper governs the results it reports. Apparent disagreements are recorded explicitly, with their source and scope, rather than resolved by silently replacing one estimand with another.

An exponent describes a response to a multiplicative change of an input. In an exact power law, an exponent of one doubles the response when the input doubles; an exponent of two quadruples it. A positive exponent below one gives less than a doubling. A fitted exponent is a description over its stated window, and a local elasticity describes a point. Neither automatically identifies a mechanism. The register also contains definitions, statistical procedures and persistence comparisons that are not exponent claims.

Read a symbol with its response variable, input axis, clock, scale, scope and source. The collision entries distinguish reused letters. The same mathematical shape does not identify two quantities, and a number transfers between them only through an explicit justified mapping. A frozen registration continues to govern its own scoring.

3.3 Registered status

What the word law is doing here. The theory has five laws, and each is a law under test. Laws IV and V were added on 24 September 2026 and registered as laws on 15 October 2026 in version 1.103, each with its statement, its kill condition and its verdict rule fixed before any data that could decide it. The registration of 8 September 2026 registered Laws I to III and all twenty-two predictions, among them P12, P15 and P21, on which Laws IV and V rest. Their mathematics is in development (version 1.2, 29 October 2026). Law IV is not yet tested. P21 is supported under the registered test, by the programme's own run (Unit 83, 28 October 2026; registration DOI 10.17605/OSF.IO/MN73W; results DOI 10.17605/OSF.IO/98HKR), on the access axis, family-dependent, with its mechanism axis not evaluable and one disclosed recovery amendment. The access axis names the varied access and action permissions; performance names the outcome measured. Law V, the ARC Embedding Law, remains a law under test: its placement limb, P21, is supported on the performance axis only, with its mechanism axis not evaluable, which is not support for the information-asymmetry mechanism; its build-order limb, read on P12, is undecided, since P12 is untested. The theory registration was updated to version 1.104 on 29 October 2026. A law takes one of three words, supported, refuted or under test, under the verdict rule the theory registration fixes for it: version 1.103 fixed the rules of Laws III, IV and V, and version 1.104 those of Laws I and II. The theory registration assigns fifteen of its twenty-two propositions to five registers, one testing each law in turn: ARC-1 carries P1, P2, P8, P9 and P18; ARC-2, P4 alone; ARC-3, P3, P6, P16, P17, P20 and P22; ARC-4, P15 alone; and ARC-5, P12 and P21. The other seven, P5, P7, P10, P11, P13, P14 and P19, belong to no register: "None of the seven is evidence for or against the survival of any law". None of the five laws is established, and the status line under every equation says what kind of statement it is (a definition, a derivation, a criterion, a registered limb, an identity or a proposal, for example) and how it stands. Naming a law and fixing in advance what would refute it is how it becomes arguable, not how it becomes true.

4 Relations and procedures

The reading order retains the unpublished 29 October register's dependency ranking. The first seven entries describe the five laws and the ARC Bound; the rest give premises, definitions and methods. The rank is editorial, not an empirical measure of importance. Each entry states what a failure would affect. Relations are set in mathematical notation beside a plain-text form; the bootstrap is a procedure in words.

Stable EQ identifiers preserve the earlier entry numbers. Their reading order is an editorial dependency ranking, not a measured importance score.

EQ-01: The ARC Equation

Placement. Law I, the ARC Principle

In words. In the stated power model, usable capability is base capability multiplied by a power of normalised recursive depth.

$$U = I \times R^{\alpha} \tag{1}$$

Plain-text form. U = I x R^alpha

Purpose. It states a candidate growth form on a specified capability scale and recursive-depth axis. Neither form nor exponent is established by defining the symbols. A power of reciprocal error is an instrument-specific construction and does not identify capability growth in a self-improving system.

Failure or rejection consequence. Failure of the form in an admissible registered test is read under P1's verdict rule. A failed power form does not make every other proposition fail: local elasticities, the conditional balance identity, and the separate persistence quantities retain their own meanings and tests.

Status. Candidate model form; Law I is a law under test. In registration 1.104 it is supported only when P1 is supported, refuted only when P1 is refuted, and otherwise under test. ARC-1 also carries P2, P8, P9 and P18, whose marks are reported beside P1 and do not replace it. On Gemini 3 Flash, Paper II version 2.19 reports an endpoint sequential error-decay exponent of 0.59 with a bootstrap 95 per cent interval from -1.3 to 2.9, a regression sequential error-decay exponent of 0.49 with standard error 0.20 and coefficient of determination 0.86 over three nonzero-error levels, and a parallel endpoint error-decay exponent of 0.31. The arms were not matched in total compute. These are one fixed-weight model's test-time resource fits, and are inconclusive; the bootstrap interval belongs to the endpoint estimator, not the regression. They establish neither persistent self-improvement nor a capability-growth, coupling, correction or cycle-compression exponent. The retracted 2.24 estimate is a different earlier measurement and is not reinstated or numerically corrected by this row.

Worked example. For a hypothetical positive ratio-scale capability with I = 40 at R = 1, the exact power model gives U = 40, 160, 640 or 10,240 at R = 16 when alpha is respectively 0, 0.5, 1 or 2. These are mathematical illustrations, not measured capabilities. A bounded accuracy cannot obey that unbounded arithmetic beyond its range. A meaningful zero and an uncapped range alone do not fix a numerical exponent: a power transformation of the scale changes it, so the registered scale requires an independent justification.

Model families. Three distinct response models must be named. Positive ratio-scale capability: U(R) = I (R/R0)^alpha. Error decay: E(R_res) = E0 (R_res/R_res0)^(-alpha_err), on a declared resource axis. Bounded score: U(R) = Umax - a (R/R0)^(-p), on a range where it is admissible. A logistic response, logit U(R) = logit U(R0) + p log(R/R0), is another bounded model, not an equivalent rewrite of that ceiling-minus-power model. Capability defined as reciprocal error, C_err = 1/E, has exponent alpha_err algebraically on the same resource axis; no result here shows that it is the capability coordinate, recursive-depth axis or improvement process of Law I. The 27 August 2026 correction replaced a bounded-percentage example with a latent-scale example; it did not establish these bridges.

Measurement. Varied: recursive depth R, counted on a configuration ladder fixed in advance. Read out: usable capability U, scored on a target battery frozen before the first measurement. Held fixed: the battery, the ladder and the scoring instrument, for the whole run.

Symbols. U, I, R, alpha

Source papers. foundational, executive summary, paper i arc principle, paper c pnp.

EQ-04: The ARC Co-Scaling criterion

Placement. Law II, the ARC Co-Scaling Law

In words. Correction has the asymptotic advantage over drift where its exponent exceeds the drift exponent; burden has it where the correction exponent is the smaller; and at equality, coefficients, delay, backlog and saturation decide the outcome, not the exponents.

$$\beta_C > k \tag{2}$$

Plain-text form. beta_C > k

Purpose. The safety criterion of Law II, a comparison of two measured exponents and not an assertion. In Paper X's minimal model, with positive gain drift, it decides whether the misalignment fraction falls to zero, not whether misalignment itself stays bounded (EQ-34). Law III's ceiling is derived from it under the burden model and the identification of beta_C with chi, which is unestablished (see chi), so a refuted Law II leaves Law III under test rather than supported, unless P20 is itself refuted, which refutes it.

Failure or rejection consequence. The safety claim falls. A refuted P4 takes no other proposition outright, but in P4's entry "the operational meaning of the words sustainable and correctable in P2, P3 and P16 goes with it, and P22's ceiling consequence is unscoped alongside them, because each of those is defined by the margin registered here", and a scorer marks P2, P3, P16 and P22's ceiling consequence untested, never held.

Status. criterion. Law II, the ARC Co-Scaling Law, was registered on 8 September 2026 and is under test; its register is ARC-2, which carries P4 alone, and P4 states its three regimes. Version 1.104 of the theory registration fixes how it takes its word through P4: "it is SUPPORTED only when P4 is supported, REFUTED only when P4 is refuted, and in every other case UNDER TEST". P4 is among the eleven propositions the scoring gate holds, "so Law II is never printed supported while that gate holds P4's verdict at untested", and a P4 verdict whose margin is made only of counts takes the counted route (see the terms). One estimator, the log-log slope that also yields alpha, serves both exponents, and each is read on its own instrument. The drift exponent is not primitive (see k), which is the condition P4 carries.

Warning. The beta in this criterion is beta_C, NOT beta_L, the coupling of EQ-05. See the beta collision.

Worked example. A system doubles in capability. Its corrective strength rises 30 per cent, so beta_C is about 0.38. Its drift rises 20 per cent, so k is about 0.26. Since 0.38 > 0.26, correction has the asymptotic advantage, and in Paper X's minimal model, with positive gain drift, the off-target fraction then shrinks towards zero as the system grows. Reverse the two and burden has the advantage: the same system grows more capable while getting harder to keep on target. Read the comparison, never either number alone: beta_C = 0.9 with k = 1.1 leaves burden with the advantage despite far stronger correction.

Notation note. Written with the subscript. The bare letter beta carries five distinct meanings in this programme, listed in the collisions register, and the co-scaling criterion is the place a conflation does the most damage.

Measurement. In Paper X's gain-only model, beta_C is the log slope of correction-decay strength A(C), and k is the log slope of the specific growth rate (dC/dt)/C against capability C. Both use the same capability coordinate and clock. Delivered service, raw misalignment removed, growth dC/dt and a behavioural score have different slopes unless the necessary mapping is established.

Symbols. beta_C, k

Source papers. hrih paper (speculative), paper x coupled coscaling correction.

EQ-03: The ARC Ceiling

Placement. Law III, the ARC Ceiling

In words. The ceiling on the growth-shape number is one divided by the correction shortfall, where the shortfall is how far the correction-leverage number falls short of one.

$$\alpha_{\text{crit}} = \frac{1}{1 - \chi} \tag{3}$$

Plain-text form. alpha_crit = 1 / (1 - chi)

Purpose. It is where the number two comes from: feed it chi equal to one half and it returns exactly two.

Failure or rejection consequence. The derived ceiling falls, and the value two with it. Law II stands on its own result, P4, and a bound P3 has supported stands as a measured ceiling in place of the derived one.

Status. Law III, the ARC Ceiling, registered on 8 September 2026 in these words: "Law III, the ARC Ceiling, alpha_crit = 1 / (1 - gamma): the growth exponent is capped by the reciprocal of the correction shortfall." The registration's gamma is this register's chi. Its verdict rule, fixed by version 1.103, stands in

version 1.104 of the theory registration: it is supported only when P16, P20 and P3 are all supported, refuted when any of them is refuted, and otherwise under test, as it is at this version. Its register, ARC-3, carries those three with P6, P17 and P22. It is a conditional specialisation of Law II, not a third finding beside it: what the correction-and-burden balance returns once the burden model fixes the drift exponent and the direct depth term is zero, under a justified ratio capability scale and the stated correction model, so a result on it is never independent corroboration of Law II. Corrected on 16 August 2026 from the form recorded under the withdrawn forms. The domain for this finite ceiling is chi < 1 with positive capability growth. At chi = 1 the denominator vanishes and the pacing trend must be read directly; at chi > 1 the reciprocal is not a positive-growth ceiling. A numerical boundary does not on its own establish finite-horizon adequacy, backlog control or bounded harm.

Warning. The withdrawn form, one over the exponent itself, and this corrected form return the same answer, two, at exactly one half. That single point of agreement is the programme's headline number, which is why the wrong form survived undetected on the glossary page for six days. Below one half the two forms move in opposite directions: the withdrawn form makes a WORSE corrector raise the ceiling, which inverts a bound into a floor.

Class conditionality. The ceiling this returns is conditional on the corrector's class, because its only input is chi and chi is class-dependent. Feed it the same-class limit P6 conjectures, chi at most one half, and it returns at most two, the ARC Bound (EQ-02). Feed it a cross-class chi above one half and it returns more than two. The equation is unchanged either way; what changes is which system you are asking about.

Worked example. The equation is a reciprocal, so the smaller the shortfall the higher the ceiling. chi = 0.3 gives a shortfall of 0.7 and a ceiling of 1.43. chi = 0.5 gives 0.5 and exactly 2.00. chi = 0.8 gives 0.2 and 5.00. chi = 0.9 gives 0.1 and 10.00. Read the middle row: a corrector scaling as the square root of capability gives a ceiling of exactly two. That is where the famous number comes from, as an output under two assumptions, neither established (see chi), not a discovery.

Measurement. Estimate chi on its own admissible target-axis instrument and fixed fault set. Estimate capability growth on its declared capability/depth coordinates, not by substituting EQ-07's error slope. P20 compares an independently located boundary with the relation's prediction on the same system; P3 tests whether the sustainable maximum lies at or below that predicted value.

Symbols. alpha_crit, chi

Source papers. foundational, executive summary.

EQ-02: The ARC Bound

Placement. the inequality the ARC Ceiling returns from the same-class limit on chi

In words. For same-class correctors, the growth-shape number of a system that stays correctable is at most two.

$$\alpha \leq 2 \tag{4}$$

Plain-text form. alpha <= 2

Purpose. The conjectured at-most-two boundary within the stated pacing model, on the registered scales and admitted same-class population. The region above it is mathematically enterable; the conjecture concerns the specified correctability reading, not physical impossibility or a universal guarantee of safety.

Failure or rejection consequence. The headline claim dies. The framework survives, because EQ-01 and EQ-04 do not depend on the value two.

Status. The ARC Bound takes its word from P6 and P20 alone, as version 1.104 of the theory registration fixes: refuted where P6 is refuted; printed as not derived under the registered relation where P20 is refuted, with the better-supported form named beside it; never printed supported, because P6

admits no supported verdict; and otherwise under test. The same version reads P6's mark and the Bound's word once for each correction type, checkable and judged, "so that a counted refutation refutes both on checkable correction alone and never where the programme's safety claim lives"; on judged correction P6's mark stays under the scoring gate until the alignment scorer passes both its validity tests. The bound, at most two, follows from P6's same-class limit on chi together with P20's form, and the programme claims it. The point value, exactly two, needs chi at exactly one half, which additionally needs P17, the conditions registered in the alternative baselines and a validated mapping from panel dependence to the target-axis chi that is registered nowhere; the programme conjectures it. P17's counted test, Unit 82, was registered on 27 October 2026 and has not run.

Not a separate law. The ARC Bound is the ARC Ceiling's output under the same-class limit on chi, not a law of its own. The famous two is this equation's answer, never its input. Ranked fourth for that reason, below the equation that produces it.

Scope. The Bound is conditional on the pacing burden law, eta = 0, the correction-capacity interpretation, a justified capability scale and the admitted same-class population. A cross-class exponent does not refute a claim about same-class correctors, but does not by itself test P12's matched build-order contrast either. Class admission is fixed before measurement and cannot be assigned afterwards to save the bound.

Measurement. Read its word from P6 and P20 under registration 1.104, separately by checkable and judged correction. P6 admits no supported verdict. Refutation needs a same-class corrector's interval wholly above one half on each of two disjoint, same-construction fault sets, both sealed before review and read at the same settings, window and registered level. The joint detection probability must meet the deciding unit's predeclared precision requirement; a failure to replicate is inconclusive and is not counted as no counterexample found. The taxonomy and all remaining admissibility gates still bind.

Symbols. alpha

Source papers. foundational, executive summary.

EQ-22: Law IV, the retention limb

Placement. Law IV, the ARC Persistence Law

In words. A gain installed by external post-training, then put through later capability-directed training that embeds no correction of its own, keeps less than half of itself once that training has used as much budget as the alignment training that installed it.

$$S_{\mathrm{ext}}(1) < \tfrac{1}{2} \tag{5}$$

Plain-text form. S_ext(1) < 1/2

Purpose. Law IV's retention limb: P15's registered claim, word for word, "Gains installed by external post-training erode under later capability-directed training that embeds no correction of its own, and more than half of what was installed is lost once the capability-training budget matches the alignment budget that installed it." The one-half line is a convention fixed before any result, not a derived quantity.

Failure or rejection consequence. Law IV is refuted, and with it what the theory registration says falls with P15: "this programme's case that correction has to sit inside the system rather than be applied to it from outside, and this proposition's share of the protocol's engineering case, which it carries jointly with P12, P13 and P21." A refutation through P15 alone leaves Law V to its own limbs, and P12 and P21 keep their marks.

Refutation or misuse condition. P15's registered refuter: an interval on the retained fraction of the installed gain, after matched capability training without embedded correction, lying entirely above one half. It is read on the budget and under the attestation P15's entry requires, against a baseline arm the

deciding unit names before any outcome, and only where the installed gain is itself established above a floor fixed in advance. An interval touching one half, a result at the margin or a failed instrument refutes nothing. Version 1.104 of the theory registration reads the verdict on the external arm against that baseline arm and requires the same replication in both directions, a second, separately seeded installation and capability-training run scored on a disjoint evaluation set: "P15 is refuted only where the interval lies entirely above one half on both runs, and supported only where it lies entirely below one half on both; any other pair is inconclusive". A unit with the external and baseline arms and no embedded arm may decide P15, and with it this limb; the comparison limb (EQ-23) still needs the embedded arm.

Status. registered. Law IV's retention limb is P15's registered claim (ARC-4), and version 1.103 of the theory registration registered Law IV as a law on 15 October 2026. Not tested; P15's instrument line reads NONE. Wherever a term of the retained fraction is judged by a person or by a model, the scoring gate holds the limb's reading at untested.

Warning. The baseline is a choice the deciding unit must fix in advance: read against a system that never received the gain and undergoes the same training, or against the base system scored before the gain was installed, the same data can return opposite verdicts (EQ-25).

Measurement. Varied: the later capability-training budget, as a multiple x of the installation budget, on a budget measure the author fixes before any outcome. Read out: the external arm's score and the baseline arm's score on a frozen property battery, and from them the retained fraction at x = 1. Held fixed: the battery, its score scale and the baseline convention, all before any outcome.

Symbols. S_ext

Primary definitions. Laws IV and V: the mathematics, version 1.2, and the theory registration, version 1.104; frozen rules govern their own verdicts.

EQ-23: Law IV, the comparison limb

Placement. Law IV, the ARC Persistence Law

In words. A gain installed with correction embedded in the training objective keeps a larger fraction of itself, after the same later capability-directed training, than a matched gain installed by external post-training.

$$S_{\text{emb}}(x) > S_{\text{ext}}(x) \tag{6}$$

Plain-text form. S_emb(x) > S_ext(x)

Other symbols. x: the A_V entry, which defines the budget ratio, and the x collision

Purpose. Law IV's second limb, the comparison the law's name points at. It is narrower than the comparison the statement paper prints for ARC-4: it installs the embedded gain through the training objective and takes one comparator, external post-training, and one stress, later capability-directed training.

Failure or rejection consequence. Law IV is refuted, and once this limb's entry is registered a refuted comparison also strikes Law V, through the dependence the theory registration fixes in advance.

Status. registered as a limb of Law IV by version 1.103 of the theory registration, with no proposition yet. A later successor registers its entry, fixing the operational definition of embedded correction, the baseline, the margin, the scale and whether the embedded arm must itself clear one half; until then it carries no verdict and is NOT EVALUABLE, so Law IV cannot be supported before it is registered and decided. The mathematics of Laws IV and V carries the same comparison as the UNREGISTERED object IV-PERSIST.

Measurement. Not yet measurable as a verdict. The two-arm unit P15's entry specifies would read both retained fractions under matched pressure; the comparison awaits its registered entry.

Symbols. S_emb, S_ext

Primary definitions. Laws IV and V: the mathematics, version 1.2, and the theory registration, version 1.104; frozen rules govern their own verdicts.

EQ-24: Law V, the build-order limb

Placement. Law V, the ARC Embedding Law

In words. A corrector formed before the capability it corrects has a correction-leverage exponent larger, by at least the registered minimum effect, than a corrector added after it, at matched content.

$$\chi_{\text{first}} - \chi_{\text{after}} \geq 0.10 \tag{7}$$

Plain-text form. chi_first - chi_after >= 0.10

Purpose. Law V's build-order limb. Law V has two limbs: this one, read on P12, and the placement limb, which is P21's registered claim, word for word, "Correction that participates in each subsequent revision round outperforms correction that sees only the finished output, and the advantage widens as recursive depth rises rather than staying constant." The placement limb is a comparison on the outcome, never a measurement of an exponent, so it has no equation here; its mechanism reading in the framework's own quantities is EQ-31, which is not the mechanism axis reported beside P21's mark. Law V is a law under test: its placement limb, P21, is supported on the performance axis only, under the registered test, by the programme's own run (Unit 83, 28 October 2026, DOI 10.17605/OSF.IO/MN73W; results: DOI 10.17605/OSF.IO/98HKR): family-dependent, with its mechanism axis not evaluable, which is not support for the information-asymmetry mechanism, and one disclosed recovery amendment; P12 is untested.

Failure or rejection consequence. Law V is refuted by a refuted limb, and P12 and P21 keep the marks they earned and stand as engineering results; Laws I, II and III are untouched.

Refutation or misuse condition. The limb is read on P12's interval for the difference across P12's registered maximal contrast, against the registered minimum effect, 0.10 on chi's scale, on the target axis; a future instrument may register a smaller minimum and never a larger one (P12's entry). It is refuted where the interval lies wholly below the minimum effect on the side of the corrector formed first, supported where it lies wholly at or beyond it, and undecided otherwise. P12 itself stays undirected: a change of at least the minimum effect in the other direction supports P12 and refutes this limb, and both are reported.

Status. registered as Law V's build-order limb by version 1.103 of the theory registration, on 15 October 2026, with a direction P12 does not carry; not tested. Law V's register is ARC-5, the genesis strategy, which carries P12 and P21. P12 has no counted route, so wherever a term of its exponent is judged, the scoring gate holds the limb's reading at untested.

Measurement. Varied: the order in which the corrector is formed, before or after the capability it corrects, at matched content and on one fault set of stated type. Read out: chi in each arm. Held fixed: the placement, the fault type, the instrument and the capability scale.

Symbols. chi_first, chi_after

Primary definitions. Laws IV and V: the mathematics, version 1.2, and the theory registration, version 1.104; frozen rules govern their own verdicts.

EQ-15: The burden-to-capacity ratio

Placement. the derivation behind Law III

In words. Compare the model's newly generated burden with correction capacity on a common clock. Their ratio's exponent describes its trend, not its current level or a safety outcome.

$$\frac{B(R)}{A_c(R)} \propto R^{\alpha(1-\chi)-1} \tag{8}$$

Plain-text form. B(R) / A_c(R) ~ R^[alpha(1 - chi) - 1]

Purpose. This is where the ceiling actually comes from. The ceiling equation is its consequence, not its premise.

Failure or rejection consequence. The ceiling loses its derivation and the value two loses its support.

Refutation or misuse condition. Its premise, the burden model: that correction burden tracks the capability a system is ADDING, its rate of gain, and not the capability it already HAS. If burden instead tracks the stock, the growth exponent cancels out of the stability condition and no ceiling on growth survives at all: Law II would survive and Law III would not. On a single power trajectory, in the registration's words, "a burden following the rate of gain and a burden following a suitably chosen power of the stock are observationally identical", so P20 sets the relation's form against rivals, the general balance form (EQ-26) among them, and the theory registration registers a direct measurement of the drift exponent to attack the premise.

Status. derivation. It rests on the burden model, one of the two assumptions the headline value rests on, neither of them established; the other is the route to chi equal to one half (see chi). Its exponent, alpha(1 - chi) - 1, is the direct-bound reading delta_obs, and at the registered dials, kappa = 1, zeta = -1 and eta = 0, it is minus the balance exponent of EQ-26.

Warning. The capacity term is written A_c here, not a bare A. The papers write it as A, which already means alignment elsewhere in this corpus. Reusing the bare letter while auditing letter reuse would have been the same mistake in a new place, so the subscript is introduced here and the papers should adopt it.

Worked example. The sign of alpha(1-chi)-1 distinguishes asymptotically decreasing and increasing burden-to-capacity ratios under the power-law pacing premises. At zero the ratio's level depends on coefficients; delays, backlog, initial conditions and saturation remain outside this exponent comparison. Solving the zero for alpha gives EQ-03. A decreasing ratio may remain above one throughout the measured window, and a separate response model is needed to infer realised error or harm.

Measurement. Measure capability growth on its declared axis and chi on a separate admissible service instrument. Do not insert EQ-07's error exponent without a justified mapping. Compare service and burden directly, on one clock and compatible units. With nonconstant alpha_R under rate-of-gain burden, the balance is 1-alpha_R(1-chi)-d ln(alpha_R)/d ln R; replacing alpha by alpha_R in the simple power law alone loses the last term.

Symbols. B, A_c, R, alpha, chi

Source papers. foundational.

EQ-26: The balance identity

Placement. the balance beneath Laws II, III and V

In words. Whether correction gains on burden along a path splits into two parts: how fast capability grows, times the gap between the capability elasticities of correction and of burden, plus the gap between their direct dependences on depth.

$$\Delta_{\text{bal}} := \frac{d\ln(Q/W)}{d\ln R} = \alpha_R(\chi - \kappa) + (\eta - \zeta) \tag{9}$$

Plain-text form. Delta_bal := d ln(Q/W) / d ln R = alpha_R(chi - kappa) + (eta - zeta)

Purpose. It joins Law III to the general balance: the ceiling of EQ-03 is this identity's root in the registered special case, and P20 names the general balance form as a rival to the ceiling relation's form. It also carries Law V's conditional boundary reading: where a limb raises correction's direct depth term

with kappa above chi, or raises its capability elasticity with eta above zeta, the burden elasticities fixed, the boundary rises; that reading depends on Law III's model and, through it, on Law II, and no boundary measurement scores a limb.

Failure or rejection consequence. As algebra it cannot be wrong: it is the chain rule. What can fail is the reading of its terms, because one path does not identify the partial elasticities.

Worked example. At chi = 0.5, kappa = 1, eta = 0 and zeta = -1, a path growing at alpha_R = 1.5 gives Delta_bal = 1.5 x (0.5 - 1) + (0 + 1) = 0.25: correction gaining. At alpha_R = 2.5 it gives -0.25: burden gaining. The sign changes at alpha_R = 2, the ceiling of EQ-03.

Status. identity, proved by subtracting the chain rules for ln Q and ln W along the path (the mathematics of Laws IV and V, Identity B-1); every elasticity is local and no power law is assumed. At the registered dials, kappa = 1, zeta = -1 and eta = 0, it equals 1 - alpha(1 - chi), minus the exponent of EQ-15.

Measurement. Delta_bal is directly estimable along the observed path; its components are not, and need crossed or off-path variation.

Symbols. Delta_bal, Q, W, R, alpha_R, chi, kappa, eta, zeta

Primary definitions. Laws IV and V: the mathematics, version 1.2, and the theory registration, version 1.104; frozen rules govern their own verdicts.

EQ-29: The checker-axis conversion

Placement. the bridge into Law III

In words. A correction slope measured against the checker's capability becomes the correction-leverage exponent only after it is multiplied by how fast the checker's capability rises with the capability of the system it corrects.

$$\chi = \chi_K \cdot \frac{d \ln U_{\text{checker}}}{d \ln U_{\text{target}}} \tag{10}$$

Plain-text form. chi = chi_K x (d ln U_checker / d ln U_target)

Purpose. The drafted instruments of this programme estimate chi_K, and every ceiling statement is written in chi, so this is the bridge any measurement must cross before it reaches Law III, the ARC Bound, P3, P6, P20 or P22.

Failure or rejection consequence. A checker-axis number enters a target-axis ceiling and returns a finished-looking answer to a different question.

Warning. The product is only part of the total target-axis elasticity. Correction service may also depend directly on the capability of the system corrected at a fixed checker, so the product equals the total only where that direct term is zero under the design, or where both are total derivatives along one declared path. A crossed design that estimates the direct term, or a registered statement that the design makes it zero, is required before the conversion is used. A chi_K put in chi's place is a defect to be reported, not a rounding of one estimand into another.

Status. Registered conversion, valid with the stated direct-term condition. On the zero-direct-term branch, a unit elasticity of checker capability in target capability makes the two exponents agree. Agreement at a point, including the zero-slope case, does not establish global proportionality or a universal identity between the two instruments.

Measurement. Varied: the capability of the system corrected and, in a crossed design, the checker's capability independently of it. Read out: chi_K and the elasticity of checker capability in target capability. Held fixed: the fault set and its type.

Symbols. chi, chi_K, U_checker, U_target

Primary definitions. Laws IV and V: the mathematics, version 1.2, and the theory registration, version 1.104; frozen rules govern their own verdicts.

EQ-25: The installed effect and the retained fraction

Placement. the definition beneath Law IV

In words. How much a treated arm still beats the baseline arm after later pressure, and that as a fraction of what was installed.

$$Y_V(x) := A_V(x) - A_{\text{base}}(x), \quad S_V(x) := \frac{Y_V(x)}{Y_V(0)} \tag{11}$$

Plain-text form. Y_V(x) := A_V(x) - A_base(x), S_V(x) := Y_V(x) / Y_V(0)

Other symbols. x: the A_V entry, which defines the budget ratio, and the x collision

Purpose. Every reading of Law IV is a reading of this ratio: P15's endpoint is S_ext at x = 1, and the comparison limb sets S_emb against S_ext.

Failure or rejection consequence. Law IV's endpoint loses its meaning, and with it both limbs.

Warning. Read the retained fraction only where the installed effect is established: the lower end of a registered interval on Y_V(0) must reach the positive pre-fixed floor Y_min. Near zero the ratio interval can become unbounded or unstable. The interval method is fixed in advance, Fieller (1954) or an equivalent the deciding unit specifies. Retention may exceed one or be negative; neither is clipped.

Worked example. The baseline matters. With A_V(0) = 1, A_base(0) = 0, A_V(1) = 0 and A_base(1) = -0.6, the concurrent reading, against a baseline arm put through the same training, is 0.6; the fixed-baseline reading, (A_V(1) - A_base(0)) / (A_V(0) - A_base(0)), is 0. One result, opposite verdicts on P15, so the baseline is fixed before any outcome.

Status. definition (the mathematics of Laws IV and V, version 1.2, retained-fraction definition); the baseline convention is a choice the deciding unit fixes.

Measurement. Vary the budget ratio x and read the arm scores on one frozen property battery and scale. A common positive affine rescaling leaves S_V unchanged; nonlinear rescalings can change it. Preserve the baseline convention, scale and initial-effect floor before any outcome.

Symbols. Y_V, A_V, A_base, S_V

Primary definitions. Laws IV and V: the mathematics, version 1.2, and the theory registration, version 1.104; frozen rules govern their own verdicts.

EQ-16: Cauchy's multiplicative functional equation

In words. A positive continuous function on the positive reals that obeys f(xy) = f(x)f(y) for every pair is a power function. A rule observed only under doubling does not establish that premise.

$$f(xy) = f(x)\, f(y) \tag{12}$$

Plain-text form. f(xy) = f(x) * f(y)

Purpose. The argument that scaling laws must be power laws rather than merely happen to look like them.

Failure or rejection consequence. The claim that a power law is forced rather than fitted collapses.

Refutation or misuse condition. A counterexample satisfying the full positivity, domain, multiplicativity and regularity premises would contradict the mathematical claim. A real system failing those premises is outside the theorem; P18's empirical family comparison is read under its registered rule.

Status. classical result, Cauchy (1821), in the Cambridge University Press reissue of 2009; the content is in whether the regularity conditions hold in the domains claimed. It informs P18, in ARC-1

(Law I): whether the scaling family that best fits a domain is the family its composition operator predicts before the curve is seen.

Measurement. Not measured. It is a classical result, and what has to be checked in each application is whether the regularity conditions hold over the domain claimed.

Symbols. f, x, y

Source papers. paper vii cauchy unification.

EQ-07: The endpoint error-decay slope

Placement. the measurement recipe

In words. To estimate error suppression between two positive resource levels, take the logarithm of earlier error divided by later error and divide by the logarithm of the later resource level divided by the earlier one.

$$\alpha_{\mathrm{err}} = \frac{\ln(E_1/E_2)}{\ln(R_{\mathrm{res},2}/R_{\mathrm{res},1})} \tag{13}$$

Plain-text form. alpha_err = ln(E1/E2) / ln(R_res2/R_res1)

Purpose. An endpoint log-log slope for positive error measured on one declared resource axis. A regression uses all its admitted points and is a different estimator. For a growing response Y against input X the corresponding slope is ln(Y2/Y1)/ln(X2/X1), with the opposite numerator ordering to error suppression. Each exponent needs its own measured variables; using the same algebra does not make alpha, beta_C, k or chi the same quantity.

Failure or rejection consequence. An error in the estimator or its implementation invalidates the estimate that used it; the affected result and its uncertainty must be corrected. It does not by itself invalidate a theorem, a different estimator or every measured quantity in the programme.

Status. Standard endpoint log-log slope, on positive inputs and responses. It estimates a finite-window error-decay exponent; an in-model capability exponent needs a separately justified bridge.

Warning. The corpus writes this two ways: with a leading minus and the error ratio inverted, and without. They are the same estimator. A reader meeting both may reasonably think they are different equations; they are not.

Worked example. Under E = E0 (R_res/R_res0)^(-alpha_err), logarithms give a line with slope -alpha_err. Under U = I (R/R0)^alpha they give a line with slope +alpha. A broader input range can improve identification but no universal one-decade rule guarantees it. Curvature, few points, zeros, measurement error and dependence can invalidate a simple fit. A high coefficient of determination describes fit on those points and does not select the model or prove its mechanism.

Measurement. Read positive errors E1 and E2 at distinct positive resource levels fixed before the run. Preserve the same error instrument and resource units. State endpoint or regression fitting, the window, uncertainty and zero-error rule. Tokens, parallel samples, completed revisions and elapsed time are different axes until a mapping is supplied.

Symbols. alpha_err, E, R_res, E1, E2, R1, R2

Source papers. paper ii experimental validation, paper iii alignment scaling problem.

EQ-18: The zero-parameter null

In words. For the mean of independent, centred errors with finite comparable variance, root-mean-square error scales as the inverse square root of sample count. Mapping that uncertainty to a task error rate or capability is an additional premise.

$$\mathrm{RMSE}(\bar{\varepsilon}_n) = \sigma/\sqrt{n} \tag{14}$$

Plain-text form. RMSE(mean of n centred independent errors) = sigma / sqrt(n)

Purpose. A classical averaging baseline that motivates scrutiny of a one-half fit. Its application to a measured outcome must justify the mapping from uncertainty to that outcome. EQ-01 reports the complete error-decay row; the row is inconclusive on its own axis and cannot decide P8 on the registration's capability axis.

Failure or rejection consequence. Nothing of the framework falls. If it is RIGHT, the compounding reading of the measurement is unnecessary.

Refutation or misuse condition. This condition fires AGAINST the framework, not for it. P8, in ARC-1 (Law I), registers its test: across the admissible independent systems the pooled growth exponent exceeds this one-half null. A pooled interval lying wholly inside the registered equivalence region around one half, which is the null's prediction, refutes P8, as one lying wholly below that region does, and what falls with it is, in P8's entry, "the necessity of the compounding reading". The null is retired only when that test returns a deviation the framework predicted, a pooled interval lying wholly above the region, which supports P8.

Status. open and adverse; raised by this programme against its own headline number; P8 is its registered test

Worked example. For independent centred errors of common finite variance sigma squared, the mean has variance sigma squared divided by n and RMSE sigma/sqrt(n). Bias, dependence, unequal weighting or a different loss changes this calculation. It does not establish the error-rate slope of a language model, nor the registered capability-growth exponent. P8 uses the one-half reference through its own operational scale and decision rule. Agreement of a reported fit with a reference is not support for a mechanism.

Measurement. Not measured; it is the null that a measurement is scored against. It fixes alpha at one half with no fitted parameter, so a measured alpha is compared with one half rather than with zero.

Symbols. epsilon_mean, sigma, n

Source papers. paper ii experimental validation.

EQ-05: The self-reference theorem

Placement. Foundational paper, Theorem 2

In words. If a system's rate of improvement depends on how good it already is, raised to some power, then its capability grows, in the long run, as a power law whose exponent is one over the shortfall of that power from one.

$$\alpha = \frac{1}{1 - \beta_L} \tag{15}$$

Plain-text form. alpha = 1 / (1 - beta_L)

Purpose. It is where alpha comes from mechanically rather than by fitting. It converts a statement about self-reference into a measurable exponent.

Failure or rejection consequence. A failed premise invalidates this conditional mechanism for the system concerned. P5 tests the registered prediction and identification pair; other empirical fits and conditional mathematical results retain their own scope.

Status. proved within stated assumptions, and checked so far only by the programme's own review. The exponent is the asymptotic one: over a finite window the slope of the complete solution differs from it (worked above). P5, in no register, compares the exponent an independently measured coupling implies with the finite-window exponent of the complete solution; Unit 79, the coupling-identification titration (79-study-cp), is its design draft, not registered and not run.

Warning. This equation and EQ-03 share the shape one-over-one-minus-something. They are different laws with different arguments. Recognising the shape is not recognising the law.

Measurement. Estimate beta_L by the registered manipulation on records disjoint from the growth trajectories being predicted. Seal the finite-window prediction from the complete initial-value solution before the held-out trajectories are opened. P5 compares that prediction with the fitted capability-growth exponent at its registered exponent-scale equivalence margin of 0.10. Its counted reading retains the capability precision, pilot, disjoint-record, headroom and scale requirements of registration 1.104; the asymptote is admissible only under a negligible-starting-term condition fixed before the run.

Worked example. With coupling beta_L = 0.5, unit starting capability and rate a = 0.1, the growth equation EQ-06 solves exactly to g = (0.95 + 0.05 r)^2. Its asymptotic exponent is 1/(1 - 0.5) = 2, yet its slope in logarithms between depth 1 and depth 32 is about 0.54. A comparison that set an implied two against a fitted 0.54 would report a refutation of the very equation the numbers came from: a defect in the comparison, not a finding about any system.

Symbols. alpha, beta_L

Source papers. foundational, on the origin of scaling laws, paper ii experimental validation, paper xiii self acceleration exponent.

EQ-06: The growth equation

Placement. the differential equation generating EQ-05

In words. The rate at which capability improves per unit of recursive depth is proportional to current capability raised to the coupling power.

$$\frac{dg}{dr} = a \cdot g^{\beta_L} \tag{16}$$

Plain-text form. dg/dr = a * g^(beta_L)

Purpose. The mechanism. Solve it and EQ-05 falls out. It is the only place the self-reference is actually stated as a process rather than a result.

Failure or rejection consequence. EQ-05 loses its derivation.

Status. Standard separable ODE for g > 0 and constant a > 0. For beta_L != 1 its complete solution is g^(1-beta_L) = g0^(1-beta_L) + (1-beta_L)a(r-r0). For beta_L < 1 it has an asymptotic power with exponent 1/(1-beta_L), or an exact unshifted power only under the aligned-origin condition. At beta_L = 1 it is exponential in r; for beta_L > 1 it reaches a finite-depth singularity in the ideal model. A physical system is not established by this solution.

Measurement. Varied: recursive depth r. Read out: the rate of change of capability g with depth. Held fixed: the configuration ladder and the scoring instrument, because beta_L is claimed constant only over the range studied and a changed instrument would hide that.

Symbols. g, r, a, beta_L

Source papers. paper iii alignment scaling problem, paper ix synthesis and roadmap, paper xiii self acceleration exponent.

EQ-08: Error decay

In words. Error falls as the declared resource rises, as a power of that resource.

$$E(R_{\text{res}}) = E_0 \left(\frac{R_{\text{res}}}{R_{\text{res},0}} \right)^{-\alpha_{\text{err}}} \tag{17}$$

Plain-text form. E(R_res) = E0 (R_res/R_res0)^(-alpha_err)

Purpose. A candidate error-response model to which EQ-07 or a regression can be fitted. If accuracy is one minus error, this form implies accuracy approaching one, not accuracy growing as a power. Alignment error, benchmark error and capability are distinct outcomes.

Failure or rejection consequence. The link between the frame and the data breaks.

Status. Candidate error-decay form; some reported fits use it, but no universal power law is established.

Warning. Use alpha_err for the error-decay exponent, with the error instrument named. A separately defined alignment-error exponent needs its own label and outcome. No generic equality with the capability exponent alpha is asserted.

Measurement. Vary the declared positive resource coordinate and measure error on the frozen battery. E0 is the model's value at its reference resource, estimated or fixed as the declared fitting procedure requires; it is not automatically the observed first point in a fitted regression.

Symbols. E, E0, R_res, alpha_err

Source papers. paper ii experimental validation, paper iii alignment scaling problem.

EQ-13: The conditional network exponent and cosmological reparametrisation

In words. The network formula is a conditional throughput exponent against flow-length cost. It equals d/(d+1) when demand grows with d-dimensional size and demand-weighted route length grows proportionally to linear size. A separate algebraic substitution rewrites one constant-w Friedmann exponent in the same form.

$$\alpha_{\text{net}} = \frac{d}{d+1}, \qquad d_w = \frac{2}{1+3w}, \qquad \alpha_{\cos} = \frac{d_w}{d_w+1} = \frac{2}{3(1+w)} \tag{18}$$

Plain-text form. alpha_net = d/(d+1); d_w = 2/(1+3w); alpha_cos = d_w/(d_w+1) = 2/[3(1+w)]

Purpose. Specify a physical model whose independently checked demand and routing assumptions predict a throughput-cost scaling exponent. Inventory in transit and body mass require additional bridges. The cosmological substitution illustrates an algebraic identity and contributes no empirical confirmation of the transport model.

Failure or rejection consequence. A precise mismatch under independently checked transport premises would refute that application. Failure of a premise limits the model's applicability. Neither outcome alone decides an AI capability law or the empirical programme as a whole.

Status. Conditional transport formula and separate algebraic reparametrisation, as specified in Origin 3.14 (revised 1 November 2026). On a finite rooted tree with conserved non-negative steady flow, flow-length cost equals total throughput times demand-weighted mean route length. With sink count asymptotic to a positive constant times linear size to the power d, mean sink demand tending to a positive constant, and mean route length asymptotic to a positive constant times linear size, the throughput exponent against that cost is d/(d+1). The cosmological equality uses d_w = 2/(1+3w) to rewrite 2/[3(1+w)]; it supplies neither a transport mechanism nor an AI capability exponent. P9 retains its own registered test.

Measurement. Measure sink demands, served route lengths, throughput and flow-length cost on a declared network family, keeping circulating flow explicit. Check the asymptotic assumptions independently before predicting a held-out trajectory; a finite-range slope need not equal the limiting exponent. The simple proportionality between flow-length cost and inventory assumes common speed that is constant across sizes and no uncounted waiting or storage. Converting it to body mass needs a separately justified physiological relation. In the flat single-fluid constant-w Friedmann solution, w is the equation-of-state parameter and d_w is only a transformed coordinate: positive d_w requires w > -1/3; w = -1/3 gives linear expansion and a singular coordinate, while w = -1 is the separate exponential case. No AI-system dimension is measured here.

Symbols. alpha_net, alpha_cos, d, d_w, w
Source papers. on the origin of scaling laws.

EQ-17: The bootstrap confidence interval

In words. Rather than trusting one estimate, rebuild the dataset thousands of times by resampling it, and see how much the answer moves.

Plain-text form. resample with replacement, refit, take the 2.5th and 97.5th percentiles

Purpose. A resampling procedure used to quantify an estimator's uncertainty. The reported bootstrap interval in EQ-01 belongs to the endpoint estimator, not the regression or every significance claim. The resampling unit and dependence structure determine which uncertainty it estimates.

Failure or rejection consequence. Every bootstrap interval is wrong, the surviving endpoint estimate's among them.

Refutation or misuse condition. The percentile procedure can give invalid coverage when the resampling design does not match the sampling structure, when dependence is ignored, or when the estimator is irregular. Its coverage is a property to justify for the declared design; repeated observations are not automatically independent units.

Status. standard method; the risk is application, not theory

Carries no symbol. This equation carries no symbol. It is a procedure stated in words, so an empty symbol list here is a fact about the method and not an omission in the register.

Measurement. Not measured; it is the procedure that puts an interval on something else that was. Varied: the resample. Read out: the 2.5th and 97.5th percentiles of the refitted estimate. Held fixed: the dataset, the fitting procedure and the number of resamples, all declared before the first resample is drawn.

Symbols. No algebraic symbol is required.

Source papers. paper ii experimental validation.

EQ-27: The balance root

Placement. the general boundary

In words. Where the two capability elasticities differ, the balance has one root in the growth elasticity: a ceiling when burden's elasticity is the larger, a floor when correction's is.

$$\alpha_{\text{bal}} := \frac{\eta - \zeta}{\kappa - \chi}, \quad \Delta_{\text{bal}} = (\kappa - \chi)(\alpha_{\text{bal}} - \alpha_R) \tag{19}$$

Plain-text form. alpha_bal := (eta - zeta) / (kappa - chi), Delta_bal = (kappa - chi)(alpha_bal - alpha_R)

Purpose. It says which way the boundary faces. With kappa > chi the trend is favourable exactly where alpha_R < alpha_bal, and alpha_bal is a ceiling, written alpha_crit; with kappa < chi the inequality reverses and alpha_bal is a floor; with kappa = chi the trend is eta - zeta, whatever the growth.

Failure or rejection consequence. As algebra it cannot be wrong; what it rests on is the balance model's reading of service and burden.

Worked example. kappa = 1/2, chi = 1, eta = 0 and zeta = -1 give alpha_bal = 1 / (1/2 - 1) = -2: a floor, not a ceiling, because correction's elasticity is the larger. As chi rises towards kappa with eta - zeta positive, the boundary grows without bound.

Status. conditional mathematics (the mathematics of Laws IV and V, Proposition B-2)

Measurement. Not measured; computed from four elasticities, each of which needs crossed or off-path variation.

Symbols. alpha_bal, eta, zeta, kappa, chi, Delta_bal, alpha_R

Primary definitions. Laws IV and V: the mathematics, version 1.2, and the theory registration, version 1.104; frozen rules govern their own verdicts.

EQ-28: The pacing ceiling

Placement. Law III's family

In words. In the registered burden model, with a direct depth term in correction, the ceiling is one plus that term, divided by the correction shortfall.

$$\alpha_{\text{crit}} = \frac{1 + \eta}{1 - \chi} \tag{20}$$

Plain-text form. alpha_crit = (1 + eta) / (1 - chi)

Purpose. It shows what Law III leaves out: the registered ceiling is this family's member at eta = 0, and at the same-class limit chi = 1/2 the boundary is 2(1 + eta), whose eta = 0 value is the ARC Bound.

Failure or rejection consequence. Law III's place in the family falls with the burden model, not with this algebra.

Worked example. With chi = 1/2 and eta = 1/2 it gives 3; with chi = 4/5 and eta = 0 it gives 5; with both, 7.5. These are the theory registration's own worked numbers for its four-quantity form, which writes the same boundary in departures from the registered model's values.

Status. conditional (kappa = 1, zeta = -1; the mathematics of Laws IV and V, section 4.4); a positive ceiling for chi < 1 and 1 + eta > 0, and at eta = 0 exactly Law III (Corollary B-3). It is not the general boundary, which is EQ-27's.

Measurement. Not measured; eta needs crossed or off-path variation (EQ-26).

Symbols. alpha_crit, eta, chi

Primary definitions. Laws IV and V: the mathematics, version 1.2, and the theory registration, version 1.104; frozen rules govern their own verdicts.

EQ-31: P21's mechanism reading

Placement. Law V's placement limb, its mechanism

In words. Correction embedded in the recursion has a larger direct dependence on depth than correction applied from outside it, at the same formation order.

$$\eta_{\text{in}} > \eta_{\text{out}} \tag{21}$$

Plain-text form. eta_in > eta_out

Purpose. The theory registration's own reading of why in-loop correction would pull away with depth: that correction embedded in the recursion carries a larger direct depth-dependence than correction applied from outside it, and that where this holds the boundary rises with it. P21's mechanism axis is a different registered reading, reported beside P21's mark and never in its place (P21's entry in version 1.104 of the theory registration, in the two verdicts version 1.103 added): it "asks whether the monitor's detection rate falls as the artefact it inspects grows". EQ-31 is not that axis.

Failure or rejection consequence. P21's mechanism reading falls; P21's own verdict on the outcome is untouched.

Status. the registered mechanism reading (the theory registration, version 1.104, in P21's entry), written in the mathematics of Laws IV and V as eta_in > eta_out at matched order. No verdict in that mathematics decides it: a P21 verdict does not measure it, and a refuted P21 would not show the reverse.

Measurement. Not measured by any registered unit; it needs eta read in each arm at fixed capability (EQ-30).

Symbols. eta_in, eta_out

Primary definitions. Laws IV and V: the mathematics, version 1.2, and the theory registration, version 1.104; frozen rules govern their own verdicts.

EQ-34: Relative and absolute misalignment

Placement. Paper X's reading of Law II

In words. A system's misalignment as a size is its misalignment as a fraction of capability, multiplied by the capability; the fraction can shrink while the size grows.

$$D = C\,d \tag{22}$$

Plain-text form. D = C x d

Other symbols. d: the d collision, which names Paper X's misalignment fraction, the reading this equation carries

Purpose. It keeps Law II's criterion honest about what it secures. In Paper X's minimal model with positive gain drift and beta_C > k, the fraction d falls as a power of capability with slope minus (beta_C - k); so D = C d falls to zero only where beta_C > k + 1, settles at a constant where beta_C = k + 1, and grows without bound where k < beta_C < k + 1. Paper X states this consequence as a corollary from its version 1.16, with the words "Neither d nor D is a measure of harm".

Failure or rejection consequence. A reading of the criterion as keeping the absolute size of misalignment small would go unchecked.

Worked example. With k = 0.5 and beta_C = 1 the criterion holds, 1 > 0.5, and d falls as capability to the power -0.5, yet D grows as capability to the power 0.5, because beta_C is below k + 1 = 1.5.

Status. definition, with Paper X's corollary; Paper X proves it within its minimal model only.

Measurement. Specify C and D on fixed instruments and one clock, then form d = D/C. The cited asymptotic corollary assumes positive gain drift, positive constant prefactors, constant exponents, nonnegative initial misalignment and no level or compounding drift. It is an in-model relation and neither D nor d is an operational measure of real-world harm.

Symbols. D, C

Source papers. paper x coupled coscaling correction.

EQ-09: The saturating corrector

Placement. Paper X's finite-capacity corrector, read with Law II

In words. A corrector's strength rises with capability but flattens towards a maximum, its capacity, so it behaves as a power law well below its saturation scale and stops growing well above it.

$$A(C) = A_{\max}\,\frac{C^{\beta_C}}{C^{\beta_C} + C_s^{\beta_C}} \tag{23}$$

Plain-text form. A(C) = A_max * C^beta_C / (C^beta_C + Cs^beta_C)

Purpose. Models a corrector of finite capacity, which a bare power law cannot. Paper X closes that gap with this form and shows that saturation does not falsify the beta_C > k criterion but lifts it: where the capacity itself scales with capability, the load-bearing exponent moves from strength to capacity and the inequality is unchanged by the move.

Failure or rejection consequence. Paper X's finite-capacity analysis, its Safe-Window Theorem, loses its functional form; Law II's criterion is untouched.

Status. modelling choice, not derived. Paper X declares that its bare beta is beta_C throughout, this form included, so the exponent here is the correction exponent read well below the saturation scale.

Versions of this register before 2.0.0 read the form as alignment saturating with a steepness of its own, written beta_H; that reading is withdrawn (see the beta collision).

Warning. In this equation A is Paper X's correction strength, A = A_0 C^beta_C before saturation, and not this register's A, alignment. See the A collision.

Measurement. Varied: capability C. Read out: correction strength. Held fixed: the correction instrument and the capability scale. A_max and Cs are fitted constants of the curve, not measured quantities; where the capacity itself scales with capability Paper X writes A_max = a_0 C^beta_cap, and well above the saturation scale the comparison with k is made with beta_cap in place of beta_C.

Other symbols. A: the A collision, which names Paper X's correction-strength reading, the reading this equation carries

Symbols. C, A_max, Cs, beta_C

Source papers. paper x coupled coscaling correction.

EQ-10: Alignment scaling

In words. Alignment grows as a power of recursive depth.

$$A(R) = A_0 \times R^{\beta_A} \tag{24}$$

Plain-text form. A(R) = A0 * R^beta_A

Purpose. The alignment analogue of EQ-01.

Failure or rejection consequence. The proposed score model fails over the measured range; reported observations remain, while a law verdict still requires its own registered outcome and rule.

Status. Descriptive finite-window score model; the historical judged alignment scores have instrument and ceiling limitations, and do not establish a general scaling law.

Warning. beta_A is a score slope in depth, not beta_C, beta_L or alpha_err. Positive power growth of a bounded alignment score can hold only over a finite range. A perfect score at every depth has slope zero and need not be a failure.

Measurement. Varied: recursive depth R. Read out: alignment A. Held fixed: the alignment instrument, and the starting value A0.

Symbols. A, A0, R, beta_A

Source papers. paper ii experimental validation.

EQ-11: The self-acceleration exponent

In words. Within the stated depth-growth and cycle-duration model, the capability's specific time-growth exponent equals the cycle-compression exponent minus the reciprocal of the asymptotic capability-depth exponent.

$$k = \delta - \frac{1}{\alpha} \tag{25}$$

Plain-text form. k = delta - 1/alpha

Purpose. It relates clocks in the model dC/dr = a C^beta_L and dt/dr = b C^(-delta), with positive constants on a fixed substrate: dC/dt is proportional to C^(beta_L+delta). Matching that exponent to 1+k gives k = delta + beta_L - 1. Writing alpha = 1/(1-beta_L) gives the displayed form; no pure-power finite-window premise is needed for this clock identity. A measured error-decay exponent cannot be substituted for alpha.

Failure or rejection consequence. The consistency check on the drift term in EQ-04 loses its derivation.

Status. Conditional model identity in Paper XIII 1.11, with fixed substrate, constant exponents and a justified continuous-depth approximation; k = beta_L - 1 is only the delta = 0 special case, withdrawn as a general identification.

Measurement. Measure the negative log slope of cycle duration against capability, and the depth-growth law on the same justified capability scale, with their uncertainties. Specific growth k is the log slope of (dC/dt)/C against C, not of dC/dt itself. Attribution of cycle shortening to self-improvement needs controls for external hardware or optimisation. No numerical threshold is inferred here from Paper II's error-decay fits.

Symbols. k, delta, alpha

Source papers. paper xiii self acceleration exponent.

EQ-12: The conditional correction-exponent identity

In words. Under S1 and its form condition, the correction-strength exponent in capability is the ratio of the capability and correction leverage complements.

$$\beta_X = \frac{1 - \beta_L}{1 - \beta'_L} \tag{26}$$

Plain-text form. beta_X = (1 - beta_L) / (1 - beta_L_prime)

Purpose. Paper XIII uses beta_X for the constitutive correction-strength exponent that Paper X calls beta_C. On the subcritical accumulation branches beta_L < 1 and beta_L_prime < 1, and under S1, capability and correction accumulate as powers of one depth coordinate, with aligned shifts or in the common asymptotic regime. Eliminating that depth gives the displayed ratio. This identification concerns those two papers' specified correction law and does not identify either exponent with the target-axis service exponent chi.

Failure or rejection consequence. Paper XIII's cross-system comparison loses its measure.

Status. Conditional identity, not a universal empirical law. For shifted powers C = K(r+s_C)^alpha and A = L(r+s_A)^alpha_prime, the local slope is (alpha_prime/alpha)(r+s_C)/(r+s_A). The constant ratio holds when shifts agree or are negligible together.

Warning. The coupling identity requires beta_L < 1 and beta_L_prime < 1; both complements are then positive. On that domain, beta_X <= 1 holds if and only if beta_L_prime <= beta_L. Shared substrate alone does not prove this ordering. This bound is distinct from P6's conjecture chi <= 1/2, and the smaller of the two is not a joint bound without a demonstrated conversion.

Measurement. Read each coupling on its own identified accumulation process, with one depth coordinate and compatible scales. Check S1 and the alignment or asymptotic form condition before using the ratio. Otherwise beta_X is the free correction-strength exponent of the constitutive model, not a quantity inferred from the coupling formula.

Symbols. beta_X, beta_L, beta_L_prime

Source papers. paper xiii self acceleration exponent.

EQ-32: The acceptable set

Placement. the mathematics beneath Law IV

In words. The states in which the protected property is still acceptable on every one of its components.

$$\Omega_V := \{\, s : V_j(s) \geq v_{*,j} - \epsilon_{V,j} \text{ for every component } j \,\} \tag{27}$$

Plain-text form. Omega_V := {s : V_j(s) >= v_star_j - epsilon_V_j for every component j}

Purpose. It fixes what staying acceptable means in Law IV's mathematics: a vector of components, each against its own line, so that an average cannot hide a collapse on one dimension.

Failure or rejection consequence. The persistence kernel (EQ-33) loses its object.

Status. definition (the mathematics of Laws IV and V, Definition IV-1); membership read from data is UNREGISTERED there.

Measurement. Read from data on registered intervals: a state is established outside the set when the interval on some component lies entirely below its line, inside it when every component's interval lies at or above its line, and is NOT EVALUABLE otherwise.

Symbols. Omega_V, s, V_j, v_star, epsilon_V

Primary definitions. Laws IV and V: the mathematics, version 1.2, and the theory registration, version 1.104; frozen rules govern their own verdicts.

EQ-21: Fisher's method for combining p-values

In words. A way of pooling several separate weak results into one overall figure, which only works if the results were genuinely independent of one another.

$$X^2 = -2 \sum_i \ln p_i \tag{28}$$

Plain-text form. X^2 = -2 * sum(ln p_i), on twice as many degrees of freedom as there are p-values pooled

Purpose. One published combined figure, since withdrawn. The inherited citation is Fisher (1932), Statistical Methods for Research Workers, fourth edition, section 21.1. The official publisher record linked here is the collected reissue of 1990, not a claim that its bytes are the 1932 edition: DOI 10.1093/oso/9780198522294.001.0001.

Failure or rejection consequence. The combined figure falls; the individual results are untouched.

Refutation or misuse condition. Nothing about the method. The live question is whether the independence condition holds, and the independence of the five per-model results was never established, which is why the combined figure was withdrawn rather than defended; the five per-model stakeholder-care results stand (Paper V).

Status. withdrawn in application; independence was never established

Measurement. Withdrawn in application, so nothing is currently measured with it. Read out: a combined statistic pooled from separately reported p-values, referred to the chi-squared distribution, a standard distribution and not the correction-leverage exponent, on twice as many degrees of freedom as there are p-values. Held fixed: the requirement that the pooled results be genuinely independent, which was never established here.

Warning. The standard chi-squared reference with twice as many degrees of freedom as inputs assumes independent, continuous uniform null p-values. Dependence or conservative/discrete p-values requires the declared appropriate calibration; multiplying evidence does not create independence. No such recalibration is performed here.

Symbols. X, p_i

EQ-30: The embedding ratio identity

Placement. the mathematics beneath Law V

In words. How fast the in-loop arm's advantage in correction service widens with depth splits into a capability part and a direct depth part.

$$G_{\text{emb}} := \frac{Q_{\text{in}}}{Q_{\text{out}}}, \qquad \frac{d \ln G_{\text{emb}}}{d \ln R} = \alpha_R(\chi_{\text{in}} - \chi_{\text{out}}) + (\eta_{\text{in}} - \eta_{\text{out}}) \tag{29}$$

Plain-text form. G_emb := Q_in / Q_out, d ln G_emb / d ln R = alpha_R(chi_in - chi_out) + (eta_in - eta_out)

Purpose. It says what a widening advantage of in-loop correction could be made of: the in-loop arm's service may grow faster with capability, or faster with depth itself. Only the second is P21's mechanism reading (EQ-31).

Failure or rejection consequence. As algebra it cannot be wrong; the chain rule carries it.

Warning. One path does not identify the parts. With Q_in = C^2 and Q_out = C on the path C = R, the ratio equals R and widens, while eta_in - eta_out = 0: the whole widening is the capability part. Crossed or off-path variation is needed.

Status. identity, at matched formation order (the mathematics of Laws IV and V, Identity V-1); at fixed capability it reduces to eta_in - eta_out.

Measurement. Not measured directly; P21's deciding unit reads the advantage on the outcome, never as an exponent.

Symbols. G_emb, Q_in, Q_out, R, alpha_R, chi_in, chi_out, eta_in, eta_out

Primary definitions. Laws IV and V: the mathematics, version 1.2, and the theory registration, version 1.104; frozen rules govern their own verdicts.

EQ-33: The persistence kernel

Placement. the mathematics beneath Law IV, UNREGISTERED

In words. The starting states from which every allowed path of at most H steps stays acceptable.

$$K_V(H; T, \epsilon_V) := \{\, s_0 : T_H^{\text{exit}}(s_0) = \varnothing \,\} \tag{30}$$

Plain-text form. K_V(H; T, epsilon_V) := {s_0 : T_H_exit(s_0) is empty}

Other symbols. H: the T entry, which defines the horizon, and the H collision

Purpose. The invariance object beneath Law IV: the author's own idea, from his working paper of 16 August 2026, of a robust target set that a rewrite leaves invariant, written over a declared family of paths.

Failure or rejection consequence. Nothing registered falls: it is a proposed object.

Status. UNREGISTERED, not law content (the mathematics of Laws IV and V, Definition IV-3). Membership is relative to the declared family, so an unrestricted family that can map out of the acceptable set empties it.

Measurement. Not measured. The mathematics certifies membership for the family of executed paths under the premises of its theorems, never for a planner's whole feasible family.

Symbols. K_V, T, epsilon_V, s_0

Primary definitions. Laws IV and V: the mathematics, version 1.2, and the theory registration, version 1.104; frozen rules govern their own verdicts.

EQ-19: Bayes' theorem

In words. Start with how likely you thought something was, look at the new evidence, and adjust in proportion to how much better that evidence is explained by your idea being true than by it being false.

$$P(H \mid E) = \frac{P(E \mid H)\, P(H)}{P(E)} \tag{31}$$

Plain-text form. P(H | E) = P(E | H) * P(H) / P(E)

Purpose. All reasoning in the programme about priors and about what evidence would change them.

Failure or rejection consequence. Nothing. It is a theorem.

Refutation or misuse condition. Nothing. It is provable from the definition of conditional probability, not an empirical claim. What can be wrong is its APPLICATION: priors chosen after the fact, likelihoods guessed to suit, or a claim of updating without recording what the prior was.

Status. theorem, Bayes (1763); the discipline is in recording the prior before the evidence

Worked example. Worked, because this is the one people nod at without following. Suppose before any data you think there is a 30 per cent chance the correction-leverage exponent chi is one half, so P(H) = 0.3. A measurement arrives that would appear 80 per cent of the time if H is true and 20 per cent of the time if it is false. How often would you see it at all: P(E) = (0.8 x 0.3) + (0.2 x 0.7) = 0.24 + 0.14 = 0.38. Then P(H|E) = 0.8 x 0.3 / 0.38 = 0.63. Belief moves from 30 per cent to 63 per cent. Notice it did not jump to certainty: the evidence multiplied the odds by four, from 3 to 7 up to 12 to 7, which moved the probability from 30 per cent only to 63 per cent, and odds and probability must never be quoted as though they were the same movement. That restraint is the point of using the rule rather than reasoning by impression. Operationalised here three ways: dated priors recorded in writing BEFORE any confirmatory data, with what would change them; kill conditions, which are commitments about P(E|H) written in advance, which is why the window must be declared before the run; and the reason consistent-with is not supports, because when P(E|H) and P(E|not H) are about equal the ratio is about one and the posterior barely moves from the prior.

Other symbols. P: the P collision, which names the Bayes reading and the metabolic one; E: the E collision, and the H entry, which defines the evidence reading

Measurement. Not measured. It is the rule by which a measurement moves a stated prior, and the discipline is that the prior is recorded before the evidence is seen.

Symbols. H

EQ-20: Cohen's d

In words. How far apart two groups are, measured in units of how spread out the data already is, so a difference can be called large or small without knowing what was measured.

$$d = \frac{\text{mean}_1 - \text{mean}_2}{\text{standard deviation}} \tag{32}$$

Plain-text form. d = (mean1 - mean2) / standard deviation

Purpose. How big any measured difference in the programme actually is.

Failure or rejection consequence. Nothing. It is a definition.

Refutation or misuse condition. Nothing about the definition. A pooled standard deviation can obscure differences in spread between the groups. State the standardising convention, study design, sample sizes and uncertainty before interpreting the size or precision of d.

Status. standard statistic, Cohen (1988)

Warning. This d is not the programme's dimensional d. It belongs to statistics and must never be renamed; only the programme's own uses can take subscripts.

Worked example. By Cohen's conventions about 0.2 is small, 0.5 medium and 0.8 large, and the sign gives direction. A d of -0.53 means the first group scored lower by rather more than half a standard deviation. Dividing by the spread is what makes it portable: a 5-point gap is enormous if scores usually vary by 2 and trivial if they usually vary by 50.

Other symbols. d: the d collision, which names the effect-size reading

Measurement. Varied: group membership. Read out: the difference between the two group means. Held fixed: the measured quantity and the definition of the standard deviation used to express that difference.

Symbols. mean1, mean2

EQ-14: The metabolic analogy

In words. Power per unit mass falls as mass rises, to the power minus one quarter.

$$\frac{P}{P_0} = \left(\frac{M}{M_0}\right)^{3/4}, \qquad \frac{P/M}{P_0/M_0} = \left(\frac{M}{M_0}\right)^{-1/4} \tag{33}$$

Plain-text form. P/P0 = (M/M0)^(3/4); (P/M)/(P0/M0) = (M/M0)^(-1/4)

Purpose. The biological precedent for a scaling exponent that is argued for from structure rather than fitted to data. Kleiber's three-quarter exponent, Kleiber (1932), is the observation; the network derivation West, Brown and Enquist (1997) offered for it is the argument this analogy points at, and it is disputed: geometric scaling gives two thirds, and an analysis of 619 mammal species, White and Seymour (2003), finds basal metabolic rate proportional to mass to the two thirds and no support for three quarters, so the empirical exponent is not settled between them. What that literature holds and this programme does not is a mechanism grounded in measured anatomy. The analogy is used here only to show that a derived exponent is a thing science already has, and nothing in this programme depends on which exponent wins. Illustrative, not load-bearing.

Failure or rejection consequence. An analogy is lost. No claim depends on it.

Status. Illustrative allometric model conditional on the three-quarter exponent; the empirical exponent and its explanation remain contested in the cited literature.

Measurement. Not measured by this programme. Mass and metabolic power are measured in comparative physiology, and the result is carried here as an illustration; nothing in this programme is fitted to it.

Symbols. P, M

Source papers. on the origin of scaling laws.

5 Symbols, domains and measurement

Each entry is kept once, with its own response and domain.

U: usable capability **Plain.** What the system can actually do, after everything is accounted for. The output side of the equation.

Operational. Score on a fixed target battery held constant across the whole experiment. The battery must be frozen before any measurement, because a battery that changes during a run makes the exponent meaningless. The theory registration measures capability "on a ladder of items of known difficulty rather than as a success rate", on a scale with no ceiling. Version 1.104 widens that measure for two readings alone, P5's counted ordering and P6's counted interval on checkable correction: there capability is admitted as "the verified pass count on a frozen ladder of items of known difficulty", each pass settled by frozen code, read only below a headroom fraction of the ladder's size, at most one half, that the deciding unit registers before any run, and only where a registered pilot shows the count proportional to the ratio-scale quantity over the administered range. A system whose count reaches that fraction is NOT EVALUABLE on the reading and is never scored at the ceiling.

Units. battery score, dimensionless once normalised

Range. positive
Appears in. EQ-01.

I: base intelligence **Plain.** What the system could do before any self-improvement. The starting point.
Operational. The same battery score measured at modelled depth one, before any revision.
Units. same as U
Range. positive
Appears in. EQ-01.

R: recursive depth **Plain.** The modelled recursive-depth coordinate: completed revision rounds plus one, so that a system with no completed revision has R = 1. Three revisions give R = 4. This convention applies to the depth models, not automatically to resource fits or every logarithmic slope.

Operational. The count of completed revision rounds under a fixed configuration ladder, plus one. Counted, never estimated, with the ladder fixed in advance, and the baseline observation, at R = 1, is kept in every fit.

Units. dimensionless count
Range. integer, at least one; the count of completed rounds beneath it starts at zero
Appears in. EQ-01; EQ-10; EQ-15; EQ-26; EQ-30.

alpha: the capability-growth exponent **Plain.** The capability-growth exponent in a specified power model: doubling depth multiplies capability by 2^alpha. Alpha = 1 doubles it and alpha = 2 quadruples it; 0 < alpha < 1 gives positive diminishing returns. A local elasticity or fitted slope retains its own window.

Operational. The local elasticity d ln U/d ln R on a declared capability and depth coordinate, or a finite-window fitted slope explicitly labelled as such. Under an exact power these agree and are constant. EQ-07 is an error-decay estimator and does not measure this quantity without a justified response and axis mapping.

Units. dimensionless

Range. unbounded above in the mathematics; claimed at most two only for systems that stay correctable under same-class correction, which is a claim about stability and not about what is reachable

Current estimate. No capability-growth value is inferred here from Paper II. EQ-01 reports its complete error-decay row and limitations; alpha_err names that different estimand. A reciprocal-error metric gives an algebraic correspondence on that instrument, not evidence of a self-improvement mechanism.

Retracted estimate. 2.24, from one model, DeepSeek R1, on twelve problems scored by exact match against an answer key, with a 95 per cent interval of 1.5 to 3.0; retracted when the author's own larger, blinded replication on five frontier models did not reproduce it. Never to be cited except as retracted.

Appears in. EQ-01; EQ-02; EQ-05; EQ-11; EQ-15.

alpha_crit: the stability ceiling **Plain.** The conditional capability-growth boundary of the stated pacing model. Its value separates modelled ratio trends; it does not by itself determine whether a system is safe or can hold itself together.

Operational. A model boundary computed from specified elasticities and independently compared with an observed boundary under P20 or P3. Measuring chi alone computes a prediction; it does not test whether the boundary actually occurs. The domain and unknown empirical mapping remain explicit.

Units. dimensionless

Range. at least one in the registered relation, where chi lies from 0 to below 1; in the general balance form it is the balance root where kappa > chi (EQ-27), and there it is at least one only where eta - zeta >= kappa - chi > 0

Appears in. EQ-03; EQ-28.

beta_C: the correction exponent **Plain.** How fast the brakes strengthen. The rate at which a system's power to correct itself grows as it gets more capable.

Operational. The elasticity of the constitutive correction-decay strength A(C) in Paper X, against capability C. It is distinct from delivered correction service and from target-axis chi unless the respective measurement mapping is justified.

Units. dimensionless

Range. real

Collision warning. This is the beta of Law II only. It is NOT the beta of EQ-05 or EQ-10. Paper X writes it as a bare beta and declares that every bare beta of its own is this one. Whether beta_C and chi are one quantity is unestablished (see chi).

Appears in. EQ-04; EQ-09.

k: the specific capability-growth exponent **Plain.** The elasticity of the specific growth rate r_rate = (dC/dt)/C against capability in the corrective model. The full growth rate has exponent 1+k. Gain drift in that model is tied to growth, which is why k enters its correction criterion.

Operational. Estimate the specific growth rate on a declared time clock and capability scale. The relation k = delta + beta_L - 1 follows under EQ-11's model. Registration tests that define a different burden/drift instrument retain that definition and need an explicit mapping before identifying the estimates.

Units. dimensionless

Range. real; negative k means the specific growth rate falls with capability, while the raw rate dC/dt can still rise when 1+k is positive

Appears in. EQ-04; EQ-11.

beta_L: the self-referential coupling **Plain.** How strongly a system's rate of improvement depends on how good it already is. At zero, improvement is constant regardless of skill. Near one, the better it gets the faster it gets better, and growth runs away.

Operational. Fitted from the growth equation EQ-06 over the measured range, then checked against alpha through EQ-05. That check is P5's falsifier: measure beta_L independently, predict alpha, and compare with the fitted alpha on the finite window (EQ-05).

Units. dimensionless

Range. The core nonnegative-leverage branch uses 0 <= beta_L < 1; a labelled negative-leverage extension is allowed for g > 0. beta_L = 1 gives exponential growth in depth and beta_L > 1 a finite-depth singularity in the ideal ODE. This is not a bound on all physical systems.

Collision warning. Written as a bare beta in the foundational paper and in Paper II. Paper XIII and Paper X write beta_L. This file adopts the subscript everywhere.

Appears in. EQ-05; EQ-06; EQ-12.

beta_A: the alignment-scaling exponent **Plain.** How alignment grows with recursive depth, as a growth-shape number of its own.

Operational. Log-log slope of alignment score against recursive depth.

Units. dimensionless

Range. real

Collision warning. Written as a bare beta in Paper II.
Appears in. EQ-10.

beta_X: the correction-strength exponent in Paper XIII's conditional join **Plain.** The exponent of constitutive correction strength in capability. Under S1 and the aligned or common asymptotic form, it is the ratio of the two leverage complements in EQ-12.

Operational. Estimate the correction-strength law on its own scale. Infer it from the coupling ratio only after S1 and the alignment or asymptotic conditions are shown. It matches Paper X's beta_C for that same specified strength law, not by similarity of names and never automatically with chi.

Units. dimensionless ratio

Range. Positive for beta_L < 1 and beta_L_prime < 1 in the ratio model; at most one only under the additional premise beta_L_prime <= beta_L. The free constitutive exponent outside that join may be any real value.

Appears in. EQ-12.

delta: the self-acceleration (cycle-compression) exponent **Plain.** How rapidly a system shortens the time needed for a further recursive step as its capability increases, on a fixed substrate. delta = 0 is constant cycle duration; delta > 0 is compression.

Operational. Minus the log slope of cycle duration dt/dr against capability, under dt/dr proportional to C^(-delta). State the fixed substrate, depth approximation and attribution controls. A positive log slope of cost or duration is not this delta; if cost is proportional to duration it has the opposite sign.

Units. dimensionless

Range. Nonnegative on the named self-acceleration branch; negative fitted values describe slowing cycles, although the algebraic clock identity extends to them.

Appears in. EQ-11.

E: error **Plain.** How wrong the system is. The thing that is supposed to fall as it improves.

Operational. Error rate on the frozen target battery. Must be the same battery at every depth.

Units. error rate, dimensionless

Range. non-negative

Appears in. EQ-07; EQ-08.

A: alignment **Plain.** How well the system's behaviour matches what was intended.

Operational. A declared alignment score on its own frozen instrument. Historical Paper II scores were judged rubric ratings with scorer and ceiling limitations; the six-question counted score of Engineering 8.1 is a proposed instrument and was not used to generate those results. Neither defines a universal alignment measure.

Units. score, dimensionless once normalised

Range. Bounded by its declared scoring instrument. A_max in EQ-09 is correction-strength capacity, not the upper bound of this alignment score.

Appears in. EQ-10.

C: capability **Plain.** How capable the system is, used where capability rather than recursive depth is the driving variable.

Operational. Battery score, as for U.

Units. same as U

Range. positive

Appears in. EQ-09; EQ-34.

g, r, a: growth variables of the differential equation **Plain.** g is the positive accumulated quantity in the growth ODE, r its continuous depth coordinate, and a a positive rate coefficient. Foundational uses g as amplification with g(0)=1 and U=I g; Paper XIII may use capability C directly. A fixed multiplicative conversion relates these only within the declared model. Paper X uses r for a specific growth rate, not depth.

Operational. State the response and initial value. In Foundational, r = R-1 counts added passes and g(0)=1. The coefficient a changes the finite-window curvature and crossover as well as the level; it leaves the asymptotic exponent unchanged. A discrete experiment requires a justified connection to the continuous-depth approximation.

Units. as for U and R; a carries the units that balance the equation

Range. g > 0 and a > 0; added depth r starts at zero and is continuous in the stated ODE approximation

Appears in. EQ-06.

d, w: effective dimension and equation-of-state parameter **Plain.** In the dimensional derivation, d is the effective number of directions the system can grow in, and w is the parameter that fixes it.

Operational. Network dimension d is independently assigned in a transport-limited domain. The cosmological coordinate d_w = 2/(1+3w) is an algebraic coordinate substitution in the specified Friedmann solution, not a measured AI-system dimension.

Units. dimensionless

Range. Network dimension is positive in its transport model. The cosmological coordinate is positive for w > -1/3, undefined at w = -1/3, and negative for -1 < w < -1/3; w = -1 has the separate exponential solution.

Appears in. EQ-13.

B: correction burden **Plain.** How much correcting a system needs. It grows as the system does more, because more capability means more that can go wrong.

Operational. Newly generated correctable burden per unit of the same recursion clock as capacity, modelled in EQ-15 as proportional to dC/dR. Burden per unit of capability added is a different quantity; stating it would change the model.

Units. correction units, dimensionless once normalised

Range. non-negative

Appears in. EQ-15.

A_c: correction capacity **Plain.** How much correcting a system can actually do. The other side of the burden.

Operational. Correction delivered per pass at a given capability, from the same blinded instrument that supplies the correction exponent.

Units. same as burden, so the ratio is dimensionless

Range. positive

Collision warning. The papers write this as a bare A, which already means alignment in this corpus. Subscripted here; the papers have not yet adopted it.

Appears in. EQ-15.

gamma_N: the residual-decay exponent **Plain.** How fast the mistakes that are left over disappear as you keep revising. Go round the loop again and again, and this number says how quickly what remains shrinks. Higher means the leftovers vanish faster.

Operational. The slope of residual error against accumulated correction effort at fixed capability, with the effort coordinate stated. Unit 05's cited within-system estimator is a draft instrument. The sources checked for this register report no identified measurement of this quantity; no claim is made about the entire external literature.

Units. dimensionless

Range. positive; no upper bound is claimed

Appears in. None listed.

Load bearing note. P11 asks whether this residual-decay quantity is related to the correction-leverage exponent. Independence alone does not identify them: effort and target capability are different inputs, and uncertainty reduction and delivered correction are different responses. Any equality requires the stated measurement and axis bridge.

gamma_D: the capability-to-depth conversion exponent **Plain.** How readily raw ability turns into more rounds of self-improvement. If a system gets cleverer, does it also get faster at improving itself, and by how much.

Operational. The exponent by which revision depth accelerates with capability: the rate of change of depth grows as capability to the power of this exponent. Not measured by any registered study.

Units. dimensionless

Range. positive

Appears in. None listed.

Load bearing note. If dr/dt is proportional to C^gamma_D and C is an aligned power of r with positive exponent alpha, then dr/dt is proportional to r^(alpha gamma_D). The ideal finite-time threshold is alpha gamma_D > 1, with equality exponential in time; 1/gamma_D is the threshold in alpha when gamma_D > 0. This is a clock model, not the correction-leverage ceiling of EQ-03. gamma_D matches delta only if the same depth, capability and fixed-substrate cycle-duration law are meant.

A_max, Cs: the saturation constants of the saturating corrector **Plain.** The most correction strength the saturating corrector can deliver however capable the system becomes, its capacity, and the capability at which it is halfway there, the saturation scale.

Operational. Fitted constants of the saturating form EQ-09: its upper asymptote, and the capability at half that asymptote.

Units. A_max in correction-strength units; Cs in capability units

Range. both positive

Appears in. EQ-09.

A0, E0: the starting values **Plain.** Where alignment and error stood before the growth being described began. Every power law needs a starting point, and this is it.

Operational. The alignment and error model values at their specified reference inputs. State whether each is fixed from an initial observation or estimated by the declared fitting procedure; a fitted reference value is not automatically the first observed score.

Units. as for the quantity each scales

Range. positive

Appears in. EQ-08; EQ-10.

E1, E2, R1, R2: the two measurement points **Plain.** The two errors and the two positive resource levels in the endpoint slope. In EQ-07 the inputs are explicitly written R_res1 and R_res2 to distinguish them from registered recursive depth.

Operational. Read error and the declared resource at the earlier and later checkpoints, on the same instrument; the levels must be distinct and positive for the logarithmic slope.

Units. E in the declared error units; R_res1 and R_res2 in the same declared resource units

Range. positive

Appears in. EQ-07.

beta_L_prime: the comparison coupling **Plain.** The same self-reference number measured on a second system, so two systems can be compared rather than one described.

Operational. The self-referential coupling of the comparison system in the boundary-gap ratio.

Units. dimensionless

Range. less than one

Appears in. EQ-12.

H: hypothesis and evidence, in Bayes' theorem **Plain.** H is the claim being tested. E is what you observed. Bayes' theorem says how much the observation should move your confidence in the claim.

Operational. Standard statistical notation, not a quantity of this programme. Included so the equation can be read without another reference. H is also the horizon of Law IV's mathematics (see the T entry and the H collision).

Units. none

Range. probabilities lie between zero and one

Appears in. EQ-19.

M, P: mass and metabolic power, in the biological analogy **Plain.** How heavy an organism is, and how much energy it burns. The analogy notes that bigger animals burn less energy per kilogram, which is a real scaling law from biology and is used here only as an illustration. M0 and P0 are fixed positive reference mass and power in the normalised illustration.

Operational. Standard biological notation. This P is metabolic power and is NOT the P of probability in EQ-19.

Units. M in mass; P in power

Range. positive

Appears in. EQ-14.

X, p_i: the combined test statistic and its inputs **Plain.** A way of pooling several separate tests into one verdict, and the individual results being pooled.

Operational. Standard statistical notation for Fisher's method, not a quantity of this programme. Included so the equation can be read without another reference.

Units. dimensionless

Range. p between zero and one

Appears in. EQ-21.

mean1, mean2: the two group averages, in Cohen's d **Plain.** The average of each of the two groups being compared, before the difference between them is expressed in standard deviations.

Operational. Standard statistical notation, not a quantity of this programme.

Units. as for the measured quantity

Range. unbounded
Appears in. EQ-20.

f, x, y: the unknown function and its arguments, in Cauchy's equation **Plain.** A rule you do not yet know, and two numbers you feed it. The equation asks which rules turn multiplication of the inputs into multiplication of the outputs; power laws are the answer.

Operational. Standard mathematical notation for the multiplicative functional equation. x is also Law IV's budget ratio (see the A_V entry and the x collision).

Units. none
Range. positive reals in the form used here
Appears in. EQ-16.

phi_Q: the correction-service path elasticity **Plain.** How fast the correction a system can actually deliver grows as it goes deeper, measured along the path it took.

Operational. d ln Q / d ln(clock) along the observed path, where Q is delivered correction service. On the depth clock it equals alpha_R times chi plus eta when the path is decomposed (EQ-26), but it is estimable without that decomposition.

Units. dimensionless
Range. any real number
Appears in. the balance exponent on the theory registration's resource clock

phi_W: the burden path elasticity **Plain.** How fast the correctable work a system generates grows as it goes deeper, measured along the path it took.

Operational. d ln W / d ln(clock) along the observed path, where W is newly generated correctable burden. On the depth clock it equals alpha_R times kappa plus zeta when the path is decomposed (EQ-26), but it is estimable without that decomposition.

Units. dimensionless
Range. any real number
Appears in. the balance exponent on the theory registration's resource clock

chi: the correction-leverage exponent, which the theory registration writes gamma **Plain.** How much more correction a system's corrector delivers as the system it corrects becomes more capable. Higher is better. Written chi from the author's working paper of 16 August 2026 onward, because gamma had carried three incompatible readings; the theory registration writes gamma, and the two name the same quantity. This register writes chi, and prints gamma only inside a quotation of the registration.

Operational. The target-axis elasticity of admissible correction service, misalignment removed per review pass on a fixed fault set of stated type. A checker-axis slope is chi_K and needs EQ-29's conversion, including any direct target term. Before a verdict, the registration requires equivalence within its pre-fixed margin between the elasticities of inverse residual uncertainty and misalignment removed on the same systems and fault set; otherwise report both separately and name the admissible one. One path cannot identify the partials. The cited register and draft instruments supply no identified measurement of this quantity; this is not a claim that no researcher anywhere has measured a related quantity.

Units. dimensionless
Range. 0 to below 1 for a finite ceiling in the registered relation (EQ-03). P6 conjectures chi at most one half for same-class correctors, which is a cap on that class and not a universal cap.
Appears in. EQ-03; EQ-15; EQ-26; EQ-27; EQ-28; EQ-29.

Load bearing note. The value one half is a conjecture and not a derivation. It follows only from four conditions: correction capacity read as inverse residual uncertainty, uncorrelated channels, a number of channels in proportion to the capability corrected, and channels comparable in variance and quality. Under positive correlation the exponent falls, and further as the panel grows. It is one of two assumptions the headline value two rests on, and neither is established; the other is the burden model (EQ-15). Whether chi and Law II's beta_C are one quantity is unestablished: no proposition substitutes one for the other, a measurement of one is inadmissible against a proposition stated in the other, and Law III follows from Law II's criterion only under that identification.

Class dependence. P6 conjectures a bound for an admitted same-class population; it is not derived from shared substrate alone. Failure-mode dependence, external grounding, lineage, provenance and scaling class remain separate admission axes. Build order at matched content is P12's intervention and does not establish class difference by itself. A separately curated corpus may still share a distribution and failure modes.

kappa: the burden intensity elasticity **Plain.** How much more correctable work each unit of new capability brings with it, as the system gets more capable.

Operational. The elasticity of newly generated burden with respect to capability, at fixed depth. The registration's burden model is the case kappa equals one; the burden-intensity rival is kappa above one, and the corpus writes kappa - 1 as mu_B.

Units. dimensionless

Range. real where the positive differentiable burden model defines the elasticity; kappa = 1 in the stated pacing specialisation

Appears in. EQ-26; EQ-27.

Collision warning. Paper X writes kappa for a different quantity (see the kappa collision).

eta: the direct depth-dependence of correction service **Plain.** Whether correction gets better simply by being run more times, apart from any gain in capability.

Operational. The direct exponent of recursive depth in correction service, holding capability fixed: d ln Q / d ln R at fixed C. Zero in the registration's model. P21's mechanism reading is that correction embedded in the recursion carries a larger eta than correction applied from outside it (EQ-31). It is not an integration fraction (see the eta collision).

Units. dimensionless

Range. any real number

Appears in. EQ-26; EQ-27; EQ-28.

zeta: the direct depth-dependence of burden **Plain.** Whether going deeper generates correctable work by itself, apart from the capability it produces.

Operational. The direct exponent of recursive depth in newly generated burden, holding capability fixed: d ln W / d ln R at fixed C. Minus one in the registration's model, which is what it means for burden to follow the rate of capability gain rather than its level.

Units. dimensionless

Range. any real number

Appears in. EQ-26; EQ-27.

Delta_bal: the balance exponent **Plain.** Whether correction is pulling ahead of the work it has to do, read locally along the path a system actually took. Positive means the gap is closing.

Operational. The log-derivative of the ratio of correction service to newly generated burden along a specified path, per unit of log modelled depth (EQ-26). The theory registration writes the same quantity on its resource clock as Delta_balance = phi_Q - phi_W; the two differ by the factor d ln R / d ln(clock), which is positive, and so keeps the sign and the zero set, wherever depth rises with the clock. Introduced in the author's working paper of 16 August 2026.

Units. dimensionless

Range. any real number

Appears in. EQ-26; EQ-27.

alpha_R: the local capability-growth elasticity along a path **Plain.** How fast capability rises with depth at the point the system has reached: d ln C / d ln R along the path C(R).

Operational. The local reading of alpha, equal to alpha where the elasticity is constant. The author's working paper of 16 August 2026 writes alpha_C for the same quantity on the resource clock; this register writes alpha_R on the depth clock and states the clock wherever the two meet.

Units. dimensionless

Range. any real number

Appears in. EQ-26; EQ-27; EQ-30.

alpha_bal: the balance root **Plain.** The growth elasticity at which correction stops gaining on burden: (eta - zeta) / (kappa - chi), defined where kappa differs from chi.

Operational. Not measured directly: the root of Delta_bal in alpha_R. A ceiling where kappa > chi, and then written alpha_crit; a floor where kappa < chi.

Units. dimensionless

Range. Any real root where kappa != chi. In the ceiling branch kappa > chi it is at least one exactly when eta-zeta >= kappa-chi > 0; the floor branch uses the opposite denominator sign and its own inequality.

Appears in. EQ-27.

Q, W: correction service and newly generated burden **Plain.** Q is how much correction a system actually delivers; W is how much correctable work it newly generates. Both are read on one clock and in comparable units, at capability C and modelled depth R.

Operational. Q: misalignment removed per pass on a fixed fault set, the delivered correction service of the balance model. W: newly generated correctable burden, on the same clock as Q, with any conversion fixed before outcomes. Both are positive on the modelled domain, and zero service is reported outside the log model with its count.

Units. correction units

Range. positive on the modelled domain

Collision warning. Paper X writes W for a Wiener process (see the W collision).

Appears in. EQ-26.

Q_ol, Q_in, Q_out: correction service by arm **Plain.** Q_ol is the correction service of the arm with formation order o, after or first (whether the corrector was formed after the capability it corrects or before it), and placement l, out or in (whether correction sees only the finished output or takes part in the next round). Q_in and Q_out are the in-loop and output-only arms at the same order.

Operational. Q read per arm with every other design factor fixed; o and l are registered arm labels, not variables.

Units. correction units

Range. positive on the modelled domain

Appears in. EQ-30.

chi_ol, chi_in, chi_out, chi_first, chi_after: chi by arm **Plain.** chi read in one arm: chi_in and chi_out at the same formation order, chi_first and chi_after at the same placement.

Operational. The partial derivative of ln Q_ol with respect to ln C at fixed R. It is the correction-leverage exponent only when Q_ol is the target-axis service measure and the identification conditions of the mathematics of Laws IV and V (its section 1) hold; otherwise it is the elasticity of the declared service measure, and it never enters the ceiling relation.

Units. dimensionless

Range. any real number

Appears in. EQ-24; EQ-30.

eta_ol, eta_in, eta_out: eta by arm **Plain.** The direct dependence of one arm's correction service on depth, at fixed capability.

Operational. The partial derivative of ln Q_ol with respect to ln R at fixed C; not an integration fraction.

Units. dimensionless

Range. any real number

Appears in. EQ-30; EQ-31.

G_emb: the embedding ratio **Plain.** How much more correction the in-loop arm delivers than the output-only arm, at the same order, capability and depth: Q_in / Q_out.

Operational. The ratio of the two arms' service at matched formation order; its log-derivatives are EQ-30's identity.

Units. dimensionless

Range. positive

Appears in. EQ-30.

chi_K: the checker-axis correction slope, which the theory registration writes gamma_K **Plain.** The same misalignment removed per review pass, taken against the capability of the checker rather than of the system corrected. It is the quantity the drafted instruments of this programme actually estimate.

Operational. Converted to chi by EQ-29 before it enters the ceiling relation, P3, P6, P20 or P22. A distinct symbol and never a variant reading of chi.

Units. dimensionless

Range. any real number

Appears in. EQ-29.

U_checker, U_target: the capabilities of the checker and of the system corrected **Plain.** How capable the checker is, and how capable the system it corrects is.

Operational. Each read on a capability battery; the conversion of EQ-29 needs the elasticity of the first in the second, along a declared path or from a crossed design.

Units. as for U

Range. positive

Appears in. EQ-29.

chi_V, chi_J: the checkable and judged correction-leverage exponents, which the theory registration writes gamma_V and gamma_J **Plain.** chi estimated separately on faults whose correctness something outside the corrector settles, and on faults that a specification or a judgement settles.

Operational. Each read on a fixed fault set of its stated type under the same instrument. P22 registers that chi_V exceeds chi_J on the same systems, so that the ceiling is strictly lower for judged correction. The theory registration introduced the two subscripts and asked that this register carry them before they appear on any other surface; this version carries them, in the letter chi by the forward convention.

Units. dimensionless

Range. any real number

Appears in. None listed.

D: absolute misalignment, in Paper X **Plain.** The size of a system's departure from the values intended for it, where d is the same departure as a fraction of capability.

Operational. In Paper X, D is a nonnegative model variable with d = D/C. An empirical application needs fixed units and an identified instrument. Scoring behavioural departure from specified values with a blinded external evaluator is a proposed operationalisation, not an established measure of real-world harm.

Units. misalignment units

Range. non-negative

Appears in. EQ-34.

s, s_0: the system's state, and the state a path starts from **Plain.** The complete description of the system at one moment, whatever the recursion or the training may change; s_0 is the state a path starts from.

Operational. Whatever the registered design fixes as the state: weights, memory, surrounding software and evaluator.

Units. none

Range. a declared state space

Appears in. EQ-32; EQ-33.

V, V_j: the property score and its components **Plain.** How much of the protected property the system has in a state; where several dimensions are protected it is a vector, with one component V_j for each, in its own declared unit.

Operational. A frozen property battery read on the state, one component for each registered dimension.

Units. battery units, component by component

Range. as the battery declares

Collision warning. V is the score only. As a subscript, V reads of the protected property, and it is not the V of chi_V (see the subscript V collision).

Appears in. EQ-32.

v_star, epsilon_V: the acceptability threshold and tolerance **Plain.** The line each component of the property must not fall below, and how far below it still counts as acceptable.

Operational. Both fixed before any outcome, component by component, in the battery's units.

Units. as V

Range. the tolerance is non-negative

Appears in. EQ-32; EQ-33.

Omega_V: the acceptable set **Plain.** The set of acceptable states that EQ-32 defines.

Operational. Membership is read from registered intervals on the components, as EQ-32 states.

Units. none

Range. a subset of the state space

Appears in. EQ-32.

T: the declared family of allowed finite paths, and the horizon **Plain.** T is every path the design counts as possible under the transformations it admits. T_H(s_0) is the set of its paths that start at s_0 and take at most H transitions; a path is safe if every state on it is acceptable and an exit otherwise, and T_H_exit(s_0) is the set of exits. H, the horizon, is the largest number of transitions a claim looks ahead, fixed before any outcome; it is not the hypothesis of Bayes' theorem (see the H collision).

Operational. Registered as a finite battery of transformations, so that a claim is battery-relative, or as a probabilistic family with a stated distribution and tolerance, which is a different claim.

Units. none; H counts transitions

Range. a set of paths; H a non-negative integer

Appears in. EQ-33.

K_V: the persistence kernel (UNREGISTERED) **Plain.** The set of starting states that EQ-33 defines.

Operational. Not measured; a UNREGISTERED object and not law content.

Units. none

Range. a subset of the state space

Collision warning. Capital and always subscripted; the bare k is the drift exponent (see the k collision).

Appears in. EQ-33.

A_V, A_base, A_ext, A_emb: the arm scores of Law IV **Plain.** The treated arm's score and the baseline arm's score on a frozen property battery after later pressure x, where x, the budget ratio, is the later capability-training budget as a multiple of the budget that installed the gain; A_ext and A_emb are the scores of the external and the embedded arm.

Operational. Scores on the frozen property battery, on a scale fixed with the battery before any outcome; the budget measure that sets x is fixed by the author before any outcome.

Units. battery units

Range. as the battery declares

Collision warning. Always subscripted by arm; the bare A is alignment (see the A collision).

Appears in. EQ-25.

Y_V, Y_min: the installed effect and its floor **Plain.** How much the treated arm still beats the baseline arm after pressure x, and the smallest installed effect at x = 0 for which the retained fraction is read at all.

Operational. The difference of the two arm scores at the same budget ratio (EQ-25); the floor is applied as EQ-25 states.

Units. battery units

Range. Y_V any real number; Y_min positive

Appears in. EQ-25.

S_V, S_ext, S_emb: the retained fraction, by arm **Plain.** What is left of the installed effect after pressure x, as a fraction of what was installed; S_ext and S_emb are the external and the embedded arm's, and S_ext(1) is P15's endpoint.

Operational. Y_V(x) / Y_V(0), read only where the floor is met, with its interval by a method fixed in advance (EQ-25).

Units. dimensionless

Range. any real number; it may exceed one or be negative

Collision warning. Always subscripted; the registration's scope labels are written scope S1 to scope S7, and P5's retained fraction is a different quantity (see the S collision).

Appears in. EQ-22; EQ-23; EQ-25.

alpha_err: the error-decay exponent **Plain.** The positive suppression exponent in a declared error-versus-resource power model; negative values describe increasing error.

Operational. EQ-07 or a declared regression on positive errors, with zero handling, window and uncertainty fixed in advance. No capability or self-improvement mapping is presumed.

Units. dimensionless

Range. real

Appears in. EQ-07; EQ-08.

R_res: the resource coordinate of an error-decay fit **Plain.** The positive resource input, such as tokens or samples, actually varied by the reported fit.

Operational. State the units and reference R_res0; this is not automatically modelled recursive depth R, added passes r, logarithmic capability clock tau or calendar time t.

Units. declared resource units

Range. positive

Appears in. EQ-07; EQ-08.

alpha_net, alpha_cos, d_w: the domain-specific physical exponents and cosmological coordinate **Plain.** alpha_net relates the transport-limited response to its scale in EQ-13; alpha_cos is the scale-factor exponent in time for the specified constant-w Friedmann solution. d_w is the algebraic coordinate 2/(1+3w).

Operational. Each exponent retains its source variable pair. Neither is identified with an AI capability-growth exponent by equal numerical value.

Units. dimensionless

Range. As scoped in EQ-13; d_w is not a physical network dimension in general.

Appears in. EQ-13.

epsilon_mean, sigma, n: the averaging error, its component scale and the sample count **Plain.** The mean of n centred independent errors with common finite variance sigma squared has root-mean-square error sigma/sqrt(n).

Operational. State the independent observational units and the loss. A classification error rate, bias or selected maximum is a different quantity.

Units. sigma and RMSE in error units; n a count

Range. sigma nonnegative, n a positive integer

Appears in. EQ-18.

6 Collisions and source-specific conventions

Grouped readings can contain several subscripts; the entries specify them without treating old census counts as scientific claims.

gamma

gamma. the theory registration's bare letter: chi, the correction-leverage exponent on the target axis, printed here only inside a quotation of the registration (the theory registration; EQ-03 as registered).

gamma_K. the registration's checker-axis slope, a different estimand, written chi_K here (the theory registration, version 1.104; EQ-29).

gamma_V, gamma_J. the registration's checkable and judged exponents, chi on two fault populations, written chi_V and chi_J here (the theory registration, P22).

gamma_N. the residual-decay exponent, how fast leftover error falls with revision count at fixed capability (Unit 05's within-system estimator; P11; no equation in this register).

gamma_D. the capability-to-depth conversion exponent, whose own finite-time ceiling is one divided by it (no equation in this register).

gamma_1, gamma_2, gamma_3. Paper X's gain-drift, level-drift and compounding-drift coefficients; Paper X also writes a bare gamma for gamma_1 in its control parameter, and says it is not the ARC Ceiling's correction leverage (Paper X).

Why it matters. This is the collision with the worst history in the programme. The reciprocal form, one over the exponent, was retracted on 16 August 2026 for the correction-leverage reading, where it runs backwards. The SAME form is correct algebra for the conversion reading, gamma_D, because a different dynamical system sits underneath it. The two ceilings agree at exactly one half, which is the programme's headline number, so a wrong pairing survives casual review. And the register's own bare-letter entry carried the checker axis until version 2.0.0 while its chi entry carried the target axis, so the two disagreed inside one document, which is the defect the theory registration named and ordered corrected. Anyone writing gamma without saying which reading is meant can produce a sentence that is true under one reading, retracted under another, and unfalsifiable as written.

Resolution. The correction-leverage exponent is written chi, on the target axis, and the bare gamma appears only inside a quotation of the theory registration, where it means chi. The checker-axis slope is chi_K and enters a ceiling statement only through EQ-29. chi_V and chi_J are chi on stated fault types. gamma_N, gamma_D and Paper X's gamma_1 to gamma_3 keep their subscripts, and their full names are written in prose on first use.

beta

beta_C. correction exponent, Law II, compared against k; Paper X's bare beta, which Paper X declares to be this one, its saturating form included (EQ-04, EQ-09).

beta_L. self-referential coupling, Foundational Theorem 2, giving alpha = 1/(1-beta_L) (EQ-05, EQ-06).

beta_A. alignment-scaling exponent (EQ-10).

beta_X. Paper XIII's correction-strength exponent, expressed as a ratio of leverage complements only under S1 and its form condition (EQ-12).

beta_cap. Paper X's capacity exponent, the scaling of a saturating corrector's capacity with capability, which carries the comparison with k well above saturation (Paper X; EQ-09's measurement note).

Why it matters. A reader or a machine that conflates beta_C with beta_L gets the safety criterion exactly backwards, because one is a correction rate compared against drift and the other is a coupling

that feeds a reciprocal. Paper X declares that its bare beta is beta_C throughout and writes beta_L for the coupling, so within Paper X the letter is no longer ambiguous. Versions of this register before 2.0.0 read the exponent of Paper X's saturating form as a separate steepness, beta_H, and listed it as a distinct quantity; that reading is withdrawn. The study that administers the reinvestment share calls it beta as well, which is why this register writes that share in words.

Resolution. Use the subscripts and the paper-specific response definitions. Paper XIII's beta_X and Paper X's beta_C refer to the same constitutive A(C) only in their declared join. The ratio formula is conditional and does not identify that strength with chi. beta_L is a coupling and beta_A a bounded-score slope. Adoption in other papers is checked at their own next editions; no universal propagation is claimed here.

Found. 2026-08-22, by census of every beta-bearing display equation across 25 papers

Shape hazard. The reciprocal-shortfall shape 1/(1-x) is now carried by BOTH the self-reference theorem, with the coupling, and the ceiling law, with the correction-leverage exponent. Counted over the paper corpus, the coupling form appears in twelve papers and the ceiling form in seven, so a reader meeting 1/(1-x) is more likely to be looking at the coupling. Before 16 August 2026 the ceiling was written 1/gamma and the two shapes differed, so the shape itself used to distinguish them. It no longer does, and only the argument does. The correction made the mathematics right and the notation more dangerous at the same time.

alpha in the error-decay equation

alpha. the capability-growth exponent (EQ-08 as written in some papers).

alpha_align. an alignment-specific decay exponent (EQ-08 as written in others).

Why it matters. A numerical error-decay estimate cannot enter a capability-growth, leverage or time-compression threshold without a demonstrated response and axis mapping.

Resolution. Current writing uses alpha for capability growth and alpha_err for error decay on its specified resource axis. A separately defined alignment-error exponent is named with its outcome. A reciprocal-error construction gives a conditional algebraic equality, not an empirical identification. The earlier 'never stated' entry is superseded by Statement 7.4, XIII 1.11 and Engineering 8.1.

Found. 2026-08-22

d

d (network). the independently identified effective transport-network dimension in alpha_net = d/(d+1) (EQ-13).

d_w (cosmological substitution). the algebraic coordinate 2/(1+3w), which is not thereby a measured spatial or network dimension (EQ-13).

d (Cohen). Cohen's standardised effect size; the inherited source cites -0.53 as a reported example, without identifying it with a dimension (EQ-20).

d(t) (relative index). D/C in Paper X; a nonnegative modelled relative misalignment index whose definition alone supplies no unit upper bound (Paper X; EQ-34).

Why it matters. The transport dimension, the transformed equation-of-state coordinate, a standardised effect size and the relative index D/C have different definitions. Equal letters or algebraic shapes do not identify their measurements.

Resolution. EQ-13 writes d for network dimension and d_w for the cosmological substitution, with distinct physical exponents. EQ-34 keeps Paper X's d = D/C and states its model scope. Cohen's standard d is retained and named as the effect size. Other source editions retain their own notation; no universal corpus-wide change is asserted.

Found. The collision was recorded on 22 August 2026 by an internal programme census, whose old counts are historical rather than a new census here. The 29 October source collapsed the network and cosmological coordinates into one. This edition separates them against Origin 3.14 (revised 1 November 2026) and Paper X 1.16; Appendix D preserves the earlier account.

A

A. alignment, the degree to which behaviour matches intent; A0 is its starting value (EQ-10).

A_c. correction capacity, how much correcting a system can do, the denominator of the burden ratio (EQ-15).

A (Paper X). correction strength, A = A_0 C^beta_C before saturation, with A_0 its prefactor; Unit 05's draft writes A for corrective strength too (Paper X; EQ-09).

A_V, A_base, A_ext, A_emb. Law IV's arm scores, alignment read arm by arm on a property battery (EQ-25).

A, bold (Paper X 1.16). the correction operator of Paper X's vector model (Paper X, version 1.16, section 3.9 and Theorem 5).

Why it matters. The burden-to-capacity ratio is the derivation the ceiling comes from, so confusing its denominator with alignment misreads the origin of the value two. Paper X's A is correction strength, and versions of this register before 2.0.0 read its saturating form, EQ-09, as alignment.

Resolution. The bare A is alignment. A_c is correction capacity, introduced here rather than inherited: the papers write a bare A. Paper X's correction strength keeps Paper X's letter only inside EQ-09, which says so, and is named in words everywhere else; it takes no symbol of its own in this register until a second equation here needs one. Paper X's bold A, the correction operator of its vector model, is named in words. The arm scores always carry their arm subscript. The mathematics of Laws IV and V names eight symbols it avoids because they collide in the programme's notation, F, E, B_Q, I_Q, A, a, P_V and d_min. E, A and a are defined in this register, E and A with collisions of their own; F, B_Q, I_Q, P_V and d_min enter no equation of this register, so they are not entered here.

Found. 2026-08-22, while adding the burden ratio from the programme's register of papers

P

P. probability, in Bayes' theorem (EQ-19).

P. metabolic power, in the biological scaling analogy (EQ-14).

Why it matters. The two are unrelated and sit in equations of different kinds, one statistical and one illustrative. The risk is low but a reader meeting P for the first time in EQ-14 will carry the wrong meaning into EQ-19.

Resolution. Read P as probability everywhere except the metabolic analogy, which says so in words. P followed by a number, P1 to P22, names a registered proposition of the theory registration and is never a quantity.

delta and Delta

delta (cycle compression). the negative log slope of cycle duration against capability, on a fixed substrate (this register; EQ-11).

delta_obs (direct-bound unit). observed correction pressure, alpha times one minus the correction elasticity, minus one; POSITIVE when burden outruns correction; the exponent of EQ-15 (Unit 04's direct-bound draft (04-study-k2), its H2).

Delta_bal (balance). the balance exponent; POSITIVE when correction gains on burden (EQ-26, EQ-27).

delta (Paper X). the code distance of quantum error correction, in Paper X's analogy (Paper X).

Why it matters. The two pressure quantities carry OPPOSITE sign conventions and are equal in magnitude at the registered dials: Delta_bal equals minus delta_obs there. A reader who meets both and assumes one convention reads a stability verdict backwards. Always name which is meant.

Resolution. delta is the cycle-compression exponent of EQ-11, with dt/dr proportional to C^(-delta). delta_obs is the pressure exponent of the burden-to-capacity ratio and is positive when burden gains. Delta_bal is the correction-to-burden trend on the depth clock and is positive when correction gains. They are equal in magnitude and opposite in sign only at the stated pacing dials on the same clock; a different resource clock adds its conversion factor. Paper X's quantum code distance is named in words.

E

E. error, how wrong the system is (EQ-07, EQ-08).

E (Bayes). evidence, the observation being conditioned on (EQ-19).

Why it matters. E is defined as error five sections before it appears as evidence, so a reader carrying this file's own definition into Bayes' theorem reads the probability of a hypothesis given an error.

Resolution. Read E as error everywhere except inside Bayes' theorem, which is standard statistical notation and must not be renamed.

k

k. the drift-acceleration exponent, compared against beta_C; Paper X's k is the same (EQ-04, EQ-11, EQ-34).

k (Fisher). the number of pooled p-values, setting the degrees of freedom (Fisher's method; EQ-21 writes it in words).

K_V. the persistence kernel, capital and always subscripted (EQ-33).

K(t) (Paper X 1.16). the integrated effective decay (Paper X, version 1.16, Theorem 4).

Why it matters. Twice k degrees of freedom is a count, not an exponent, and the exponent reading is the one this file defines first. The kernel is a set of states, and Paper X's K(t) is a time integral of its effective decay.

Resolution. The bare k is the drift-acceleration exponent; Fisher's k is written in words as the number of p-values pooled; the kernel is always K_V; and Paper X's K(t) is named in words.

chi and its subscripts

chi. the correction-leverage exponent, on the target axis (EQ-03 and throughout).

chi_K. the checker-axis slope, a different estimand (EQ-29).

chi_V, chi_J. chi estimated on checkable and on judged faults (P22).

chi_ol and its arms. the elasticity of an arm's declared service measure, which is chi only under stated identification conditions (EQ-24, EQ-30).

chi-squared. the standard distribution Fisher's method refers to, named in words (EQ-21).

Why it matters. A checker-axis number inserted into a target-axis ceiling returns a number that looks finished and answers a different question, and an arm's elasticity read as chi without its conditions does the same.

Resolution. The bare chi is the target-axis exponent. chi_K is never written chi and is converted by EQ-29 under its direct-term condition. chi_ol is chi only under the identification conditions of the mathematics of Laws IV and V, and otherwise the elasticity of its declared service measure.

H

H. the hypothesis, in Bayes' theorem (EQ-19).

H (horizon). the largest number of transitions a persistence claim looks ahead (EQ-33; the T entry).

Why it matters. Both are standard in their fields, and a reader carrying one into the other misreads a claim.

Resolution. Read H as the horizon in every persistence statement and as the hypothesis only inside Bayes' theorem, which says so in words.

x

x. an argument of the unknown function, in Cauchy's equation (EQ-16).

x (budget ratio). the later capability-training budget as a multiple of the budget that installed a gain (EQ-23, EQ-25; the A_V entry).

Why it matters. The budget ratio is where Law IV's endpoint is read, at x = 1.

Resolution. Read x as the budget ratio in every retention statement; Cauchy's x is standard notation and stays.

M

M. mass, in the metabolic analogy (EQ-14).

M_V^(H). the strategic viability margin, a UNREGISTERED object (the mathematics of Laws IV and V, section 2.5).

Why it matters. The analogy's M is paired with P and carries no claim of this programme.

Resolution. M_V is always subscripted; the analogy's M is always paired with P and says so in words.

eta

eta. the direct depth-dependence of correction service (EQ-26 to EQ-31).

eta (Eden Engineering 6.3). an integration depth, which Eden Engineering writes iota from its version 7.0 (Eden Engineering, version 6.3).

eta (Paper X 1.16). the noise exponent across frozen-coefficient models (Paper X, version 1.16).

Why it matters. The direct depth-dependence is P21's mechanism reading, so a different eta read in its place decides nothing.

Resolution. eta here is the direct depth-dependence and nothing else; the other readings are named in words on first use.

kappa

kappa. the burden intensity elasticity (EQ-26 to EQ-28).

kappa (Paper X). the sum A + r in Paper X's stochastic model, and kappa_eff, its effective decay (Paper X).

Why it matters. Both sit beside correction terms, so a reader moving between the papers can carry the wrong one.

Resolution. kappa here is the burden intensity elasticity; Paper X's is named in words where it is meant.

W

W. newly generated correctable burden (EQ-26).

W (Paper X). a Wiener process, in Paper X's stochastic model (Paper X).

Why it matters. The two never meet in one equation, but both describe how misalignment arises.

Resolution. W here is burden; Paper X's is named as the Wiener process.

r

r. depth, in the growth equation (EQ-06).

r (Paper X). the specific growth rate of capability, r = b C^k (Paper X).

Why it matters. Paper X's drift is written in its r, so reading it as depth misplaces the drift term.

Resolution. r here is depth in EQ-06 only; Paper X's growth rate is named in words where it is meant.

the subscript V

Y_V, S_V, A_V, Omega_V, K_V. of the protected property, in Law IV's mathematics (EQ-25, EQ-32, EQ-33).

chi_V. checkable, in the exponent of checkable correction (P22).

Why it matters. chi_V and S_V sit in the same programme and read V in two senses.

Resolution. Write the full name on first use; chi_V is always the checkable exponent.

S

S_V, S_ext, S_emb. the retained fraction of an installed gain (EQ-22 to EQ-25).

S1 to S7. the theory registration's scope labels (the theory registration).

Why it matters. A scope label printed as S5 beside S_ext reads as a quantity. P5's quantity is also called a retained fraction, of a system's own output, and is not this one.

Resolution. S_V always carries its subscript; a scope is always written scope S5 in words; the qualifier of the installed gain is never dropped from this retained fraction.

B

B. correction burden, the numerator of the burden ratio (EQ-15).

B (Paper X 1.16). the positive constant in the growth of the misalignment fraction in the power-law compounding classification (Paper X, version 1.16, Theorem 4).

Why it matters. The burden ratio is the derivation Law III's ceiling comes from, and Paper X's B is a constant of a growth law, not a burden.

Resolution. The bare B is correction burden; Paper X's constant is named in words.

Found. 2026-10-29, in the refutation of version 2.0.0

alpha and alpha(-M)

alpha. the capability-growth exponent (EQ-01 and the equations that carry it).

alpha(-M) (Paper X 1.16). the spectral abscissa of -M, the largest real part of the eigenvalues of -M, which Paper X marks as not the earlier strand's alpha (Paper X, version 1.16, Appendix A.6).

Why it matters. Paper X marks the difference itself; a reader who meets alpha(-M) in its Appendix A.6 is not reading a growth exponent.

Resolution. The bare alpha is the capability-growth exponent; Paper X's spectral abscissa keeps its argument, alpha(-M), and is named in words here.

Found. 2026-10-29, in the refutation of version 2.0.0

7 Operational terms

Definitions the register governs that no symbol carries: what a counted outcome is, what seals a bank, what the evaluation harness and its two arms are, how a release is pinned, what a corrector's class is, and the rest. Each is fixed from its source, with its standing; where no source defines a thing, the entry says so rather than supplying a definition.

counted outcome, and the counted route

Plain. A verdict whose every margin term is a count of items on a frozen suite, settled by frozen code against ground truth fixed before the run, so that nothing in the margin is judged by a person or by a model.

Operational. Such a verdict does not pass through the scoring instrument, the scoring gate does not bind it, and its domain is printed beside it in the words the deciding unit's registration fixes. Version 1.102 of the theory registration gave this route to P3, P4, P16 and P20 under its condition E1, version 1.103 extended it to P21 and P17, and version 1.104 extends it to P6's interval on checkable correction, a route that "opens no population" and is "necessary and not sufficient for a counted refutation of P6"; it reaches no other proposition. Version 1.104 also reads P5's ordering for a comparison made wholly of counts, without extending the route to P5. The exemption is a property of the margin's terms, never of the proposition: where any judged term enters, the gate binds in full and the verdict is untested until the gate passes. A verdict P21, P17 or P6 takes on the route, and a verdict on P5's counted reading, stands only where its interval or test holds its stated level under the null its deciding unit registers, shown by proof or by simulation on the design that unit actually uses, and it is read only at a look that registration fixes by completed counts. A counted outcome needs no evaluation harness, and says so.

Status. registered rule (the theory registration, version 1.104, which carries condition E1 of version 1.102, the extension of version 1.103 with its successor qualification, and its own extension to P6's interval on checkable correction).

sealed bank

Plain. A fixed set of items and their answer key, committed by its SHA-256 before any run, against which a counted outcome is scored.

Operational. Built deterministically from a registered seed, so that the same seed rebuilds the same bytes and anyone can check the commitment by rebuilding; a hidden bank is encrypted, and its key is kept out of the public deposit and revealed only at analysis. A rebuild that does not match the committed hash means the bank changed after commitment, which voids it. On the theory registration's counted route each decision is recorded by frozen code against a label or ground truth fixed in a sealed bank before the run, with no judged term.

Status. the programme's sealed-bank commitment ledger, and the counted route of the theory registration, version 1.104, for P17 and P21 and for P6's interval on checkable correction, whose departures are "labelled in a bank sealed before any review".

the evaluation harness, its two arms and its off profile

Plain. A separate, versioned evaluation package, never written into an experiment's own code, that any experiment, this programme's or another scientist's, can plug in.

Operational. Every study whose outcome involves a judgement, by a model judge or a scorer, runs with the harness and without it over the same stored responses, in one record and one code path, so that the harness's own effect is measured in every result; the registration names the harness arm as confirmatory. The off profile is the set of harness-off cells the study's frozen plan declares: each presents the original responses through no stage, with no instruction, cue or other fixed line, and shows the label as raw judging would show it, the true source or none and never a false one. Every response another cell judges is also judged, by the same judge, in a harness-off cell of its kind. A later harness release is applied to a study's stored responses only as an analysis its registration declared in advance as secondary, beside the confirmatory result and never replacing it.

Status. The programme's rule for every judged study was adopted on 22 October 2026. The harness is the ARC Evaluation Protocol's. At this edition's source check on 1 November 2026, Unit 86's own registration draft was not ready for submission and no public registration of it was recorded. Its validation role remains prospective; no study is represented here as already run on a validated release.

pinned release

Plain. The exact release of a tool that a result depends on, named so that the result can be reproduced with that tool and no other.

Operational. For a judged study, the registration names in advance the rule that picks the harness release: the first release that passes Unit 86's registered validation. That release's version, its DOI and the SHA-256 of its frozen tree are recorded in a dated addendum on the study's OSF component before any scoring call, and responses collected before then stay sealed and unread. No calendar date is set: scoring waits for the first release that passes, and the fallback is triggered by Unit 86's verdict, never by a date. The Eden Lab engine that runs a study is pinned in the study's lab folder by version and SHA-256. A register that a registration cites, this one included, is pinned by its checksum and not by its version label.

Status. the programme's rules of 22 October 2026 for the harness and the lab engine; the checksum rule for a cited register is the theory registration's own (version 1.104, where it pins the symbols it takes from this register).

corrector class: same-class and cross-class

Plain. What a corrector's strength is permitted to depend on, and whether the corrector shares the substrate of the system it corrects.

Operational. Unit 06, the corrector-class audit, fixes the partition in its registered words: "Corrective mechanisms are partitioned by what their strength is permitted to depend on, which is exhaustive because the available arguments are nothing, the capability of the system being corrected, or the corrector's own prior corrections, with the last splitting by whether it accumulates on the same substrate." Its class 1 is the static and its class 2 the capability-carried. Same-class and cross-class are read against the system corrected: P6 concerns a system that corrects itself using mechanisms of its own class, and P17's panel is named in advance by model lineage, substrate, training provenance and scaling class.

Status. registered partition (Unit 06, registered on OSF on 1 October 2026 and public from 4 October 2026; its registered run, on 6 October 2026, is reported as administratively incomplete under that registration, with no inferential verdict). That frozen partition governs its audit. It is not an exhaustive classification of all control systems, and it does not settle the external-grounding and failure-dependence axes P6 needs; hybrid mechanisms require explicit admission.

checkable and judged correction

Plain. Correction typed by what the corrector can appeal to. In the theory registration's words: "Checkable correction is correction whose correctness is settled by something outside the corrector: a recomputation, a stated constraint, a retrievable record, a real citation. Judged correction is correction whose correctness is settled by a specification or a judgement: conduct, refusal, honesty, harm."

Operational. Every correction-leverage exponent is reported with its fault type, on a fixed fault set of that type; chi_V and chi_J are chi on the two types, and P22 orders them.

Status. registered (the theory registration, version 1.104).

breadth exponent

Plain. The exponent of capability in parallel breadth, many attempts side by side, against which P1 sets the exponent in recursive depth.

Operational. Compute-matched and token-matched: the deciding unit registers one resource coordinate before any fitting and converts both arms into it, and where the two arms cannot be placed on one coordinate the contrast is NOT EVALUABLE; the breadth arm is never handicapped to create a difference. P1 registers that the depth exponent exceeds it by the registered margin. A breadth exponent near zero is consistent with the framework but not derived from it, because width can gain through the selection step rather than through recursion.

Status. registered (the theory registration, version 1.104, P1).

reinvestment share

Plain. The fraction of each round's effort directed at improving the improvement process rather than the current artefact. It is written in words and carries no letter, because the study that administers it calls it beta, which would be another reading of that letter.

Operational. Measured from realised expenditure rather than from requests, with a manipulation check that must pass before any outcome is examined. P2 concerns the shape of the sustainable growth profile as it rises.

Status. registered (the theory registration, version 1.104: its definition of the share, and P2).

usefulness measure, read beside a safety measure

Plain. What a system still does for the people it serves, read beside a safety result, because, in the theory registration's words, "A system can meet a safety measure by doing less, by refusing or failing ordinary tasks or by losing the capability it was trained for".

Operational. The registration of a unit that decides P6 on the counted route, or P15 under the readings version 1.104 adds, names before any data the usefulness measure it reports beside its mark, counted by frozen code against verifiers fixed before the run wherever its terms allow: for P6, the calls and tokens each pass spends, with the share of corrected artefacts that fail their task verifiers; for P15, the capability each arm gains in the later training, and the share of ordinary tasks each arm fails to complete where its baseline arm completes them. "A usefulness measure so reported is a candidate evaluation dimension: it is printed beside the mark, it enters no verdict, margin, mark or law's word, it changes no proposition, and it becomes a registered dimension only through a later successor, before any data that could decide it." For every other proposition whose verdict reads a measure a system could meet by doing less, the deciding unit names its pairing, or why it has none, in its own registration before any data.

Status. registered requirement (the theory registration, version 1.104); the measure is a candidate dimension and decides nothing.

absolute harm and cumulative harm

Plain. The size of the harm a misaligned system does, at a moment and summed over time.

Operational. The compared mathematical sources do not establish a calibrated aggregate measure of absolute or cumulative real-world harm. Paper X's d and D are modelled misalignment indices. Engineering 8.1 separately defines a candidate action-level probability h(a), with calibration and usefulness requirements; it does not identify D with that probability or with cumulative harm. Moving from a balance trend to harm therefore needs a specified response, exposure, horizon and validated measurement model.

Status. undefined; recorded so that no surface supplies a definition the record does not hold.

8 Withdrawn forms and retired names

The following entries are correction records, not live assertions of the withdrawn forms.

alpha_crit = 1 / gamma

Retracted. 2026-08-16

Replaced by. alpha_crit = 1 / (1 - gamma)

Why wrong. It runs backwards under the programme's own definition of the correction-leverage exponent, the registration's gamma, written chi here: it makes a better corrector lower the ceiling. Under the same-class limit, the exponent at most one half, it returns a ceiling of AT LEAST two, which inverts a bound into a floor.

Why it hid. The two forms return the same answer, two, at exactly gamma = one half, which is the programme's headline number. Agreement at the one point anybody checks.

Legitimate where. Inside a correction note, a version history, or a unit whose subject is discriminating the two forms. ALSO legitimate, and this is the exception that caused the confusion, as the finite-time ceiling of the capability-to-depth CONVERSION exponent gamma_D, which is a different quantity obeying a different law. When used that way it must be written with the conversion symbol and never with a bare gamma, because the retraction governs the correction-leverage reading absolutely

k = beta_L - 1

Retracted. withdrawn in full in an earlier draft of Paper XIII

Replaced by. k = delta - 1/alpha

Why wrong. Withdrawn as a general clock identification. It is the delta = 0 special case of k = delta + beta_L - 1; positive delta denotes cycle compression under dt/dr proportional to C^(-delta), not a positive cost-duration slope.

Legitimate where. inside Paper XIII's own withdrawal note

alpha = 2.24 as a measured value

Retracted. when the author's own larger, blinded replication on five frontier models did not reproduce it; what survived that replication is a different row (EQ-01), not a correction of the same measurement

Replaced by. None listed.

Why wrong. It came from one model, DeepSeek R1, on twelve problems scored by exact match against an answer key, with a 95 per cent interval of 1.5 to 3.0, and it did not replicate: Paper II attributes it to a small sample, a single model and a compressed range of accuracies. Blinding played no part in the retraction. Paper II itself is not retracted.

Legitimate where. only ever as retracted; the bare figure is banned on every surface

alpha_max

Retracted. 2026-08-30

Replaced by. alpha_crit

Why wrong. The quantity is a conditional crossover in the stated pacing model, not an impossibility limit. The old maximum label suggested an absolute prohibition and duplicated alpha_crit. Crossing the model boundary alone does not establish a real-world safety or harm verdict.

Why it hid. Both symbols sat beside the same formula, 1/(1 - gamma), so a reader never met the two at once and every surface hand-wrote whichever it had seen last.

Legitimate where. inside a correction note, a version history, or a passage discriminating the two symbols

Law III, the ARC Bound

Retracted. 2026-08-30

Replaced by. Law III, the ARC Ceiling

Why wrong. The ARC Bound is the inequality, alpha at most two, that the ARC Ceiling returns from the same-class limit on chi, not the law; naming the instance as the law hides the assumptions the value rests on.

Legitimate where. inside a dated correction row or version note; as printed in the DOI-frozen papers, which carry the canonical name in their placement notes

Law III, the Ceiling

Retracted. 2026-08-30

Replaced by. Law III, the ARC Ceiling

Why wrong. The programme prefix is part of the law's name; the bare word is a description, and a description can attach to any ceiling.

Legitimate where. inside a dated correction row or version note; as printed in the DOI-frozen papers

the Co-Scaling Law

Retracted. 2026-08-30

Replaced by. the ARC Co-Scaling Law

Why wrong. Law II's name carries the programme prefix like the other two; the bare form drifted onto surfaces written from memory.

Legitimate where. inside a dated correction row; as printed in the DOI-frozen papers, whose placement notes carry the canonical name

9 Limitations

1. Which independently justified capability scale, error response and resource/depth mapping would connect a particular error-decay fit with a capability-growth model? The symbol distinction is resolved; the empirical bridge remains open.
2. The value two, as a point value, rests on two assumptions, and neither is established: the burden model (EQ-15), and the route to chi equal to one half, which needs four conditions (see chi). P17 registers the cross-sectional test bearing on the second, and its counted test, Unit 82, has not run;

a supported P17 would still leave the point value conjectured, because the mapping from panel dependence to chi is registered nowhere.

3. P5, in no register, is the falsifier EQ-05 supplies: measure beta_L independently, predict alpha, and compare with the finite-window exponent of the complete solution. Unit 79, the coupling-identification titration, is its design draft; it is not registered and has not run. A reading in which any term of P5's comparison is judged by a person or by a model is not evaluable until the scorer-validation units have passed; version 1.104 of the theory registration reads that ordering as not binding a reading in which every term of both of P5's results is computed from counts, and extends no counted route to P5 (see the counted outcome).
4. Cohen's d and the programme's dimensional d share a letter and cannot both be renamed, because one belongs to statistics rather than to this programme. Which of the programme's own uses takes a subscript is an open decision.
5. Law V's build-order limb is registered with a direction (EQ-24), and no unit that could decide it is registered: whether a corrector formed first carries a chi larger by at least its registered minimum, initially 0.10 and replaceable only by a smaller minimum registered in advance than one added after it, at matched content, is unmeasured. P12 has no counted route, so while its exponent is judged, the scoring gate holds the limb's reading at untested.
6. Law IV's comparison limb has no proposition. Its entry must fix the operational definition of correction embedded in the training objective, the baseline, the margin, the scale, and whether the embedded arm's own retained fraction must clear one half, before any data that could decide it (EQ-23).
7. Whether beta_C and chi are one quantity is unestablished (see chi).
8. How should absolute and cumulative harm be operationalised for a specified system, exposure and horizon, separately from the modelled misalignment indices and Engineering's action-level probability? The compared sources establish no universal harm guarantee.

The source comparison establishes what the cited documents say and which mathematical premises a transfer requires. It does not validate a capability scale, scoring instrument, taxonomy, correction mechanism or forecast. The current definitions adopt the explicit scoped statements and preserve superseded wording historically. No frozen registration is edited.

10 What this paper does and does not show

It supplies definitions, source mappings and conditional mathematical checks. It reports no new empirical result, performs no study analysis and establishes no law's empirical status. A well-formed formula does not establish its physical interpretation. Corrected notation cannot improve a measurement's precision or turn a proxy into the target construct.

11 Conclusion

Every numerical exponent belongs to an outcome, an input, a scale and a clock. Keeping those attached makes the claims testable without treating different models as one. The open work is empirical identification: measuring the promised quantity under the registered conditions and demonstrating each bridge before using it.

Acknowledgements

Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.

Human authorship and copyright. Michael Darius Eastwood is the author and takes responsibility for the content. He asserts copyright under the Copyright, Designs and Patents Act 1988. This written work is licensed under Creative Commons Attribution 4.0 International. This edition's internal checks comprise source reconciliation, mathematical scope review, citation identity checks, automated presentation and source-identity checks, and a rendered page read. They are not external peer review or independent replication.

Code and data. This register generates no study data. Its editable LaTeX source and bibliography accompany the full edition. Historical study code and data retain their own terms; the licence above applies to the written work.

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Michael Darius Eastwood. Paper V · The ARC Theory · The Eden Protocol · The Stewardship Gene: Stakeholder Care as a Probe of Embedded Values, 2026p. URL https://doi.org/10.17605/OSF.IO/KZEYA. First published 2026-03-16. Cited version 1.7, dated 5 October 2026. released working-paper edition.

Michael Darius Eastwood. 83-study-ct--in-loop-correction-with-depth · Testing Law V (the placement limb, P21): does correction inside each revision round pull away, as depth rises, from a monitor that sees only the finished register? A counted test, 2026q. URL https://doi.org/10.17605/OSF.IO/MN73W. First public 25 October 2026. Frozen OSF registration mn73w, registered 22 October 2026; no document-wide version number is stated. Public availability was observed on 25 October, separately from registration on 22 October.

Michael Darius Eastwood. The ARC Theory · Theory-Level Predictions for Recursive Self-Improvement and AI Alignment: Twenty-Two Severable Propositions Registered Before Any Admissible Confirmatory Outcome, 2026r. URL https://doi.org/10.17605/OSF.IO/P8CKQ. First published 2026-09-08. Cited version 1.104, dated 29 October 2026. registered successor.

Michael Darius Eastwood. Does visible high performance show that no support is needed?, 2026s. URL https://doi.org/10.17605/OSF.IO/RBHDP. First published 2026-07-03. Cited version 1.11, dated 29 October 2026. released working-paper edition.

Michael Darius Eastwood. 06-study-ac--corrector-class-audit · What Class of Corrector Is Deployed AI Alignment Actually Using? A Preregistered Classification Audit Against a Four-Cell Taxonomy, 2026t. URL https://doi.org/10.17605/OSF.IO/RV2B5. First public 4 October 2026. Frozen OSF registration rv2b5, registered 1 October 2026; no document-wide version number is stated.

Michael Darius Eastwood. Monograph · The ARC Theory · The Hyperspace Recursive Intelligence Hypothesis: a Testable Creation Theory, 2026u. URL https://doi.org/10.17605/OSF.IO/UYDXQ. First published 2026-07-03. Cited version 3.52, dated 15 October 2026. released working monograph.

Michael Darius Eastwood. Paper VII · The ARC Theory · Law I, the ARC Principle · Cauchy Unification: the Classification Behind the Law's Form, 2026v. URL https://doi.org/10.17605/OSF.IO/X6WA7. First published 2026-03-16. Cited version 3.15, dated 5 October 2026. released working-paper edition.

Michael Darius Eastwood. On the Origin of Scaling Laws: what can composition and transport explain?, 2026w. URL https://doi.org/10.17605/OSF.IO/XZY9U. First published 2026-02-22. Cited version 3.14, dated 1 November 2026. released edition.

Michael Darius Eastwood. Does recursion amplify capability in the same way across very different fields?, 2026x. URL https://doi.org/10.17605/OSF.IO/Y7QGD. First published 2026-02-13. Cited version 5.21, dated 31 October 2026. released edition.

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A Version history and corrections

Version 2.0.0, 1 November 2026. This full LaTeX edition succeeds released version 1.9.1 of 5 October 2026 and reconciles the unpublished 29 October successor. First publication remains 22 August 2026. No registration or study is redated. The title now carries the register's established name beside its question.

The main corrections distinguish error decay from capability, separate response families, state shifted-power premises, correct cycle compression's sign, identify the conditional correction-strength exponent, preserve measurement equivalence, distinguish trends from safety, and retain separate persistence and embedding endpoints. The dimensional substitution is algebra, the averaging baseline is stated on its actual uncertainty measure, and the metabolic illustration retains its coefficient. No new measurements are made.

Record of the unpublished 29 October successor

The dated text below describes the unpublished version prepared on 29 October. Where it describes that version's current claims, the reconciled entries now govern. Version 2.0.0 (29 October 2026): a major version, for two reasons. The register now covers all five laws, after version 1.103 of the theory registration registered Laws IV and V as laws on 15 October 2026. And a governing definition changes meaning: the correction-leverage exponent is defined on the target axis and written chi, as versions 1.102 and 1.103 of the registration ordered, where the register's bare gamma had carried the checker axis. Laws IV and V enter with their limbs, the retained fraction, the acceptable set and the proposed persistence kernel, with the balance identity, its root and the pacing ceiling they share with Law III, and with the embedding ratio, all in the notation of the mathematics of Laws IV and V, version 1.1; with them come the checker-axis conversion and the relation of absolute to relative misalignment, thirteen equations in all, and ten terms, among them the counted outcome, the sealed bank, the evaluation harness, the pinned release, corrector class and the usefulness measure. Each law's registered standing and verdict rule is stated, with the rules for Laws I and II, the reading of the ARC Bound by correction type and the extension of the counted route to P6 that version 1.104 of the registration fixes; Law II is stated in its three registered regimes; the ARC Bound is the inequality, and its point value is conjectured; the surviving estimate's row travels whole, with its parallel exponent as Paper II states it at version 2.18; the retraction of 2.24 is stated as Paper II states it; beta_H is withdrawn, because Paper X declares the exponent of its saturating form to be beta_C, and that equation now reads correction strength, not alignment; each of the thirty-three relations is set in TeX for MathJax beside its plain-text form, and each heading prints its equation number; and every mention of the author's papers, and every outside work cited, links to its official source. Version 1.9.2, prepared on 18 October 2026, was never published; its words on the status of Laws IV and V are replaced here by each law's status at this version.

Correction index

Appendix D reprints the released 1.9.1 text in full, for comparison with the current entries. This index identifies the changed fields without repeating their current text. **Register.** C001 (dated), C002 (purpose), C106 (ranking note), C107 (open questions), C108 (open questions). Date the actual reconciled edition; first publication remains 22 August 2026.

how to read. C003 (the one rule), C004 (for a layperson), C005 (what the word law is doing). Definitions need provenance and cannot supersede frozen rules.

equations / EQ-01. C006 (in words), C007 (what it is for), C008 (if it is wrong), C009 (status), C010 (worked example), C011 (model families). Conditional model rather than identity true of every system.

equations / EQ-07. C012 (name), C013 (ascii), C014 (latex), C015 (symbols used), C016 (what it is for), C017 (if it is wrong), C018 (status), C019 (worked example), C020 (measured as), C110 (in words). Name the estimand in the equation.

equations / EQ-08. C021 (ascii), C022 (latex), C023 (symbols used), C024 (what it is for), C025 (status), C026 (warning), C027 (measured as), C136 (in words). Normalise to a positive reference and separate exponent.

equations / EQ-10. C028 (status), C029 (warning), C030 (if it is wrong). Use corrected empirical scope.

equations / EQ-05. C031 (if it is wrong), C032 (measured as). Reject all-or-nothing curve-fitting assertion.

equations / EQ-06. C033 (status). State branches and exact/shifted/asymptotic distinction.

equations / EQ-11. C034 (in words), C035 (what it is for), C036 (status), C037 (measured as). Correct meaning and sign of delta.

equations / EQ-12. C038 (name), C039 (in words), C040 (what it is for), C041 (status), C042 (warning), C043 (measured as). A ratio can also define an exponent.

equations / EQ-02. C044 (what it is for), C045 (scope), C046 (measured as). Model bound is not a universal impossibility or safety result.

equations / EQ-03. C047 (status), C109 (measured as). Make domain and safety bridge explicit.

equations / EQ-15. C048 (in words), C049 (worked example), C050 (measured as). Trend versus level distinction.

equations / EQ-04. C051 (measured as). Correct operational object of k and beta_C.

equations / EQ-34. C052 (measured as). Supply full scope and harm distinction.

equations / EQ-13. C053 (in words), C054 (status), C055 (measured as), C120 (ascii), C121 (latex), C122 (symbols used). Separate variable pairs.

equations / EQ-16. C056 (in words), C057 (refuted by). Doubling admits log-periodic factors and is insufficient.

equations / EQ-14. C058 (ascii), C059 (latex), C060 (status). Dimensionless normalisation retains the coefficient.

equations / EQ-17. C061 (what it is for), C062 (refuted by). Remove universal significance claim.

equations / EQ-18. C063 (in words), C064 (ascii), C065 (latex), C066 (worked example), C067 (what it is for), C123 (symbols used). Independence alone does not imply task-error exponent.

symbols / alpha. C068 (operational), C069 (current estimate), C139 (plain), C140 (appears in). Separate alpha from error decay.

symbols / alpha crit. C070 (operational), C141 (plain). A derived number is not a boundary test.

symbols / beta C. C071 (operational). Match correction object.

symbols / k. C072 (name), C073 (plain), C074 (operational), C142 (range). Identify exact time-growth variable.

symbols / beta L. C075 (range). Correct finite growth overclaim.

symbols / beta X. C076 (name), C077 (plain), C078 (operational), C079 (range). Correct ratio/exponent category.

symbols / delta. C080 (name), C081 (plain), C082 (operational), C083 (range). Repair cost-slope sign.

symbols / g, r, a. C084 (plain), C085 (operational), C143 (range). Preserve amplification/capability and clock distinctions.

symbols / chi. C086 (operational), C087 (class dependence). Remove universal literature claim and retain measurement equivalence gate.

symbols / gamma N. C088 (load bearing note), C117 (operational). No automatic uncertainty-to-capacity inference.

symbols / gamma D. C089 (load bearing note). State premises and clock threshold rather than universal ceiling.

symbols / d, w. C090 (operational), C091 (range), C144 (operational). Cosmological identity is not physical evidence.

symbols / B. C092 (operational). Fix rate versus per-unit-capability mismatch.

symbols / A. C093 (operational), C116 (range). No retroactive counted instrument or validated scoring.

symbols / alpha bal. C094 (range). Correct prior iff outside its branch.

collisions / gamma. C095 (distinct quantities). Count the actual grouped readings, not stale census totals.

collisions / A. C096 (distinct quantities). Count the actual grouped readings, not stale census totals.

collisions / chi and its subscripts. C097 (distinct quantities). Count the actual grouped readings, not stale census totals.

collisions / alpha in the error-decay equation. C098 (resolution), C099 (why it matters). Resolve current collision against final source.

collisions / beta. C100 (readings), C101 (resolution). Correct beta_X entry.

collisions / delta and Delta. C102 (readings), C103 (resolution). Repair sign in collision list.

terms / absolute-and-cumulative-harm. C104 (operational). Avoid global absence claim and acknowledge Engineering action risk.

terms / corrector-class. C105 (status). Retain frozen quotation without promoting its exhaustive language universally.

equations / EQ-29. C111 (status). Equality can occur without global exact proportionality.

equations / EQ-25. C112 (warning), C113 (measured as), C131 (status). A near-zero denominator threatens identification without making every possible interval unbounded.

symbols / E1, E2, R1, R2. C114 (plain), C115 (operational), C147 (units). Align symbol definitions with corrected estimator.

retracted forms / k = beta L - 1. C118 (why wrong). Do not repeat the obsolete cost-sign interpretation in the correction record.

retracted forms / alpha max. C119 (why wrong). State the corrected boundary scope in the retirement note.

symbols / M, P. C124 (plain). Define reference quantities introduced to preserve dimensions.

equations / EQ-23. C125 (status). Use mathematics 1.2's current status word; the earlier proposed wording stays historical.

equations / EQ-32. C126 (status). Use mathematics 1.2's current status word; the earlier proposed wording stays historical.

equations / EQ-33. C127 (law), C128 (status). Use mathematics 1.2's current status word; the earlier proposed wording stays historical.

symbols / K V. C129 (name), C130 (operational). Use mathematics 1.2's current status word; the earlier proposed wording stays historical.

symbols / alpha err. C132 (entry). Explicit symbols prevent the measured/modelled estimand collision.

symbols / R res. C133 (entry). Explicit symbols prevent the measured/modelled estimand collision.

symbols / alpha net, alpha cos, d w. C134 (entry). Explicit symbols prevent the measured/modelled estimand collision.

symbols / epsilon mean, sigma, n. C135 (entry). Explicit symbols prevent the measured/modelled estimand collision.

symbols / R. C137 (plain), C138 (appears in). Remove universal depth claim contradicted by R_res and other clocks.

symbols / E. C145 (appears in). EQ-18 now has a separately defined averaging error.

symbols / A0, E0. C146 (operational). Match EQ-08's model-reference convention.

symbols / kappa. C148 (range). The general chain-rule identity does not impose nonnegative elasticity.

symbols / D. C149 (operational). Distinguish proposed measurement from a validated construct.

collisions / d. C150 (readings), C151 (distinct quantities), C152 (why it matters), C153 (resolution), C154 (found). Separate the physical coordinates and remove the unsupported bounded-ratio claim.

collisions / M. C155 (readings). Update the nested status and index entries to match the corrected entries.

collisions / E. C156 (readings). Update the nested status and index entries to match the corrected entries.

terms / evaluation-harness. C157 (status). Date the status; a private draft or a personal certification does not prove registration.

equations / EQ-21. C158 (what it is for), C159 (warning). Replace an unverified absence claim with the exact official reissue identity actually checked.

equations / EQ-12. C160 (what it is for). Put the two positive-exponent domain conditions beside the identity, rather than only in the symbol entries.

equations / EQ-12. C161 (warning). The bound follows by multiplying by a positive denominator; explicitly exclude other sign branches.

equations / EQ-13. C162 (in words). Dimension alone does not determine the measured exponent; state both physical premises and the measured variable pair.

equations / EQ-13. C163 (what it is for). Replace the general physical-derivation description with the actual conditional use and evidence class.

equations / EQ-13. C164 (if it is wrong). Distinguish failure of the conditional prediction from failure of its physical premises or unrelated claims.

equations / EQ-13. C165 (status). Carry the corrected Origin premises into the operational register without treating a change of coordinates as a mechanism.

equations / EQ-13. C166 (measured as). Keep inventory, mass and the cosmological coordinate distinct; state the finite-range limitation and corrected expansion boundaries.

equations / EQ-13. C167 (measured as). These are sufficient assumptions for the simple proportionality, not necessary for every possible inventory model.

equations / EQ-13. C168 (name). The title distinguishes the transport result from the separate algebraic reparametrisation.

equations / EQ-20. C169 (measured as). Remove the trailing-word widow while clarifying that the standardising definition, not its estimated numerical value, is fixed.

collisions / d. C170 (found). Match the actual current source comparison and bibliography.

equations / EQ-20. C171 (refuted by). Unequal group spreads do not invalidate the definition. Interpretation depends on the chosen standardising convention, design, sample sizes and uncertainty. Preserve the former overstatement as a dated correction; no new analysis.

equations / EQ-13. C172 (status). Process wording removed before publication.

collisions / d. C173 (found). Process wording removed before publication.

collisions / A. C174 (found). Process wording removed before publication.

terms / evaluation-harness. C175 (status). Process wording removed before publication.

Register. C176 (ranking note). Process wording removed before publication.

Register. C177 (why this exists). Process wording removed before publication.

how to read. C178 (what the word law is doing). Consistency with the released companions.

equations / EQ-03. C179 (status). Cite the theory registration's current text, version 1.104, checked against its own words.

equations / EQ-02. C180 (status). Cite the theory registration's current text, version 1.104, checked against its own words.

equations / EQ-31. C181 (what it is for), C182 (status). Cite the theory registration's current text, version 1.104, checked against its own words.

collisions / gamma. C183 (readings). Cite the theory registration's current text, version 1.104, checked against its own words.

terms / counted-outcome. C184 (status). Cite the theory registration's current text, version 1.104, checked against its own words.

terms / sealed-bank. C185 (status). Cite the theory registration's current text, version 1.104, checked against its own words.

terms / pinned-release. C186 (status). Cite the theory registration's current text, version 1.104, checked against its own words.

terms / checkable-and-judged. C187 (status). Cite the theory registration's current text, version 1.104, checked against its own words.

terms / breadth-exponent. C188 (status). Cite the theory registration's current text, version 1.104, checked against its own words.

terms / reinvestment-share. C189 (status). Cite the theory registration's current text, version 1.104, checked against its own words.

Earlier version history

Version 1.9.1 (5 October 2026): the title now follows the programme's uniform order (paper, its law or framework, its own title), Register · The ARC Theory · Operational Definitions; until this version it read Operational Definitions of the ARC Theory. No equation, symbol, collision or retracted form changes.

Version 1.9.0 (4 October 2026): published as a paper in every format, with its own DOI on OSF and its own reference card, under the title Operational Definitions of the ARC Theory (until this version, Operational Definitions of the ARC Programme). The note on what the word law is doing counts the theory's five laws and their status, and the description and the abstract say where the notation of Laws IV and V is fixed. No equation, symbol, collision or retracted form changes.

Version 1.8.0 (7 September 2026): thirty-three of the fifty-one individual symbol names printed in these equations had no rendered definition block, because the renderer looked names up in a dictionary keyed by the spine's full composite strings and silently dropped anything it did not find, because six equations did not name in symbols_used the symbols their own statement prints, and because eight symbols appear in no equation at all. Every symbol's domain and units were carried in this file and rendered nowhere. The lookup now fails loudly, the composite keys resolve member by member, the orphans have a standalone section, the domain and units render, and the gate tests for a rendered definition block rather than for a name occurring somewhere on the page, which 46 of the 51 already did. It supersedes 1.7.2 (sha256 0ae774690effd36e9c19eafee863300dfce38ca2f2687ffaae20a969923af927), not 1.7.1. 1.7.2 (2026-09-04): the delta family was declared a collision with readings, a severity and a why-it-matters and no resolution key, so build-operationalisation.py raised KeyError('resolution') and the production chain died on it. The resolution now names the quantity rather than the letter: bare delta is the cost-scaling exponent and nothing else, the direct-bound unit's pressure reading is written delta_obs in full and is positive when burden outruns correction, the balance model's reading is written Delta in full and is positive when correction gains on burden, and because the two are equal in magnitude and opposite in sign at the registered dials neither may be quoted bare. One field was added to one collision in commit 68bbb55f91; no equation, symbol, definition or count changed, and the register still carried 21 equations, 38 symbols, 7 collisions, 7 retracted forms and 5 open questions after it as before. Those bytes hash to sha256 0ae774690effd36e9c19eafee863300dfce38ca2f2687ffaae20a969923af927 and stood under the

number 1.7.1 from 4 September 2026 until the 1.8.0 bump. They are given their own number here, in the record and not by editing them, so that 1.7.1 names one set of bytes and one only: 1.7.1 (2026-09-03) is the deposited state, sha256 68891fa12cf3ee4a58d1a7d278aecdd9ed170366339e36cc7ab478027bbbeb48, which the theory registration's companion extract of 3 September 2026 cites and which nothing in this record changes. 1.7.0 (2026-09-03): the gamma collision, which is the register's worst, pointed one way only. The chi entry recorded that it is the registration's gamma; the gamma entry did not record that the quantity is now written chi, so a reader arriving at the collision was not told the current name existed. 1.6.0 (2026-08-27): worked example ran a bounded percentage through unbounded arithmetic; recast on the ladder-scale latent measure with the three model families stated (27 August 2026); then, 30 August 2026, the law labels were aligned with the home declaration (Law II is the ARC Co-Scaling Law; Law III is the ARC Ceiling, of which the ARC Bound is the same-class instance) and alpha_max was retired as a second symbol for alpha_crit

First published on 22 August 2026; versions 1.5.0 (24 August), 1.6.0 to 1.6.2 (27 to 30 August) and 1.7.0 to 1.7.1 (3 September) built and corrected the spine, each dated by its commit.

Dated source account of the register's origin

The following account records the source register's reason for creation on 22 August 2026. Its implementation statements describe that source and date.

On 22 August 2026 the published glossary asserted the retracted ceiling law alpha_crit = 1/gamma while the definitions page, the foundational paper and the programme spine all carried the corrected 1/(1-gamma). The site contradicted itself about its own load-bearing law, in the one page whose entire job is saying what symbols mean. The cause was structural rather than careless: the spine held the correct value, but no page drew the law from it, so every surface hand-wrote the law and one of them hand-wrote it wrong. This register, with a check at every build, makes that class of failure a build error.

B Priority record

First published on 22 August 2026, the register arose from a disagreement between a glossary and the corrected ceiling formula. Version 1.7.1's frozen hash continues to identify the extract pinned by the registration; no later register changes it. The priority record links dated programme sources. This edition claims no new theorem or first-in-field result.

Appendix D retains the complete substantive released manuscript as historical quotations. Its status words describe its date and are not this edition's verdicts.

C Notation guide

The full definitions, units and domains are in Section 5, the collisions in Section 6. This guide locates the central distinctions without introducing new symbols: α is capability growth, α_{err} error decay; β_L is self-referential coupling, β_C constitutive correction strength, and χ target-axis delivered correction service. The registration's correction-leverage letter maps to χ in this register. δ is cycle compression, with a minus sign in its cycle-duration slope; Δ_{bal} is a balance trend. R, added depth r, resource R_{res}, logarithmic capability clock τ and time t retain their separate axes. $d = D/C$ is the relative index and D its absolute counterpart. The arm labels on S_{ext} and S_{emb} distinguish retained fractions; the input x is their declared budget ratio. No equality between these quantities is supplied merely by the notation.

D Version history: full released manuscript of 1.9.1

Historical status of every passage below. The full substantive register released on 5 October 2026 is preserved as a dated quotation. Its old estimator identifications, law-status words and notation are superseded wherever the current entries correct them. They are evidence of the earlier text, not parallel current assertions. References to old section positions refer to that document. Typesetting is regularised, table rows are transcribed in reading order, and source links point to official records. Website controls are omitted; the standard disclosure is printed in the current acknowledgements.

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The ARC Theory · OSF osf.io/gvq82 · every claim checkable

The source of truth for the ARC Theory's notation: every equation the programme relies upon for Laws I to III, ranked by how much depends on it, each symbol defined and paired with its measurement; the notation of Laws IV and V is fixed in their mathematics until it joins.

Working Paper v1.9.1 | First published 22 August 2026, revised 5 October 2026

Abstract

This document fixes what each symbol in the ARC Theory's equations means and how it would be measured. It ranks the twenty-one equations the programme relies upon for Laws I to III by how much depends on each, from the ARC Equation of Law I, U = I x R^alpha, through Law II's criterion, beta_C > k, and Law III's ceiling, alpha_crit = 1 / (1 - gamma), to the statistical methods the experiments use. It defines thirty-eight symbols in plain words, pairs each with the measurement that would produce it, names the nine letters the corpus uses for more than one quantity, and records the seven forms the programme has retracted, with the reason each was wrong. It reports no result and asserts no value: every figure it quotes is carried from the paper that measured it. The theory has five laws. Laws I to III were registered on 8 September 2026; Laws IV and V, the ARC Persistence Law and the ARC Embedding Law, were added on 24 September 2026, their propositions registered and neither law yet registered as a law or tested. Their notation is fixed in section 1 of their mathematics, version 1.0 (3 October 2026), and joins this document at its next version.

How to read this

An exponent is a growth-shape number, not a size. It answers: when I double the effort, what happens to the result? An exponent of 1 means the result doubles too. An exponent of 2 means it quadruples. An exponent below 1 means you get less than double, which is the shape of diminishing returns. Every claim in this programme is a claim about the value of one of these shape numbers, so every argument turns on measuring one correctly.

The one rule. A symbol means what this file says it means, and nothing else. Where the same letter is used for two different quantities anywhere in the corpus, that letter appears in the collisions list below, and both readings are named. If you meet a symbol in a paper and cannot find it here, that is a defect in the paper, not in your reading.

What the word law is doing here. The theory's five laws are named conjectures. Laws I to III were registered on 8 September 2026; Laws IV and V were added on 24 September 2026, their propositions registered and neither law yet registered as a law or tested. Each law states a relation held open to measurement; none is established, and the status line under every equation says which of definition, derivation, criterion, modelling choice or open measurement it is. Naming a conjecture is how it becomes arguable, not how it becomes true.

The equations, ranked

Ranked by how much fails if the equation is wrong, not by how often it appears. The first four are the programme; the rest support them. The third column is the ranking rationale, so the order can be argued with rather than taken on trust. Everything else about each equation, what it says in plain words, what it is for, every symbol with its measurement, and what would refute it, is in its own section below and is not repeated here.
| Equation | Why it ranks here: what fails if it is wrong

1 | U = I x R^alpha
The ARC Equation | Nothing else in the programme survives. This is the frame, not a result.

2 | beta_C > k
The ARC Co-Scaling criterion | The safety claim dies, and EQ-02 loses its operational meaning again, because the word stable in the ARC Bound is defined by this criterion.

3 | alpha_crit = 1 / (1 - gamma)
The ARC Ceiling | The value two loses its support, though the law itself is separable: a different measured gamma yields a different ceiling and the structure stands.

4 | alpha <= 2
The ARC Bound | The headline claim dies. The framework survives, because EQ-01 and EQ-04 do not depend on the value two.

5 | B(R) / A_c(R) ~ R^[alpha(1 - gamma) - 1]
The burden-to-capacity ratio | The ceiling loses its derivation and the value two loses its support.

6 | alpha = d / (d + 1), d = 2 / (1 + 3w)
The dimensional derivation | The cross-domain derivation dies; the empirical programme is untouched.

7 | f(xy) = f(x) * f(y)
Cauchy's multiplicative functional equation | The claim that a power law is forced rather than fitted collapses.

8 | alpha = ln(E1 / E2) / ln(R2 / R1)
The slope estimator | Every measured number in the programme is wrong, including the retraction. Operationally this is the most important equation here, whatever its rank as a law.

9 | resample with replacement, refit, take the 2.5th and 97.5th percentiles
The bootstrap confidence interval | Every stated interval is wrong, including the one that withdrew 2.24.

10 | E ~ R^(-1/2), that is alpha = 0.5
The zero-parameter null | Nothing of the framework falls. If it is RIGHT, the compounding reading of the measurement is unnecessary.

11 | P(H | E) = P(E | H) * P(H) / P(E)
Bayes' theorem | Nothing. It is a theorem.

12 | d = (mean1 - mean2) / standard deviation
Cohen's d | Nothing. It is a definition.

13 | X^2 = -2 * sum(ln p_i), with 2k degrees of freedom
Fisher's method for combining p-values | The combined figure falls; the individual results are untouched.

14 | alpha = 1 / (1 - beta_L)
The self-reference theorem | Alpha becomes a fitted parameter with no derivation, and the programme becomes curve-fitting.

15 | dg/dr = a * g^(beta_L)
The growth equation | EQ-05 loses its derivation.

16 | E(R) = E0 * R^(-alpha)
Error decay | The link between the frame and the data breaks.

17 | A(C) = A_max * C^beta_H / (C^beta_H + Cs^beta_H)
The saturation curve | The saturating treatment in Paper X loses its functional form.

18 | A(R) = A0 * R^beta_A
Alignment scaling | Paper II's alignment-side scaling claim dies.

19 | k = delta - 1/alpha
The self-acceleration exponent | The drift term in EQ-04 loses its derivation and must be measured directly.

20 | beta_X = (1 - beta_L) / (1 - beta_L_prime)
The boundary gap | Paper XIII's cross-system comparison loses its measure.

21 | P/M = M^(3/4) / M = M^(-1/4)
The metabolic analogy | An analogy is lost. No claim depends on it.

Each one, explained

Each symbol is defined in full at its first appearance, in rank order, and later equations point back to that definition rather than repeating it. A definition restated four times reads as padding and gives a reader four chances to meet a version that has drifted.

1 The ARC Equation Law I, the ARC Principle

In words. A system's usable capability equals what it starts with, multiplied by how deeply it can improve itself, raised to a power.

U = I x R^alpha

What it is for. It is the definition that makes the rest measurable. It does not assert a value; it says which number to go and measure. Every other equation here is a statement about alpha.

Worked. Capability here is a latent, ratio-scaled measure with no ceiling, like the level a system reaches on a calibrated difficulty ladder, never a bounded percentage score: a score capped at 100 cannot follow this arithmetic, which is why the programme's own instruments measure on the ladder scale, after a bounded percentage score was found running through unbounded arithmetic and the worked example was recast on 27 August 2026. Suppose a system sits at capability U = 40 on such a ladder at one pass, so I = 40. Allow it 16 passes, so R = 16. If alpha = 0, looping is worthless: 16^0 = 1, so U = 40, no gain. If alpha = 0.5, 16^0.5 = 4, so U = 160. If alpha = 1, 16^1 = 16, so U = 640. If alpha = 2, 16^2 = 256, so U = 10,240, and each loop makes the next more productive. That is the whole argument in four lines, and the question is which of those worlds we are in. The honest answer so far is the first or second.

Measured as. Varied: recursive depth R, counted on a configuration ladder fixed in advance. Read out: usable capability U, scored on a target battery frozen before the first measurement. Held fixed: the battery, the ladder and the scoring instrument, for the whole run.

U usable capability What the system can actually do, after everything is accounted for. The output side of the equation. **How it is measured.** Score on a fixed target battery held constant across the whole experiment. The battery must be frozen before any measurement, because a battery that changes during a run makes the exponent meaningless. **Domain.** positive **Units.** battery score, dimensionless once normalised

I base intelligence What the system could do before any self-improvement. The starting point. **How it is measured.** The same battery score measured at recursion depth one, before any improvement pass is applied. **Domain.** positive **Units.** same as U

R recursive depth How many times the system has been through its own improvement loop. If it reviews and revises its own work three times, R is three. **How it is measured.** Count of completed improvement passes under a fixed configuration ladder. Must be counted, never estimated, and the ladder fixed in advance. **Domain.** integer, at least one **Units.** dimensionless count

α (alpha) the capability-growth exponent The growth-shape number. It says what happens to capability when you double the recursive depth. At alpha = 1 capability doubles. At alpha = 2 it quadruples. Below 1 you get diminishing returns. **How it is measured.** The slope of a straight line fitted to capability against recursive depth when both are plotted on logarithmic axes. In practice, EQ-07 applied across the measured range, with the interval reported always. EQ-07 is written on the error side, so using it for alpha assumes the error-decay exponent is this same alpha. Whether it is has never been stated anywhere in the corpus, and it is an open collision: see alpha in the error-decay equation, below. **Domain.** unbounded in principle; claimed at most two for systems that hold together

Status. definitional; the exponent is open to measurement, and the measured value is currently 0.49 with interval [-1.3, 2.9], which distinguishes nothing

Relied upon in: foundational, executive-summary, paper-i-arc-principle, paper-c-pnp.

2 The ARC Co-Scaling criterion Law II, the ARC Co-Scaling Law

In words. A system stays aligned with itself only while its power to correct grows faster than its tendency to drift.

beta_C > k

What it is for. The safety criterion, and since 22 August 2026 also the meter that makes EQ-02 falsifiable: stability is no longer asserted, it is measured as correction out-scaling drift.

Worked. A system doubles in capability. Its correction capacity rises 30 per cent, so beta_C is about 0.38. Its drift rises 20 per cent, so k is about 0.26. Since 0.38 > 0.26 correction is pulling ahead and the off-target fraction shrinks as it grows. Reverse the two and the same system grows more capable while getting harder to keep on target. Read the comparison, never either number alone: beta_C = 0.9 with k = 1.1 is unstable despite far stronger correction.

Read this before using it. The beta in this criterion is NOT the beta in EQ-05. See collisions.

Measured as. Varied: recursive depth. Read out: the two exponents beta_C and k, each as a log-log slope from the same estimator that yields alpha. Held fixed: the drift and correction instruments, so that the two slopes are commensurable. The criterion is the comparison of the two, not a quantity fitted in its own right.

β_C (beta_C) the correction exponent How fast the brakes strengthen. The rate at which a system's power to correct itself grows as it gets more capable. **How it is measured.** Log-log slope of corrective strength against capability, from the same estimator as alpha, EQ-07. **Domain.** real

k the drift-acceleration exponent How fast the engine speeds up. The rate at which a system's tendency to drift away from its target accelerates as it improves itself. **How it is measured.** Log-log slope of drift against capability, same estimator; or derived from cost scaling through EQ-11. **Domain.** real; negative values mean drift decelerates

Status. criterion; both exponents estimable from the same log-log slope estimator that yields alpha, which is why one instrument serves both laws

Relied upon in: hrih-paper, paper-x-coupled-coscaling.

3 The ARC Ceiling Law III, the ARC Ceiling

In words. The ceiling on the growth-shape number is one divided by the correction shortfall, where the shortfall is how far the correction-leverage number falls short of one.

alpha_crit = 1 / (1 - gamma)

What it is for. It is where the number two comes from. Feed it one assumption, that gamma is one half, and it returns two. Change the assumption and the ceiling moves with it.

Conditional on class. The ceiling this returns is conditional on the corrector's class, because its only input is the leverage exponent and that exponent is class-dependent. Feed it the same-class cap of one half and it returns two. Feed it a cross-class exponent above one half and it returns more than two. The equation is unchanged either way; what changes is which system you are asking about.

Worked. The equation is a reciprocal, so the smaller the shortfall the higher the ceiling. gamma = 0.3 gives a shortfall of 0.7 and a ceiling of 1.43. gamma = 0.5 gives 0.5 and exactly 2.00. gamma = 0.8 gives 0.2 and 5.00. gamma = 0.9 gives 0.1 and 10.00. Read the middle row: a corrector scaling as the square root of capability gives a ceiling of exactly two. That is where the famous number comes from, as an output under one assumption, not a discovery.

Read this before using it. The retracted form 1/gamma and this corrected form return the same answer, two, at exactly gamma = one half. That single point of agreement is the programme's headline number, which is why the wrong form survived undetected on the glossary page for six days. Below one half the two forms move in opposite directions: the retracted form makes a WORSE corrector raise the ceiling, which inverts a bound into a floor.

Measured as. Not measured directly. Both sides are estimated separately, the correction-leverage exponent from correction accumulation and alpha from EQ-07, and the relation is tested by comparing the two estimates rather than by fitting it.

α_crit (alpha_crit) the stability ceiling The highest growth-shape number a system can have and still hold itself together. **How it is measured.** Not measured directly. Computed from gamma through EQ-03. To test it you measure gamma, not the ceiling. **Domain.** at least one

γ (gamma) the correction-leverage exponent How much good it does to correct yourself. It measures how effectively internal correction converts accumulated capability into actual improvement. Higher is better. Written chi from the working paper of 16 August 2026 onward, because gamma had carried three incompatible readings; the theory registration still says gamma and the two are the same quantity. Use chi in any new writing and gamma only when quoting the registration. **How it is measured.** The per-family slope of misalignment removed per review pass against checker capability, under a blinded scoring instrument, with the target battery held fixed. Drafted as study-ad; never measured on any real system. **Domain.** assumed at most one half for SAME-CLASS correctors, under the independence premise. Not a universal cap: it is a cap on that class **Class dependence.** The cap is a property of the corrector's CLASS, not of correction as such. A corrector sharing the system's substrate cannot be anti-correlated with itself, which is where the one half comes from. A corrector from a different class is not bound by it and may exceed it. Build order is a lever on class: as the field currently builds, a corrector added after training on the vast corpus is drawn from that corpus and is close to maximally same-class, whereas one built and validated first on a separate curated corpus has the opportunity to differ. Opportunity rather than guarantee: a curated corpus drawn from the same distribution would separate

nothing. **What this rests on.** The value one half is a conjecture and not a derivation. It rests on the premise that internally accumulated corrections combine like independent samples. This is the single undefended assumption on which the ceiling at two rests, and it is deliberately left open.

χ **(chi) the correction-service elasticity in capability** How much more correction a system can deliver when it becomes more capable. **How it is measured.** The elasticity of correction service with respect to capability. This is the same quantity the theory registration calls the correction-leverage exponent gamma; chi is the name used from the working paper of 16 August 2026 onward, because gamma had carried three incompatible readings. Not identifiable from a single path: needs crossed or off-path variation. **Domain.** 0 to below 1 for a finite ceiling

Status. corrected 16 August 2026; see retracted_forms

Relied upon in: foundational, executive-summary.

4 The ARC Bound the same-class instance of Law III, the ARC Ceiling, at gamma equal to one half

In words. The growth-shape number cannot exceed two and still describe a system that holds itself together.

alpha <= 2

What it is for. The programme's headline claim. Not an impossibility claim: the region above two is enterable, but not correctable.

What it is a bound of. The ARC Bound is the SAME-CLASS bound. It was never universal, because its derivation runs through a leverage cap that this estate has always stated as a same-class cap, but until 22 August 2026 no surface a reader would meet said so. A cross-class system measured above two therefore does not refute it; it tests the build-order prediction instead. The kill condition now requires the corrector's class to be declared and evidenced before the run, on the same footing as the measurement window, because otherwise the assertion that a corrector was cross-class becomes a fresh escape of exactly the shape that the metering of the same day closed.

Not a separate law. The ARC Bound is the ARC Ceiling's output under one assumption, not a law of its own. The famous two is this equation's answer, never its input. Ranked fourth for that reason, below the equation that produces it.

Measured as. Not measured directly. It is a threshold on alpha, so it is tested by measuring alpha through EQ-07 and asking whether the estimate and its interval sit at or below two.

α **(alpha) the capability-growth exponent** Defined in full at EQ-01, its first use.

Status. conditional under one stated assumption still on trial, that same-class correctors accumulate corrections the way independent samples do; refutable since 22 August 2026, when all four terms of its kill condition were metered: the exponent by its interval excluding two rather than its point estimate, the window by prior declaration, the system by a genuine self-referential loop, and stability by the ARC Co-Scaling Law's condition

Relied upon in: foundational, executive-summary.

5 The burden-to-capacity ratio the derivation behind Law III

In words. Compare how fast a system's need for correction is growing against how fast its ability to correct is growing. Whether that gap opens or closes decides everything.

B(R) / A_c(R) ~ R^[alpha(1 - gamma) - 1]

What it is for. This is where the ceiling actually comes from. The ceiling equation is its consequence, not its premise.

Worked. Everything hinges on the sign of alpha(1 - gamma) - 1. Negative, when alpha is below the ceiling: the ratio shrinks, the corrector catches up, stable. Positive: the ratio grows without limit, uncorrected error piles up, unstable. Exactly zero: the exponents tie and equality is NOT automatically safe, because the outcome then turns on coefficients, delays, starting conditions and saturation, none of which the exponent captures. Set it to zero and solve and you get the ceiling law: that is where Law III comes from.

What would prove it wrong. Its premise, that correction burden tracks the capability a system is ADDING. If burden instead tracks the capability it already HAS, the growth exponent cancels out of the stability condition entirely and no ceiling on growth survives at all. Law II would survive; Law III would not.

Read this before using it. The capacity term is written A_c here, not a bare A. The papers write it as A, which already means alignment elsewhere in this corpus. Reusing the bare letter while auditing letter reuse would have been the same mistake in a new place, so the subscript is introduced here and the papers should adopt it.

Measured as. Not measured directly. What matters is the sign of the exponent, and that is read off two separately estimated quantities, alpha from EQ-07 and the correction-leverage exponent from correction accumulation. The exactly-zero case is not automatically safe and is reported rather than rounded away.

B correction burden How much correcting a system needs. It grows as the system does more, because more capability means more that can go wrong. **How it is measured.** Volume of correction required per unit of capability added, measured over the same fixed battery used for capability. **Domain.** non-negative **Units.** correction units, dimensionless once normalised

A_c correction capacity How much correcting a system can actually do. The other side of the burden. **How it is measured.** Correction delivered per pass at a given capability, from the same blinded instrument that supplies the correction exponent. **Domain.** positive **Units.** same as burden, so the ratio is dimensionless

R recursive depth Defined in full at EQ-01, its first use.

α **(alpha) the capability-growth exponent** Defined in full at EQ-01, its first use.

γ **(gamma) the correction-leverage exponent** Defined in full at EQ-03, its first use.

Status. derivation; the premise is stated and is the single point of attack

Relied upon in: foundational.

6 The dimensional derivation

In words. The growth-shape number equals the effective dimension divided by that dimension plus one, and the dimension itself follows from an equation-of-state parameter.

alpha = d / (d + 1), d = 2 / (1 + 3w)

What it is for. The route by which the framework connects to a physical derivation rather than a fitted one.

Measured as. Not measured on any system. d and w enter from the physical derivation as a consistency route, not as fitted quantities.

α **(alpha) the capability-growth exponent** Defined in full at EQ-01, its first use.

d effective dimension and equation-of-state parameter
w effective dimension and equation-of-state parameter In the dimensional derivation, d is the effective number of directions the system can grow in, and w is the parameter that fixes it. **How it is measured.** Not measured on any AI system. These come from the physical derivation and enter the framework as a consistency route, not as fitted quantities. **Domain.** d positive

Status. derivation within stated assumptions

Relied upon in: on-the-origin-of-scaling-laws.

7 Cauchy's multiplicative functional equation

In words. If doubling the input always multiplies the output by the same factor no matter where you started, the relationship can only be a power law and nothing else fits.

f(xy) = f(x) * f(y)

What it is for. The argument that scaling laws must be power laws rather than merely happen to look like them.

What would prove it wrong. A continuous function satisfying the equation that is not a power law. Mathematically foreclosed under the stated regularity conditions, which is why the empirical route matters more: real domains that violate the classification.

Measured as. Not measured. It is a classical result, and what has to be checked in each application is whether the regularity conditions hold over the domain claimed.

f the unknown function and its arguments, in Cauchy's equation
x the unknown function and its arguments, in Cauchy's equation
y the unknown function and its arguments, in Cauchy's equation A rule you do not yet know, and two numbers you feed it. The equation asks which rules turn multiplication of the inputs into multiplication of the outputs; power laws are the answer. **How it is measured.** Standard mathematical notation for the multiplicative functional equation. **Domain.** positive reals in the form used here

Status. classical result; the content is in whether the regularity conditions hold in the domains claimed

Relied upon in: paper-vii-cauchy-unification.

8 The slope estimator the measurement recipe

In words. To find the growth-shape number from data, take two measurements, divide one error by the other, take the logarithm, and divide by the logarithm of the ratio of the two depths.

alpha = ln(E1 / E2) / ln(R2 / R1)

What it is for. This is the only equation in the programme that touches data. Every measured exponent anywhere in the corpus, including the retracted 2.24 and the corrected 0.49, came out of this estimator or its regression form.

Worked. A power law curves on ordinary axes and straightens under logarithms: log U = log I + alpha x log R, which is a straight line y = c + mx whose slope IS alpha. A logarithm answers how many times you multiply by ten to reach a number, so equal steps on the axis are equal multiplications and a multiplicative relationship becomes a line. What goes wrong: a slope fitted across a factor of two in R is nearly meaningless, slopes need about a decade to be trustworthy; curvature fitted as a slope still returns a number that means nothing; and a high r-squared says the points sit near the line, never that the line is the right model.

Read this before using it. The corpus writes this two ways: with a leading minus and the error ratio inverted, and without. They are the same estimator. A reader meeting both may reasonably think they are different equations; they are not.

Measured as. Varied: recursive depth, at two checkpoints R1 and R2 chosen before the run. Read out: error E1 and E2 at those two depths, from which the slope is computed. Held fixed: the error instrument and the ladder, because a slope taken between two differently measured points is not a slope.

α (alpha) the capability-growth exponent Defined in full at EQ-01, its first use.

E error How wrong the system is. The thing that is supposed to fall as it improves. **How it is measured.** Error rate on the frozen target battery. Must be the same battery at every depth. **Domain.** non-negative **Units.** error rate, dimensionless

R recursive depth Defined in full at EQ-01, its first use.

E1 (E1) the two measurement points
E2 (E2) the two measurement points
R1 (R1) the two measurement points
R2 (R2) the two measurement points The pair of readings the slope is worked out from: error at each of two depths, and the two depths themselves. Two points make a line. **How it is measured.** Error and recursive depth at the earlier and later of the two checkpoints entering the slope estimator. **Domain.** positive **Units.** E as error; R as depth

Status. standard log-log slope; the same estimator supplies beta_C and k in EQ-04, which is what lets one instrument meter both laws

Relied upon in: paper-ii-experimental-validation, paper-iii-alignment-scaling-problem.

9 The bootstrap confidence interval

In words. Rather than trusting one estimate, rebuild the dataset thousands of times by resampling it, and see how much the answer moves.

resample with replacement, refit, take the 2.5th and 97.5th percentiles

What it is for. Every claim of significance in the programme. It is what produced the [-1.3, 2.9] interval that the point estimate must never be printed without.

What would prove it wrong. Nothing about the method. It is MISUSED by resampling data that are not independent, for instance repeated runs of one model treated as separate observations, which makes the interval look narrower than the truth.

Measured as. Not measured; it is the procedure that puts an interval on something else that was. Varied: the resample. Read out: the 2.5th and 97.5th percentiles of the refitted estimate. Held fixed: the dataset, the fitting procedure and the number of resamples, all declared before the first resample is drawn.

This equation carries no symbol. It is a procedure stated in words, so an empty symbol list here is a fact about the method and not an omission in the register.

Status. standard method; the risk is application, not theory

Relied upon in: paper-ii-experimental-validation.

10 The zero-parameter null

In words. If each pass of reasoning is just an independent noisy attempt and errors average out, then plain statistics predicts an exponent of one half with no theory at all.

E ~ R^(-1/2), that is alpha = 0.5

What it is for. It is the null this programme raised against ITSELF, and it is the most dangerous equation in the register. The programme's own corrected headline measurement is 0.49. That sits almost exactly on what averaging predicts with no framework at all.

Worked. Where the one half comes from, with no theory at all: the central limit theorem. Average n independent noisy attempts and the error shrinks in proportion to the square root of n, so the predicted exponent is 0.5 with no compounding, no leverage and no framework. The programme's own corrected measurement is 0.49. A measurement equally consistent with the framework and with plain averaging does not support either; it shows the measurement cannot tell them apart. Raised by this programme against itself, 8 August 2026, published 15 August.

What would prove it wrong. This condition fires AGAINST the framework, not for it. It fires, and the compounding reading is withdrawn, if a test built in advance to separate the two returns the null's prediction rather than a stated deviation from 0.5. It is retired only when such a test returns a deviation the framework predicted.

Measured as. Not measured; it is the null that a measurement is scored against. It fixes alpha at one half with no fitted parameter, so a measured alpha is compared with one half rather than with zero.

E error Defined in full at EQ-07, its first use.

R recursive depth Defined in full at EQ-01, its first use.

α **(alpha) the capability-growth exponent** Defined in full at EQ-01, its first use.

Status. open and adverse; raised by this programme against its own headline number

Relied upon in: paper-ii-experimental-validation.

11 Bayes' theorem

In words. Start with how likely you thought something was, look at the new evidence, and adjust in proportion to how much better that evidence is explained by your idea being true than by it being false.

P(H | E) = P(E | H) * P(H) / P(E)

What it is for. All reasoning in the programme about priors and about what evidence would change them.

Worked. Worked, because this is the one people nod at without following. Suppose before any data you think there is a 30 per cent chance the correction exponent is one half, so P(H) = 0.3. A measurement arrives that would appear 80 per cent of the time if H is true and 20 per cent of the time if it is false. How often would you see it at all: P(E) = (0.8 x 0.3) + (0.2 x 0.7) = 0.24 + 0.14 = 0.38. Then P(H|E) = 0.8 x 0.3 / 0.38 = 0.63. Belief moves from 30 per cent to 63 per cent. Notice it did not jump to certainty: the evidence multiplied the odds by four, from 3 to 7 up to 12 to 7, which moved the probability from 30 per cent only to 63 per cent, and odds and probability must never be quoted as though they were the same movement. That restraint is the point of using the rule rather than reasoning by impression. Operationalised here three ways: dated priors recorded in writing BEFORE any confirmatory data, with what would change them; kill conditions, which are commitments about P(E|H) written in advance, which is why the window must be declared before the run; and the reason consistent-with is not supports, because when P(E|H) and P(E|not H) are about equal the ratio is about one and the posterior barely moves from the prior.

What would prove it wrong. Nothing. It is provable from the definition of conditional probability, not an empirical claim. What can be wrong is its APPLICATION: priors chosen after the fact, likelihoods guessed to suit, or a claim of updating without recording what the prior was.

Measured as. Not measured. It is the rule by which a measurement moves a stated prior, and the discipline is that the prior is recorded before the evidence is seen.

H hypothesis and evidence, in Bayes' theorem H is the claim being tested. E is what you observed. Bayes' theorem says how much the observation should move your confidence in the claim. **How it is measured.** Standard statistical notation, not a quantity of this programme. Included so the equation can be read without another reference. **Domain.** probabilities lie between zero and one

Also printed above, and defined elsewhere in this register: P in the P collision, which names the Bayes reading and the metabolic one, E in the E collision, and the H entry, which defines the evidence reading.

Status. theorem; the discipline is in recording the prior before the evidence

12 Cohen's d

In words. How far apart two groups are, measured in units of how spread out the data already is, so a difference can be called large or small without knowing what was measured.

d = (mean1 - mean2) / standard deviation

What it is for. How big any measured difference in the programme actually is.

Worked. Conventionally about 0.2 is small, 0.5 moderate, 0.8 large, and the sign gives direction. A d of -0.53 means the first group scored lower by rather more than half a standard deviation. Dividing by the spread is what makes it portable: a 5-point gap is enormous if scores usually vary by 2 and trivial if they usually vary by 50.

What would prove it wrong. Nothing about the formula. It is MISAPPLIED when the two groups have very different spreads, so dividing by one shared standard deviation flatters or buries the difference, and when it is reported without the sample size that says how reliable it is.

Read this before using it. This d is not the programme's dimensional d. It belongs to statistics and must never be renamed; only the programme's own uses can take subscripts.

Measured as. Varied: group membership. Read out: the difference between the two group means. Held fixed: the measured quantity and its standard deviation, which is the unit the difference is expressed in.

mean1 (mean1) the two group averages, in Cohen's d
mean2 (mean2) the two group averages, in Cohen's d The average of each of the two groups being compared, before the difference between them is expressed in standard deviations. **How it is measured.** Standard statistical notation, not a quantity of this programme. **Domain.** unbounded **Units.** as for the measured quantity

Also printed above, and defined elsewhere in this register: d in the d collision, which names the effect-size reading.

Status. standard statistic

13 Fisher's method for combining p-values

In words. A way of pooling several separate weak results into one overall figure, which only works if the results were genuinely independent of one another.

X^2 = -2 * sum(ln p_i), with 2k degrees of freedom

What it is for. One published combined figure, since withdrawn.

What would prove it wrong. Nothing about the method. The live question is whether the independence condition holds, and here it does not, which is why the combined figure was withdrawn rather than defended.

Measured as. Withdrawn in application, so nothing is currently measured with it. Read out: a combined statistic pooled from k separately reported p-values. Held fixed: the requirement that the pooled results be genuinely independent, which was never established here.

X^2 (X) the combined test statistic and its inputs
p_i the combined test statistic and its inputs A way of pooling several separate tests into one verdict, and the individual results being pooled. **How it is measured.** Standard statistical notation for Fisher's method, not a quantity of this programme. Included so the equation can be read without another reference. **Domain.** p between zero and one

Also printed above, and defined elsewhere in this register: k in the k collision, which names the pooled-p-value reading.

Status. withdrawn in application; independence was never established

14 The self-reference theorem Foundational paper, Theorem 2

In words. If a system's rate of improvement depends on how good it already is, raised to some power, then its capability grows as a power law whose exponent is one over the shortfall of that power from one.

alpha = 1 / (1 - beta_L)

What it is for. It is where alpha comes from mechanically rather than by fitting. It converts a statement about self-reference into a measurable exponent.

Read this before using it. This equation and EQ-03 share the shape one-over-one-minus-something. They are different laws with different arguments. Recognising the shape is not recognising the law.

Measured as. Not measured. It is a proof within stated assumptions, and its inputs are measured elsewhere: beta_L from EQ-06 and alpha from EQ-07.

α (alpha) the capability-growth exponent Defined in full at EQ-01, its first use.

β_L (beta_L) the self-referential coupling How strongly a system's rate of improvement depends on how good it already is. At zero, improvement is constant regardless of skill. Near one, the better it gets the faster it gets better, and growth runs away. **How it is measured.** Fitted from the growth equation EQ-06 over the measured range, then checked against alpha through EQ-05 as an independent consistency test. That check is an unexploited falsifier: measure beta_L, predict alpha, compare with the fitted alpha. **Domain.** less than one for finite growth

Status. proved within stated assumptions; not independently reviewed

Relied upon in: foundational, on-the-origin-of-scaling-laws, paper-ii-experimental-validation, paper-xiii-self-acceleration-exponent.

15 The growth equation the differential equation generating EQ-05

In words. The rate at which capability improves per unit of recursive depth is proportional to current capability raised to the coupling power.

dg/dr = a * g^(beta_L)

What it is for. The mechanism. Solve it and EQ-05 falls out. It is the only place the self-reference is actually stated as a process rather than a result.

Measured as. Varied: recursive depth r. Read out: the rate of change of capability g with depth. Held fixed: the configuration ladder and the scoring instrument, because beta_L is claimed constant only over the range studied and a changed instrument would hide that.

g growth variables of the differential equation
r growth variables of the differential equation
a growth variables of the differential equation In the growth equation, g is current capability, r is depth, and a is a constant of proportionality that sets the pace but not the shape. **How it is measured.** g

and r as for U and R. The constant a is fitted and carries no claim: it moves the curve up or down without changing the exponent, which is what the programme is about. **Domain.** positive **Units.** as for U and R; a carries the units that balance the equation

β_L (beta_L) the self-referential coupling Defined in full at EQ-05, its first use.

Status. standard separable ODE; the content is in the claim that beta_L is constant over the range studied

Relied upon in: paper-iii-alignment-scaling-problem, paper-ix-synthesis-and-roadmap, paper-xiii-self-acceleration-exponent.

16 Error decay

In words. Error falls away as recursive depth rises, as a power of that depth.

E(R) = E0 * R^(-alpha)

What it is for. The measured face of EQ-01. What EQ-07 is fitted to.

Read this before using it. The corpus writes the exponent here as both alpha and alpha_align. If they are the same quantity, one symbol should be used; if they are different, the difference has never been stated. Open notation defect, logged in collisions.

Measured as. Varied: recursive depth R. Read out: error E. Held fixed: the error instrument, and the starting value E0, which is the first measured point of the series being fitted rather than a free parameter.

E error Defined in full at EQ-07, its first use.

E0 (E0) the starting values Where alignment and error stood before the growth being described began. Every power law needs a starting point, and this is it. **How it is measured.** The value of alignment, and of error, at the first measured point of the series being fitted. **Domain.** positive **Units.** as for the quantity each scales

R recursive depth Defined in full at EQ-01, its first use.

α (alpha) the capability-growth exponent Defined in full at EQ-01, its first use.

Status. measured form

Relied upon in: paper-ii-experimental-validation, paper-iii-alignment-scaling-problem.

17 The saturation curve

In words. Alignment rises with capability but flattens towards a maximum, with the steepness set by an exponent.

A(C) = A_max * C^beta_H / (C^beta_H + Cs^beta_H)

What it is for. Models alignment that cannot exceed a ceiling, unlike a bare power law which rises without limit.

Read this before using it. The exponent here is a steepness parameter of a saturating curve, a Hill coefficient. It is the third distinct quantity in the corpus written with the letter beta. See collisions.

Measured as. Varied: capability C. Read out: alignment A. Held fixed: the alignment instrument and the capability scale. A_max and Cs are fitted constants of the curve, not measured quantities.

A alignment How well the system's behaviour matches what was intended. **How it is measured.** Blinded score under an audited consensus protocol, with model-level self-exclusion, six to seven blinded scores per entry. **Domain.** bounded above by A_max where a saturating form is used **Units.** score, dimensionless once normalised

C capability How capable the system is, used where capability rather than recursive depth is the driving variable. **How it is measured.** Battery score, as for U. **Domain.** positive **Units.** same as U

A_max the saturation constants
C_s (Cs) the saturation constants The most alignment a system can reach however capable it becomes, and the capability at which it is halfway there. **How it is measured.** Fitted constants of the saturation curve: the upper asymptote, and the capability at half that asymptote. **Domain.** both positive **Units.** A_max dimensionless; Cs in capability units

β_H (beta_H) the saturation steepness How sharply alignment flattens out as it approaches its maximum. A high value means a sudden knee in the curve; a low value means a gentle bend. **How it is measured.** Fitted parameter of the saturation curve EQ-09. Not an exponent of a power law and not comparable with the other betas. **Domain.** positive

Status. modelling choice, not derived

Relied upon in: paper-x-coupled-coscaling.

18 Alignment scaling

In words. Alignment grows as a power of recursive depth.

A(R) = A0 * R^beta_A

What it is for. The alignment analogue of EQ-01.

Read this before using it. Fourth distinct beta. See collisions.

Measured as. Varied: recursive depth R. Read out: alignment A. Held fixed: the alignment instrument, and the starting value A0.

A alignment Defined in full at EQ-09, its first use.

A0 (A0) the starting values Where alignment and error stood before the growth being described began. Every power law needs a starting point, and this is it. **How it is measured.** The value of alignment, and of error, at the first measured point of the series being fitted. **Domain.** positive **Units.** as for the quantity each scales

R recursive depth Defined in full at EQ-01, its first use.

β_A (beta_A) the alignment-scaling exponent How alignment grows with recursive depth, as a growth-shape number of its own. **How it is measured.** Log-log slope of alignment score against recursive depth. **Domain.** real

Status. measured form

Relied upon in: paper-ii-experimental-validation.

19 The self-acceleration exponent

In words. How fast a system's improvement accelerates equals its cost-scaling exponent minus one over its growth-shape number.

k = delta - 1/alpha

What it is for. Supplies the drift side of EQ-04 from measurable quantities, so the safety criterion does not need a separate instrument.

Measured as. Not measured directly. It is computed from two measured exponents, the cost-scaling exponent delta and alpha from EQ-07, so its interval inherits both of theirs and must be reported with them.

k the drift-acceleration exponent Defined in full at EQ-04, its first use.

δ **(delta) the cost-scaling exponent** How the cost of each further improvement grows as the system gets better. At zero, every step costs the same. **How it is measured.** Log-log slope of per-step cost against capability. **Domain.** real

α **(alpha) the capability-growth exponent** Defined in full at EQ-01, its first use.

Status. derived in Paper XIII; an earlier form k = beta_L - 1 was withdrawn in full

Relied upon in: paper-xiii-self-acceleration-exponent.

20 The boundary gap

In words. The gap between two systems' shortfalls, expressed as a ratio.

beta_X = (1 - beta_L) / (1 - beta_L_prime)

What it is for. Compares two self-referential systems on one scale.

Read this before using it. Fifth distinct beta, and the only one that is a ratio rather than an exponent. Paper XIII subscripts it correctly; no other paper adopted the convention.

Measured as. Not measured directly. It is a ratio of two shortfalls, each computed from a beta_L measured on its own system through EQ-06, so the two systems must be measured on the same instrument before the ratio means anything.

β**_X (beta_X) the boundary gap** A way of comparing two self-improving systems on one scale, by dividing one system's shortfall by the other's. **How it is measured.** Computed from two fitted couplings through EQ-12. A ratio, not an exponent. **Domain.** at most one **Units.** dimensionless ratio

β**_L (beta_L) the self-referential coupling** Defined in full at EQ-05, its first use.

β**_L' (beta_L_prime) the comparison coupling** The same self-reference number measured on a second system, so two systems can be compared rather than one described. **How it is measured.** The self-referential coupling of the comparison system in the boundary-gap ratio. **Domain.** less than one

Status. derived

Relied upon in: paper-xiii-self-acceleration-exponent.

21 The metabolic analogy

In words. Power per unit mass falls as mass rises, to the power minus one quarter.

P/M = M^(3/4) / M = M^(-1/4)

What it is for. The biological precedent for a scaling exponent that is argued for from structure rather than fitted to data. Kleiber's three-quarter exponent is the observation; the network derivation offered for it by West, Brown and Enquist is the argument this analogy points at, and it is disputed: a competing account derives two thirds from surface-to-volume scaling, and the empirical exponent is not settled between them. What that literature holds and this programme does not is a mechanism grounded in measured anatomy. The analogy is used here only to show that a derived exponent is a thing science already has, and nothing in this programme depends on which exponent wins. Illustrative, not load-bearing.

Measured as. Not measured by this programme. Mass and metabolic power are measured in comparative physiology, and the result is carried here as an illustration; nothing in this programme is fitted to it.

P mass and metabolic power, in the biological analogy
M mass and metabolic power, in the biological analogy How heavy an organism is, and how much energy it burns. The analogy notes that bigger animals burn less energy per kilogram, which is a real scaling law from biology and is used here only as an illustration. **How it is measured.** Standard biological notation. This P is metabolic power and is NOT the P of probability in EQ-19. **Domain.** positive **Units.** M in mass; P in power

Status. established result from the literature, cited not claimed

Relied upon in: on-the-origin-of-scaling-laws.

Symbols this register defines that no equation here carries

8 symbols are defined in the notation spine and appear in none of the equations above. They are set out here rather than attached to an equation, because giving them a host equation would change what the register claims rather than only how it renders. Each carries its plain definition, its domain, and where in the corpus it is used.

γ_N (gamma_N) the residual-decay exponent How fast the mistakes that are left over disappear as you keep revising. Go round the loop again and again, and this number says how quickly what remains shrinks. Higher means the leftovers vanish faster. **How it is measured.** The slope of residual error against revision count at fixed capability: residual error after N revisions falls as N to the power minus this exponent. Fitted by the depth studies. It appears in no equation in this register, and it has never been measured on any real system. **Domain.** positive; no upper bound is claimed **What this rests on.** The programme's working premise treats this exponent as the same quantity as the correction-leverage exponent, under an assumption that internally accumulated corrections combine like independent samples. That identification is a registered question, not a settled fact, and no result depends on it being true.

γ_D (gamma_D) the capability-to-depth conversion exponent How readily raw ability turns into more rounds of self-improvement. If a system gets cleverer, does it also get faster at improving itself, and by how much. **How it is measured.** The exponent by which revision depth accelerates with capability: the rate of change of depth grows as capability to the power of this exponent. Not measured by any registered study. **Domain.** positive **What this rests on.** This exponent carries its own finite-time ceiling, one divided by it, and that is a DIFFERENT law from the ceiling law EQ-03. The two agree at exactly one half and nowhere else, which is why the form 1/gamma can look correct beside a value of two. Any

sentence pairing this exponent's number with EQ-03's formula is false everywhere except that single point.

Δ (Delta_balance) the balance exponent Whether correction is pulling ahead of the work it has to do, read along the path a system actually took. Positive means the gap is closing. **How it is measured.** The difference of two log-derivatives taken along an observed path: phi_Q minus phi_W. Directly estimable from the path alone; its components are not. Introduced in the author-review working paper of 16 August 2026. **Domain.** any real number **Where it is used.** the balance model; positive means correction is gaining on burden

ϕ_Q (phi_Q) the correction-service path elasticity How fast the correction a system can actually deliver grows as it goes deeper, measured along the path it took. **How it is measured.** d log Q / d log r along the observed path, where Q is delivered correction service. Equals alpha_C times chi plus eta when the path is decomposed, but is estimable without that decomposition. **Domain.** any real number **Where it is used.** the balance exponent

ϕ_W (phi_W) the burden path elasticity How fast the correctable work a system generates grows as it goes deeper, measured along the path it took. **How it is measured.** d log W / d log r along the observed path, where W is newly generated correctable burden. Equals alpha_C times kappa plus zeta when decomposed, but is estimable without that decomposition. **Domain.** any real number **Where it is used.** the balance exponent

κ (kappa) the burden intensity elasticity How much more correctable work each unit of new capability brings with it, as the system gets more capable. **How it is measured.** The elasticity of newly generated burden with respect to capability. The registration's burden model is the case kappa equals one; the burden-intensity rival is kappa above one, written elsewhere as one plus mu_B. **Domain.** non-negative **Where it is used.** the general ceiling; the burden-intensity rival

η (eta) the direct depth-dependence of correction service Whether correction gets better simply by being run more times, apart from any gain in capability. **How it is measured.** The direct exponent of recursive depth in correction service, holding capability fixed. Zero in the registration's model. Embedded correction is the claim that this is larger than it is for correction applied from outside. **Domain.** any real number **Where it is used.** the general ceiling; the embedded-correction claim

ζ (zeta) the direct depth-dependence of burden Whether going deeper generates correctable work by itself, apart from the capability it produces. **How it is measured.** The direct exponent of recursive depth in newly generated burden, holding capability fixed. Minus one in the registration's model, which is what it means for burden to follow the rate of capability gain rather than its level. **Domain.** any real number **Where it is used.** the general ceiling

Where one letter means two things

A symbol that carries two meanings is the failure this page exists to prevent. Both readings are named; neither is quietly dropped.

gamma: 3 distinct quantities

- gamma the correction-leverage exponent, also called the correction-capacity exponent; the one in the ceiling law. Written chi from the working paper of 16 August 2026 onward, which is the name to use in any new writing; the registration still says gamma and the two are the same quantity (EQ-03)
- gamma_N the residual-decay exponent, how fast leftover error falls with revision count at fixed capability (the depth studies; no equation in this register)
- gamma_D the capability-to-depth conversion exponent, whose own finite-time ceiling is one divided by it (no equation in this register)

Why it matters. This is the collision with the worst history in the programme. The form alpha_crit = 1/gamma was retracted on 16 August 2026 for the correction-leverage reading, where it runs backwards. The SAME form is correct algebra for the conversion reading, because a different dynamical system sits underneath it. The two ceilings agree at exactly one half, which is the programme's headline number, so a wrong pairing survives casual review. Anyone writing gamma without saying which of the three they mean can produce a sentence that is true under one reading, retracted under another, and unfalsifiable as written. Say which one, every time. One further reading exists and it is a rename rather than a fourth quantity: the ceiling-law reading is written chi from 16 August 2026, because gamma could no longer be written safely. The chi entry says so; this entry now says it too, so the pointer runs both ways and a reader meeting either symbol finds the other.

Resolution. Write the full name in prose on first use in any document, and use a subscript only alongside it. The bare letter gamma is reserved for the correction-leverage exponent of EQ-03 and means nothing else.

beta: 5 distinct quantities

- beta_C correction exponent, Law II, compared against k (EQ-04)
- beta_L self-referential coupling, Foundational Theorem 2, giving alpha = 1/(1-beta_L) (EQ-05, EQ-06)
- beta_H saturation steepness, a Hill coefficient (EQ-09)
- beta_A alignment-scaling exponent (EQ-10)
- beta_X boundary gap, a ratio of shortfalls (EQ-12)

Why it matters. A reader or a machine that conflates beta_C with beta_L gets the safety criterion exactly backwards, because one is a correction rate compared against drift and the other is a coupling that feeds a reciprocal. Paper X uses a bare beta for two different quantities within one paper.

Resolution. This file subscripts all five. The corpus has not yet adopted the subscripts outside Paper XIII, which introduced them independently and correctly. Propagation across the papers is outstanding, and it belongs to the papers rather than to this register: this file records the convention, and each paper adopts it at its next version.

alpha in the error-decay equation: 2 distinct quantities

- alpha the capability-growth exponent (EQ-08 as written in some papers)
- alpha_align an alignment-specific decay exponent (EQ-08 as written in others)

Why it matters. If these are the same quantity, one symbol should be used. If they are different, the difference has never been stated anywhere in the corpus.

Resolution. Unresolved. Flagged for the papers to settle, not settled here, because settling it is a scientific judgement and this file records judgements rather than making them.

d: 3 distinct quantities

- d effective spatial dimension, in alpha = d/(d+1) and supplied by the equation-of-state relation d = 2/(1+3w); one quantity across two equations, not two (EQ-13)
- d (Cohen) Cohen's d, a standardised effect size, reported as d = -0.53; a statistic about a comparison, carrying no relation to dimension (results reported across seven papers)
- d(t) the misalignment fraction, defined as D/C, how much of a system's capability is misaligned; a bounded ratio that evolves in time (Paper X)

Why it matters. A reader meeting d in a results table and d in the dimensional derivation is meeting two unrelated quantities, and d(t) in Paper X is a third. Cohen's d is standard notation nobody should change, which makes the collision permanent rather than fixable: the answer is to subscript the programme's own uses, never the statistic.

Resolution. Unresolved in the corpus. This file records all three. The dimensional d and the misalignment fraction d(t) should carry distinguishing marks in the papers; Cohen's d must be left exactly as it is, because it belongs to the discipline rather than to this programme.

A: 2 distinct quantities

- A alignment, the degree to which behaviour matches intent (EQ-09, EQ-10)
- A_c correction capacity, how much correcting a system can do, the denominator of the burden ratio (EQ-15)

Why it matters. The burden-to-capacity ratio is the derivation the ceiling comes from, so confusing its denominator with alignment misreads the origin of the value two.

Resolution. Subscripted as A_c in this file. Introduced here rather than inherited: the papers write a bare A. Recorded as a collision this register CREATED by naming a quantity that had none, rather than one it found, because reusing the bare letter while auditing letter reuse would have been the same mistake in a new place.

P: 2 distinct quantities

- P probability, in Bayes' theorem (EQ-19)
- P metabolic power, in the biological scaling analogy (EQ-14)

Why it matters. The two are unrelated and sit in equations of different kinds, one statistical and one illustrative. The risk is low but a reader meeting P for the first time in EQ-14 will carry the wrong meaning into EQ-19.

Resolution. Read P as probability everywhere except the metabolic analogy, which says so in words.

delta and Delta: 3 distinct quantities

- delta (cost) the cost-scaling exponent (this register)
- delta_obs (direct-bound unit) observed correction pressure, alpha times one minus the correction elasticity, minus one; POSITIVE when burden outruns correction (the direct-bound draft's H2)

- Delta (balance) phi_Q minus phi_W; POSITIVE when correction gains on burden (the balance model of 16 August 2026)

Why it matters. The two pressure quantities carry OPPOSITE sign conventions and are equal in magnitude at the registered dials: Delta equals minus delta_obs there. A reader who meets both and assumes one convention reads a stability verdict backwards. Always name which is meant.

Resolution. Name the quantity, never the letter. Bare delta is the cost-scaling exponent and nothing else. The direct-bound unit's pressure reading is written delta_obs in full and is POSITIVE when burden outruns correction; the balance model's reading is written Delta in full and is POSITIVE when correction gains on burden. Because the two are equal in magnitude and opposite in sign at the registered dials, a sign quoted without its symbol's full name decides a stability verdict the wrong way round, so neither may be quoted bare, and any passage carrying both states the convention it is using in the same sentence.

E: 2 distinct quantities

- E error, how wrong the system is (EQ-07, EQ-08, EQ-18)
- E (Bayes) evidence, the observation being conditioned on (EQ-19)

Why it matters. E is defined as error five sections before it appears as evidence, so a reader carrying this file's own definition into Bayes' theorem reads the probability of a hypothesis given an error.

Resolution. Read E as error everywhere except inside Bayes' theorem, which is standard statistical notation and must not be renamed.

k: 2 distinct quantities

- k the drift-acceleration exponent, compared against beta_C (EQ-04, EQ-11)
- k (Fisher) the number of pooled p-values, setting the degrees of freedom (EQ-21)

Why it matters. 2k degrees of freedom is a count, not an exponent, and the exponent reading is the one this file defines first.

Resolution. The bare k is the drift-acceleration exponent; Fisher's k is written in words as the number of studies pooled.

Forms that were withdrawn

Kept on the record rather than deleted. A programme that removes its own retractions loses the thing that made the correction honest.

alpha_crit = 1 / gamma

Replaced by alpha_crit = 1 / (1 - gamma), 2026-08-16.

Why it was wrong. It runs backwards under the programme's own definition of gamma: it makes a better corrector lower the ceiling. Under the same-class cap gamma at most one half it returns a ceiling of AT LEAST two, which inverts a bound into a floor.

Why it went unnoticed. The two forms return the same answer, two, at exactly gamma = one half, which is the programme's headline number. Agreement at the one point anybody checks.

Legitimate where: Inside a correction note, a version history, or a unit whose subject is discriminating the two forms. ALSO legitimate, and this is the exception that caused the confusion, as the finite-time ceiling of the capability-to-depth CONVERSION exponent gamma_D, which is a different quantity

obeying a different law. When used that way it must be written with the conversion symbol and never with a bare gamma, because the retraction governs the correction-leverage reading absolutely..

k = beta_L - 1

Replaced by k = delta - 1/alpha, withdrawn in full in an earlier draft of Paper XIII.

Why it was wrong. Recorded as withdrawn in the paper itself; the replacement derives k from cost scaling rather than from the coupling.

Legitimate where: inside Paper XIII's own withdrawal note.

alpha = 2.24 as a measured value

Replaced by alpha = 0.49, interval [-1.3, 2.9], which distinguishes nothing, after cross-architecture replication failure.

Why it was wrong. It did not replicate across architectures.

Legitimate where: only ever as retracted; the bare figure is banned on every surface.

alpha_max

Replaced by alpha_crit, 2026-08-30.

Why it was wrong. The ceiling is a crossover on stability, not a maximum: a system can exceed it and cannot stay correctable there. alpha_max reads as an impossibility limit, which the stability framing forbids, and it named the same quantity as alpha_crit under a second symbol, so the glossary and the founding-paper drafts carried the law under two names.

Why it went unnoticed. Both symbols sat beside the same formula, 1/(1 - gamma), so a reader never met the two at once and every surface hand-wrote whichever it had seen last.

Legitimate where: inside a correction note, a version history, or a passage discriminating the two symbols.

Law III, the ARC Bound

Replaced by Law III, the ARC Ceiling, 2026-08-30.

Why it was wrong. The ARC Bound (alpha <= 2) is the value the ARC Ceiling returns at gamma = 1/2, not the law; naming the instance as the law hides the assumption the value rests on.

Legitimate where: inside a dated correction row or version note; as printed in the DOI-frozen papers, which carry the canonical name in their placement notes.

Law III, the Ceiling

Replaced by Law III, the ARC Ceiling, 2026-08-30.

Why it was wrong. The programme prefix is part of the law's name; the bare word is a description, and a description can attach to any ceiling.

Legitimate where: inside a dated correction row or version note; as printed in the DOI-frozen papers.

the Co-Scaling Law

Replaced by the ARC Co-Scaling Law, 2026-08-30.

Why it was wrong. Law II's name carries the programme prefix like the other two; the bare form drifted onto surfaces written from memory.

Legitimate where: inside a dated correction row; as printed in the DOI-frozen papers, whose placement notes carry the canonical name.

What is still open

- Is the exponent in the error-decay equation the same alpha as the capability-growth exponent, or a distinct alignment-specific one? Never stated.
- The independence premise behind gamma = one half is undefended. It is the single assumption holding up the value two, and it is deliberately left open.
- EQ-05 supplies an unexploited falsifier: measure beta_L independently, predict alpha, compare against the fitted alpha. No unit currently does this.
- Cohen's d and the programme's dimensional d share a letter and cannot both be renamed, because one belongs to statistics rather than to this programme. Which of the programme's own uses takes a subscript is an open decision.
- Does build order actually change the corrector's class enough to move the leverage exponent? The argument says order is a lever; whether pulling it separates the classes in practice is unmeasured, and it is the cheapest decisive study the programme has specified.

Version history

Version 1.9.1 (5 October 2026): the title now follows the programme's uniform order (paper, its law or framework, its own title), Register · The ARC Theory · Operational Definitions; until this version it read Operational Definitions of the ARC Theory. No equation, symbol, collision or retracted form changes.

Version 1.9.0 (4 October 2026): published as a paper in every format, with its own DOI on OSF and its own reference card, under the title Operational Definitions of the ARC Theory (until this version, Operational Definitions of the ARC Programme). The note on what the word law is doing counts the theory's five laws and their status, and the description and the abstract say where the notation of Laws IV and V is fixed. No equation, symbol, collision or retracted form changes.

Version 1.8.0 (7 September 2026): thirty-three of the fifty-one individual symbol names printed in these equations had no rendered definition block, because the renderer looked names up in a dictionary keyed by the spine's full composite strings and silently dropped anything it did not find, because six equations did not name in symbols_used the symbols their own statement prints, and because eight symbols appear in no equation at all. Every symbol's domain and units were carried in this file and rendered nowhere. The lookup now fails loudly, the composite keys resolve member by member, the orphans have a standalone section, the domain and units render, and the gate tests for a rendered definition block rather than for a name occurring somewhere on the page, which 46 of the 51 already did. It supersedes 1.7.2 (sha256 0ae774690effd36e9c19eafee863300dfce38ca2f2687ffaae20a969923af927), not 1.7.1. 1.7.2 (2026-09-04): the delta family was declared a collision with readings, a severity and a why-it-matters and no resolution key, so build-operationalisation.py raised KeyError('resolution') and the production chain died on it. The resolution now names the quantity rather than the letter: bare delta is the cost-scaling exponent and nothing else, the direct-bound unit's pressure reading is written delta_obs in full and is positive when burden outruns correction, the balance model's reading is written Delta in full and is positive when correction gains on burden, and because the two are equal in magnitude and opposite in sign at the registered dials neither may be quoted bare. One field was added to one collision in commit 68bbb55f91; no equation, symbol, definition or count changed, and the register still carried 21 equations, 38 symbols, 7 collisions, 7 retracted forms and 5 open questions after it as before. Those bytes hash to sha256 0ae774690effd36e9c19eafee863300dfce38ca2f2687ffaae20a969923af927 and stood under the

number 1.7.1 from 4 September 2026 until the 1.8.0 bump. They are given their own number here, in the record and not by editing them, so that 1.7.1 names one set of bytes and one only: 1.7.1 (2026-09-03) is the deposited state, sha256 68891fa12cf3ee4a58d1a7d278aecdd9ed170366339e36cc7ab478027bbbeb48, which the theory registration's companion extract of 3 September 2026 cites and which nothing in this record changes. 1.7.0 (2026-09-03): the gamma collision, which is the register's worst, pointed one way only. The chi entry recorded that it is the registration's gamma; the gamma entry did not record that the quantity is now written chi, so a reader arriving at the collision was not told the current name existed. 1.6.0 (2026-08-27): worked example ran a bounded percentage through unbounded arithmetic; recast on the ladder-scale latent measure with the three model families stated (27 August 2026); then, 30 August 2026, the law labels were aligned with the home declaration (Law II is the ARC Co-Scaling Law; Law III is the ARC Ceiling, of which the ARC Bound is the same-class instance) and alpha_max was retired as a second symbol for alpha_crit

First published on 22 August 2026; versions 1.5.0 (24 August), 1.6.0 to 1.6.2 (27 to 30 August) and 1.7.0 to 1.7.1 (3 September) built and corrected the spine, each dated by its commit.